

Shatter Clustering: Search Radius (r) Sensitivity Analysis

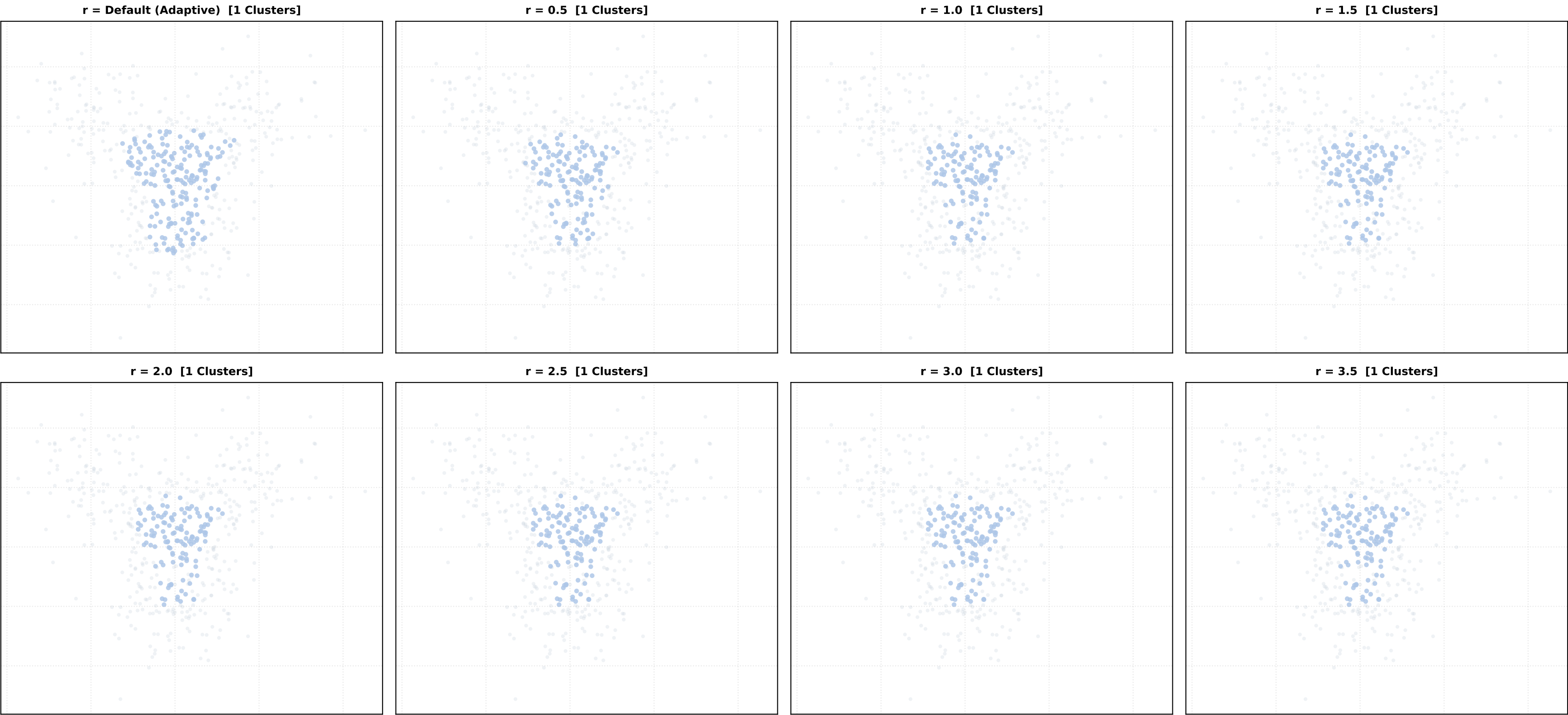
Sweep over Default Adaptive r vs Explicit $r \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5\}$

Experimental Configuration

- Total Evaluated Datasets: 30
- Tested Radius Values: Default (Adaptive), 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5
- Total Executions: 240 (8 runs per dataset)
- Fixed Default Parameters: alpha_MAD = 0.5, alpha_accept = 0.5, t_percentile = 0.15
- Target Output PDF: benchmark_report_r.pdf

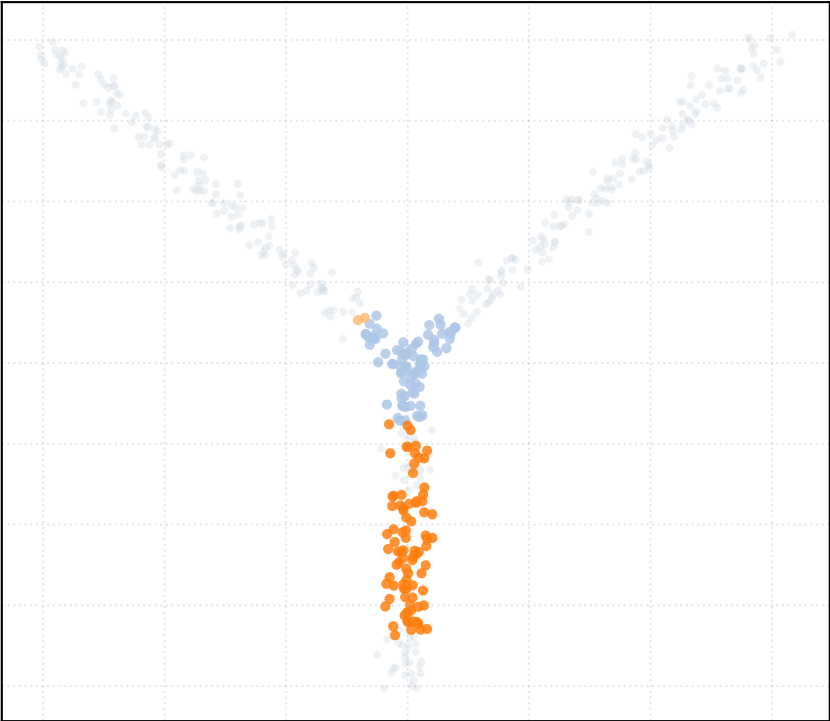
Parameter	Value / Range	Algorithmic Role
Radius r	{Default, 0.5, 1.0, 1.5, 2.0, 2.5}	Search sphere radius passed to grow_ensemble
alpha_MAD	0.50 (Default)	MAD multiplier for surface compatibility
alpha_accept	0.50 (Default)	Log2 fold-change acceptance cutoff
t_percentile	0.15 (Default)	Top 15% densest seeds
min_intersection	1 (Default)	Overlapping vectors to trigger merge

Dataset: bifurcation_2way_noise_high | Search Radius Sweep (r in grow_ensemble)

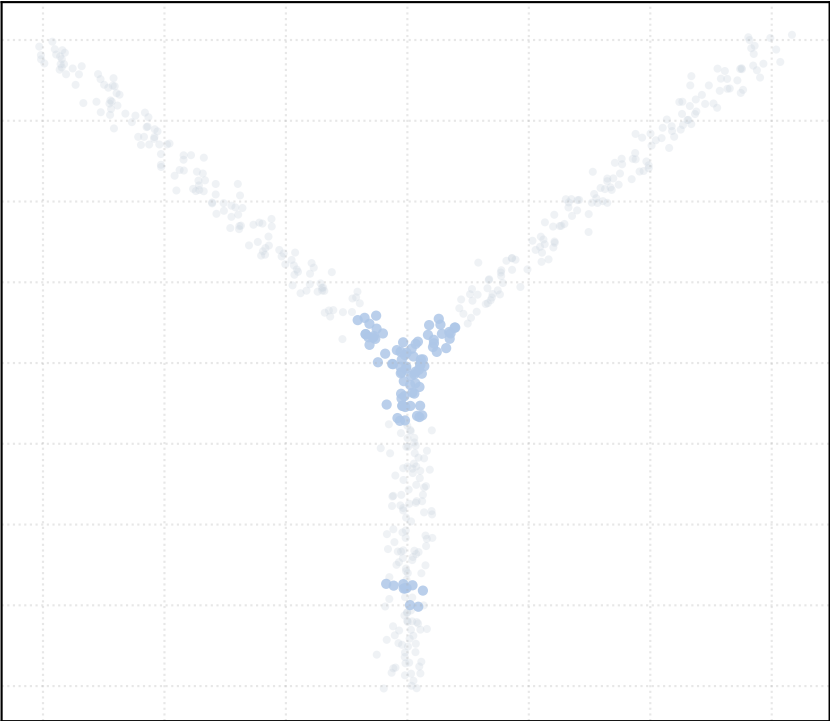


Dataset: bifurcation_2way_noise_low | Search Radius Sweep (r in grow_ensemble)

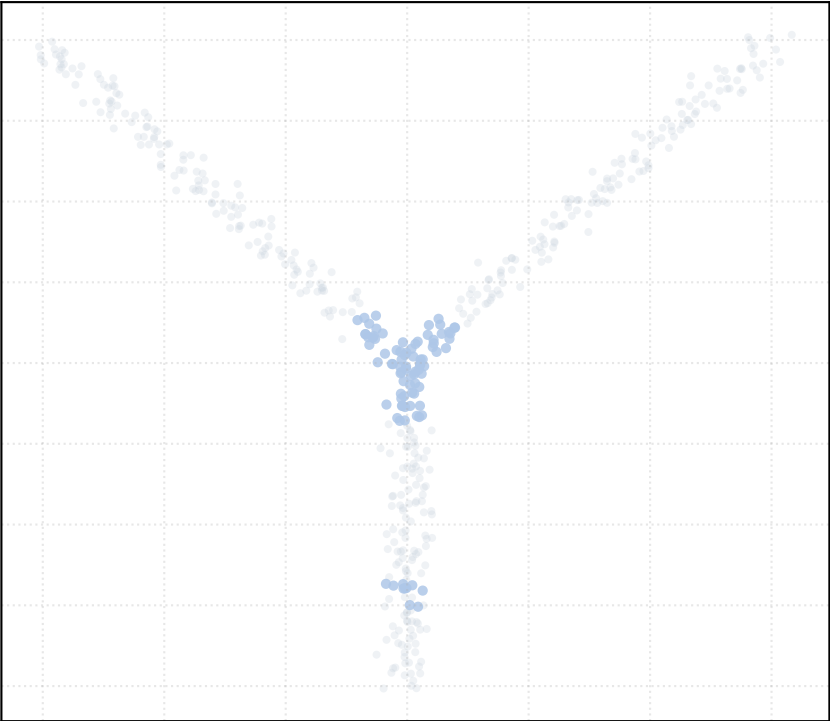
r = Default (Adaptive) [3 Clusters]



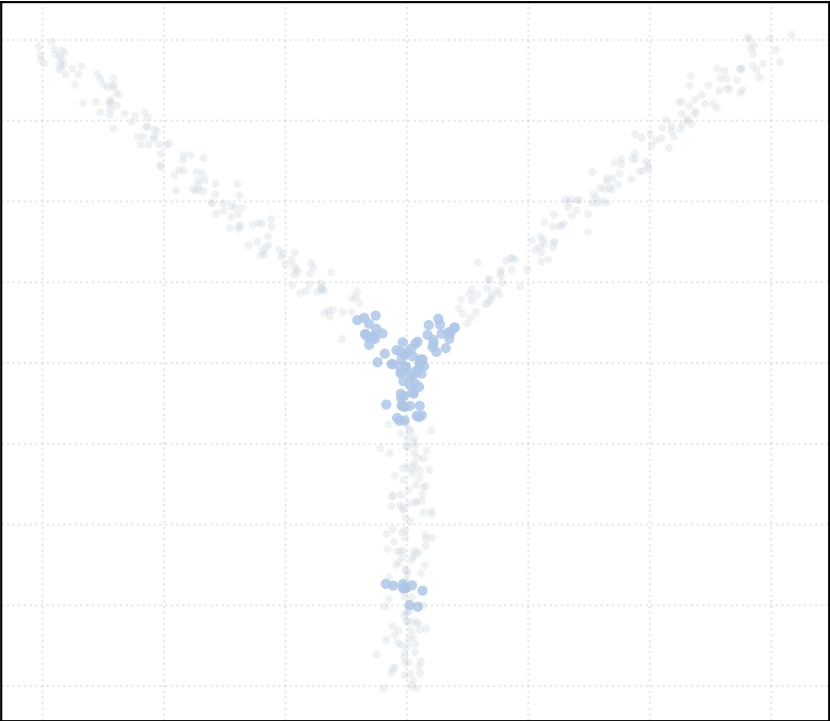
r = 0.5 [1 Clusters]



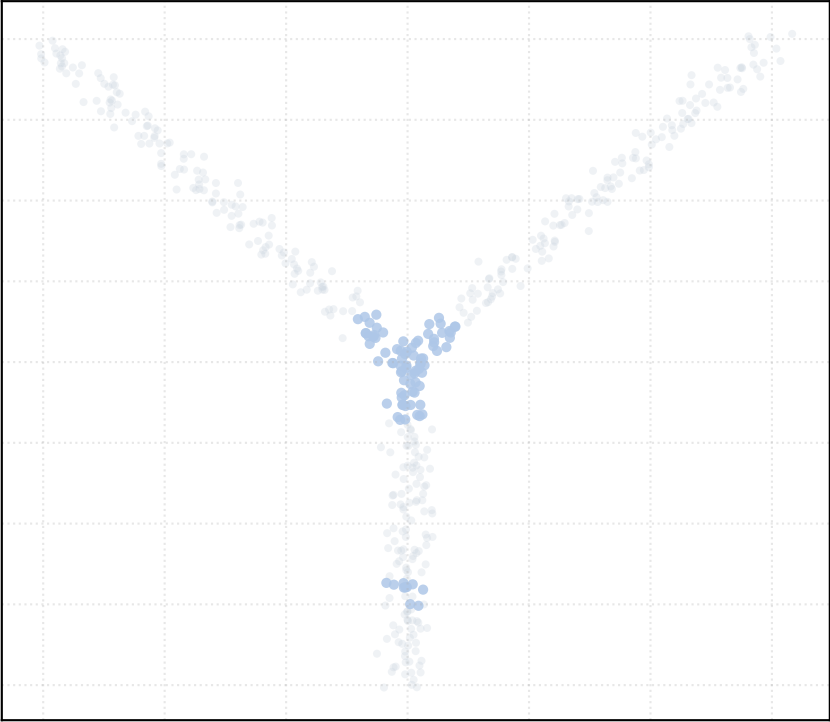
r = 1.0 [1 Clusters]



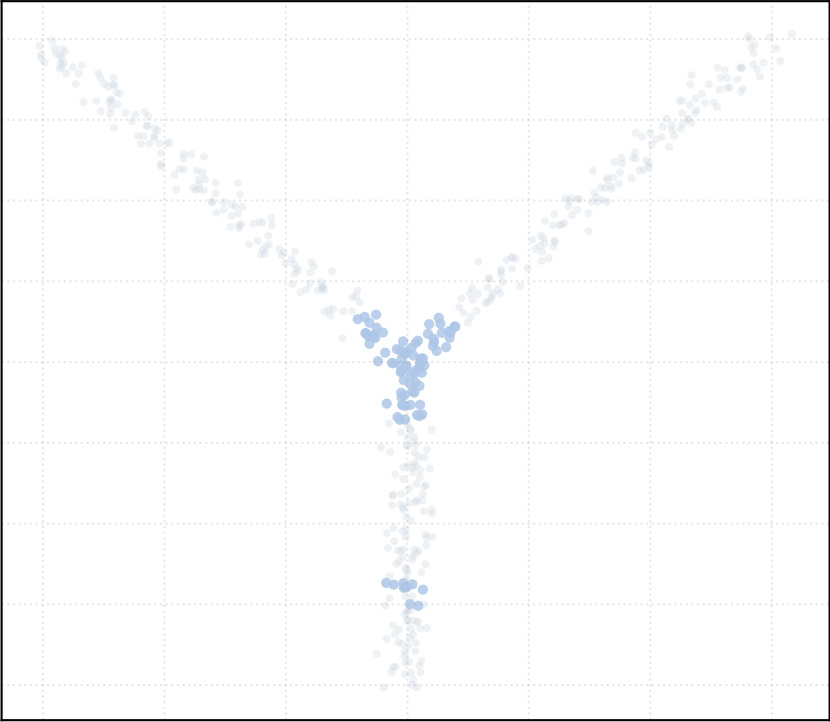
r = 1.5 [1 Clusters]



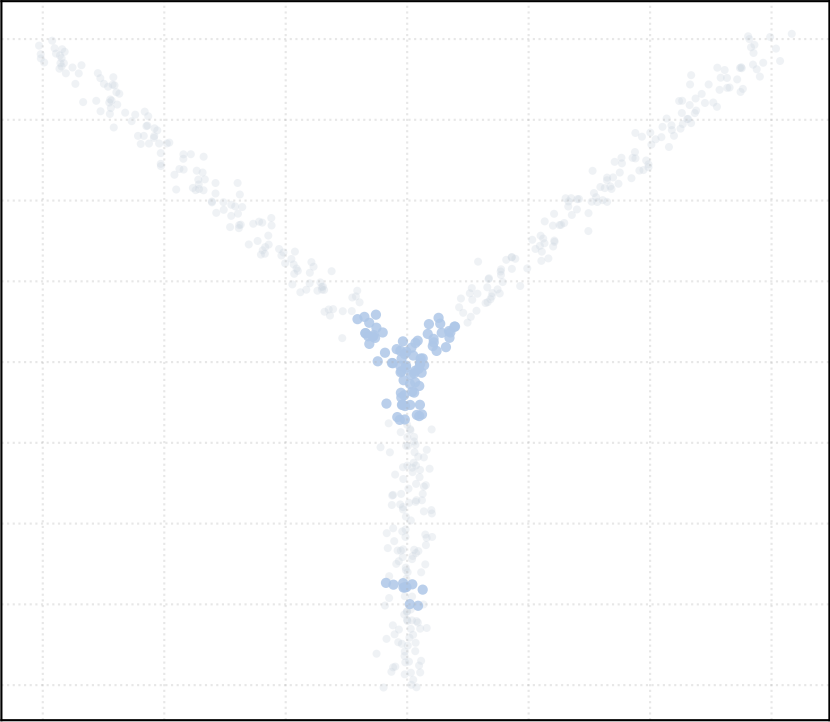
r = 2.0 [1 Clusters]



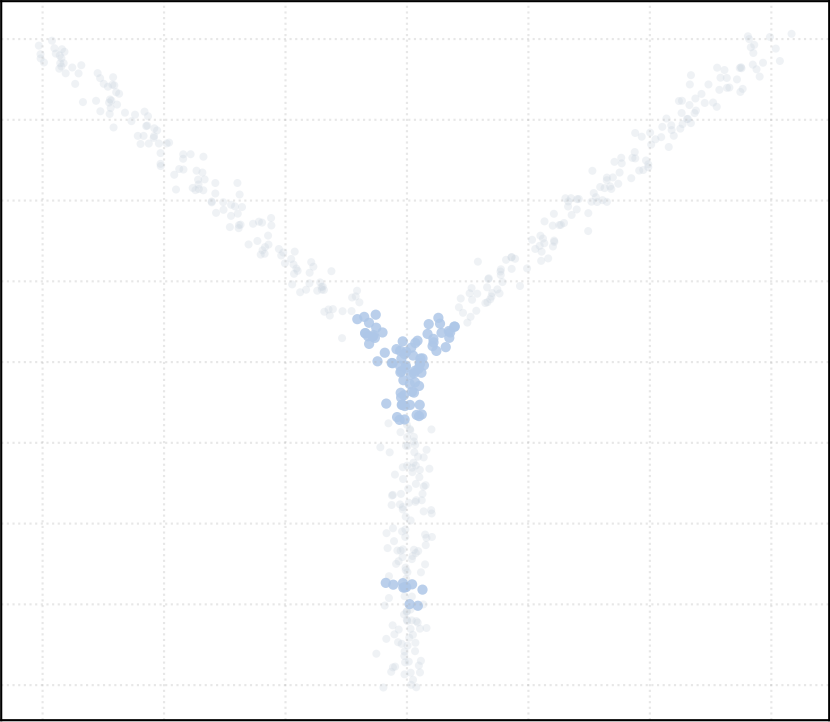
r = 2.5 [1 Clusters]



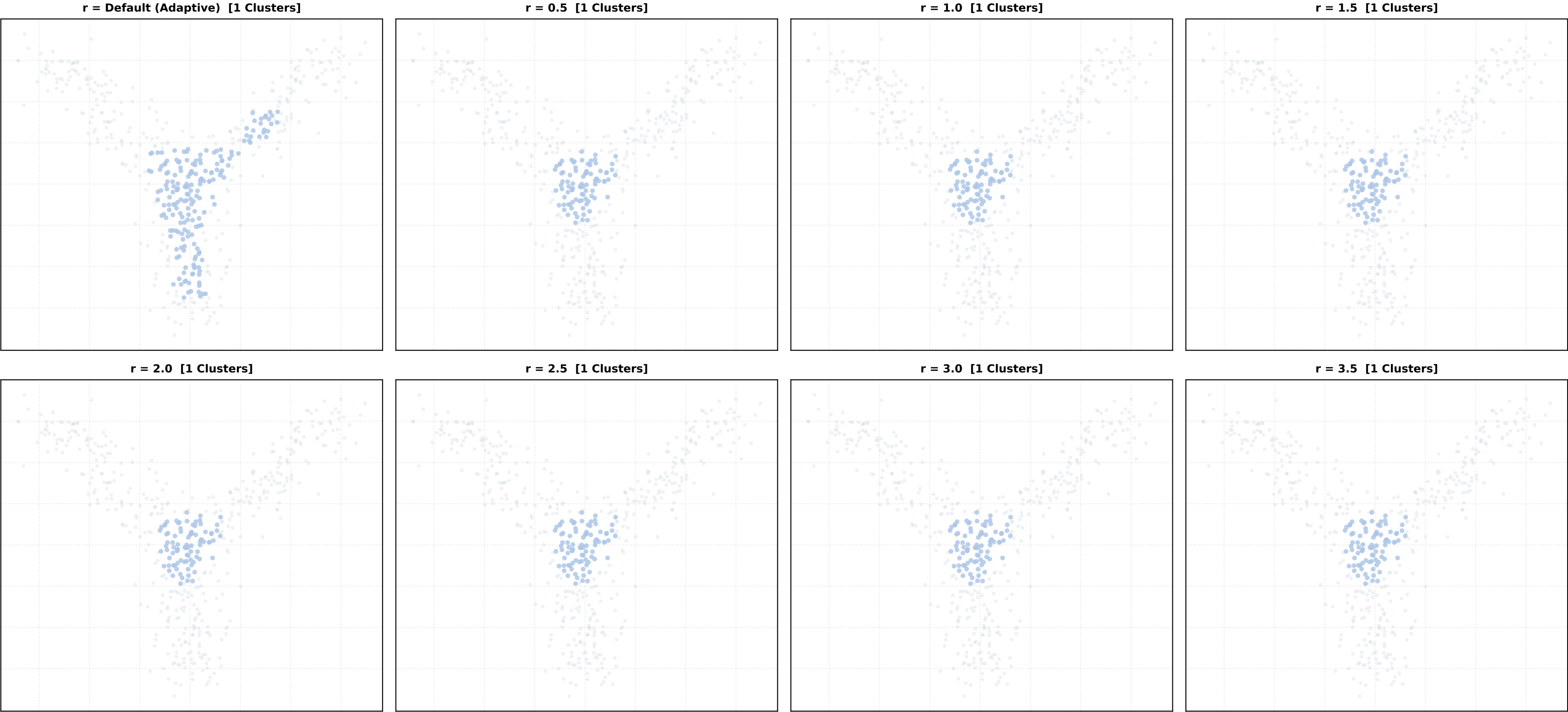
r = 3.0 [1 Clusters]



r = 3.5 [1 Clusters]

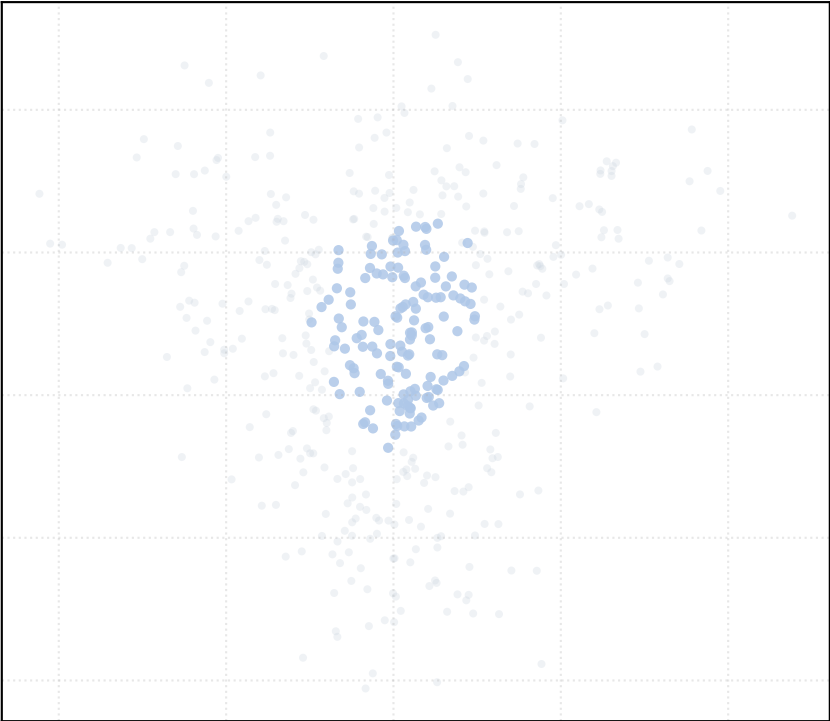


Dataset: bifurcation_2way_noise_medium | Search Radius Sweep (r in grow_ensemble)

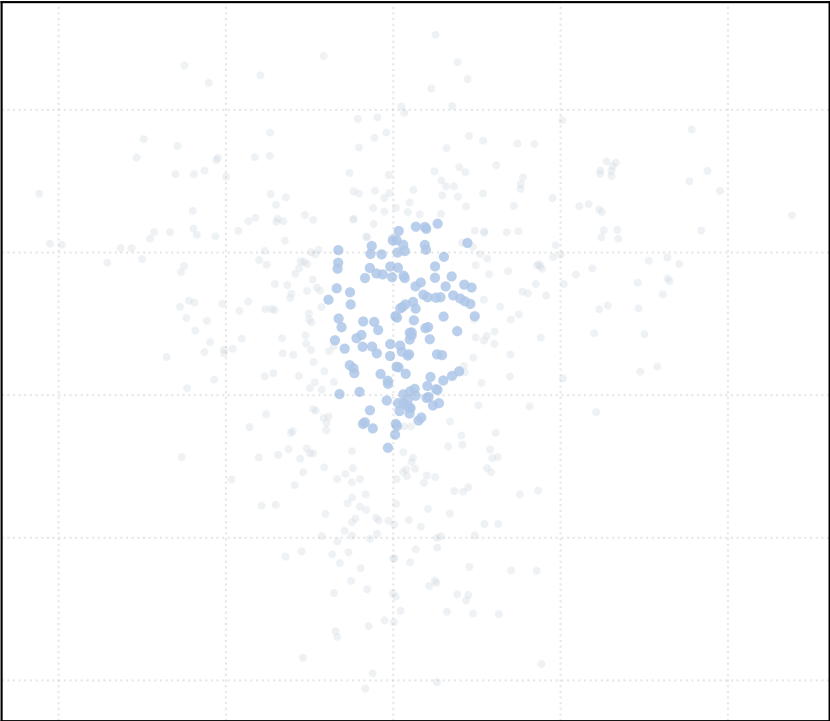


Dataset: bifurcation_3way_noise_high | Search Radius Sweep (r in grow_ensemble)

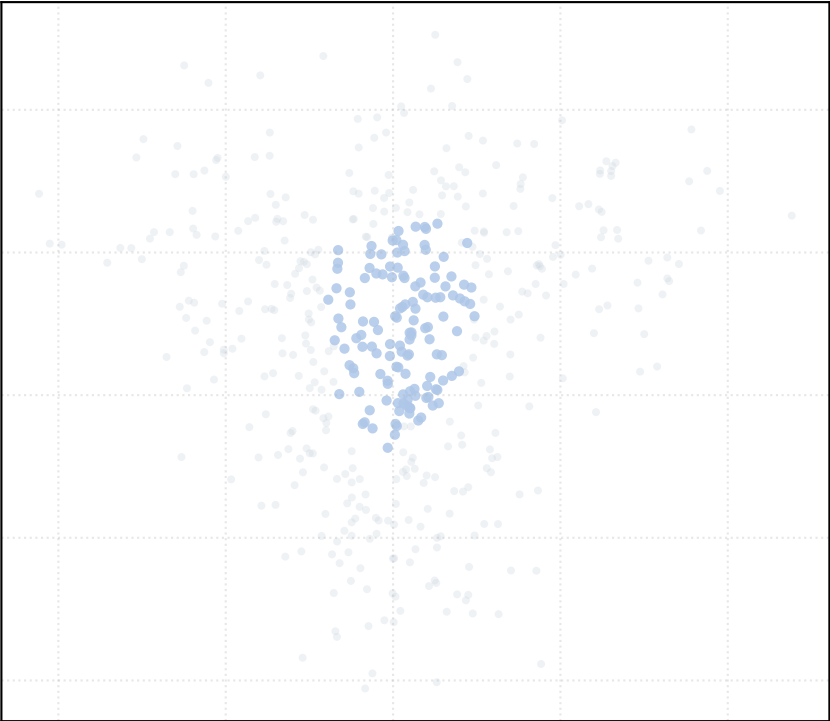
r = Default (Adaptive) [1 Clusters]



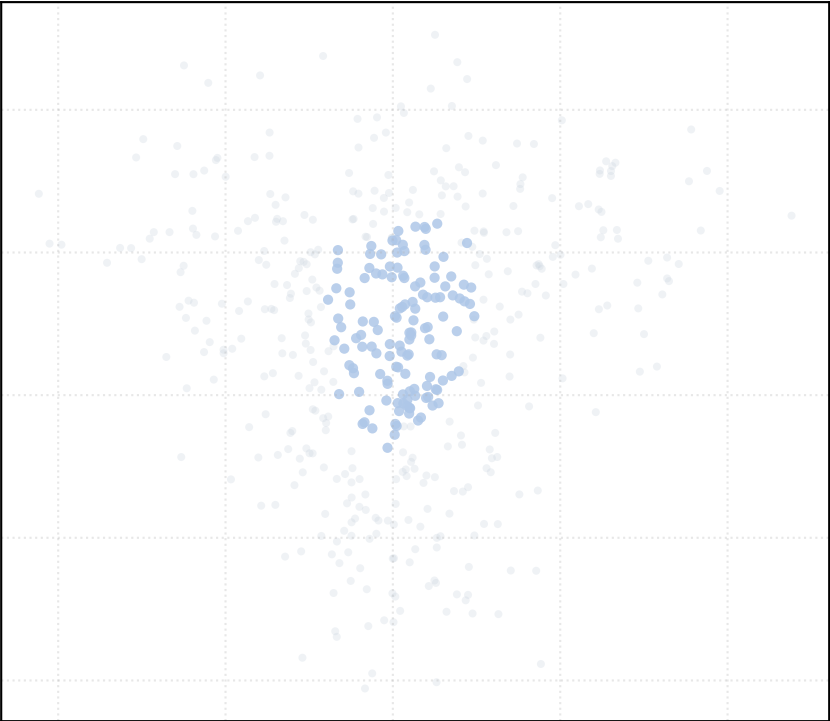
r = 0.5 [1 Clusters]



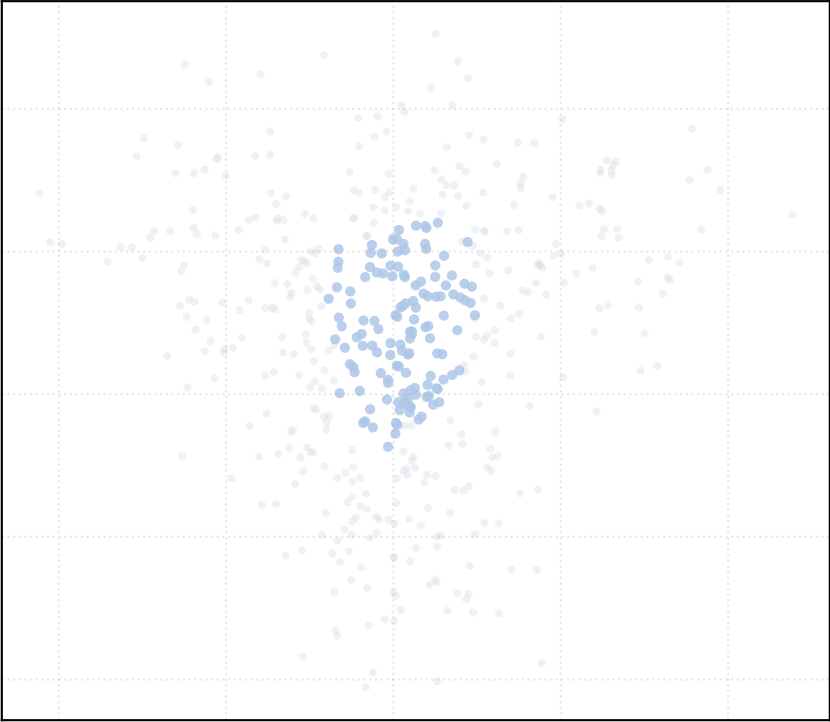
r = 1.0 [1 Clusters]



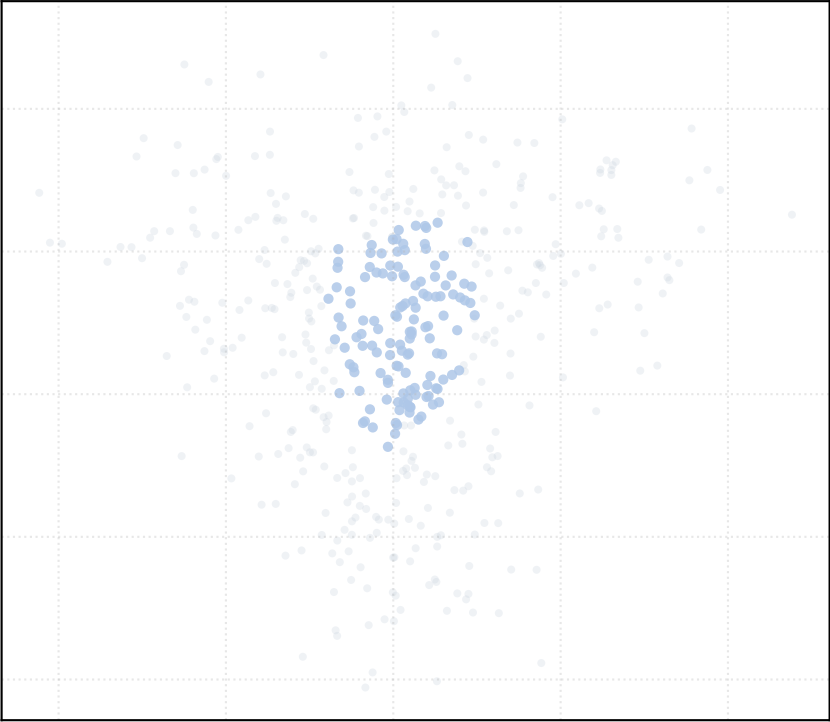
r = 1.5 [1 Clusters]



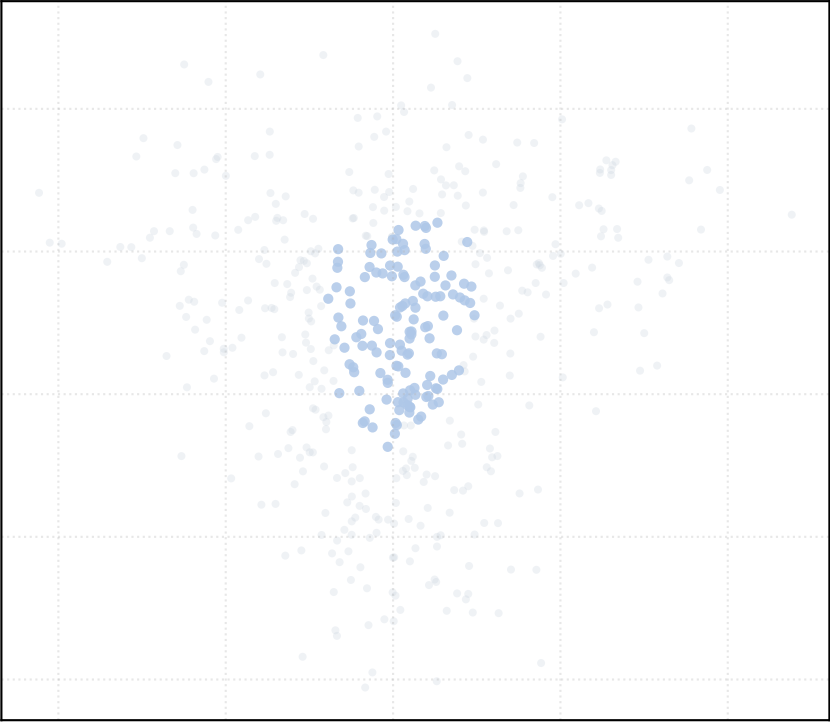
r = 2.0 [1 Clusters]



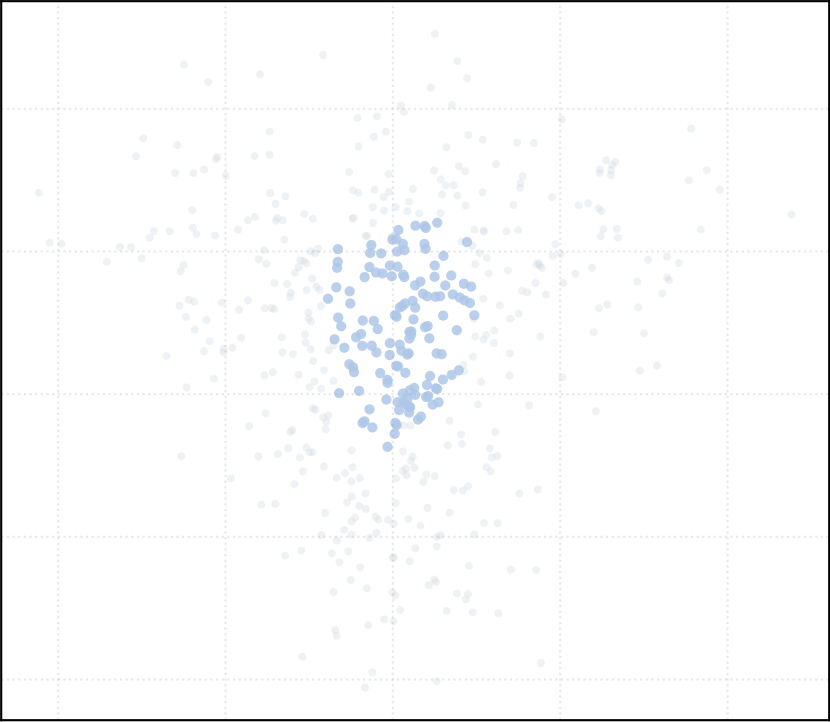
r = 2.5 [1 Clusters]



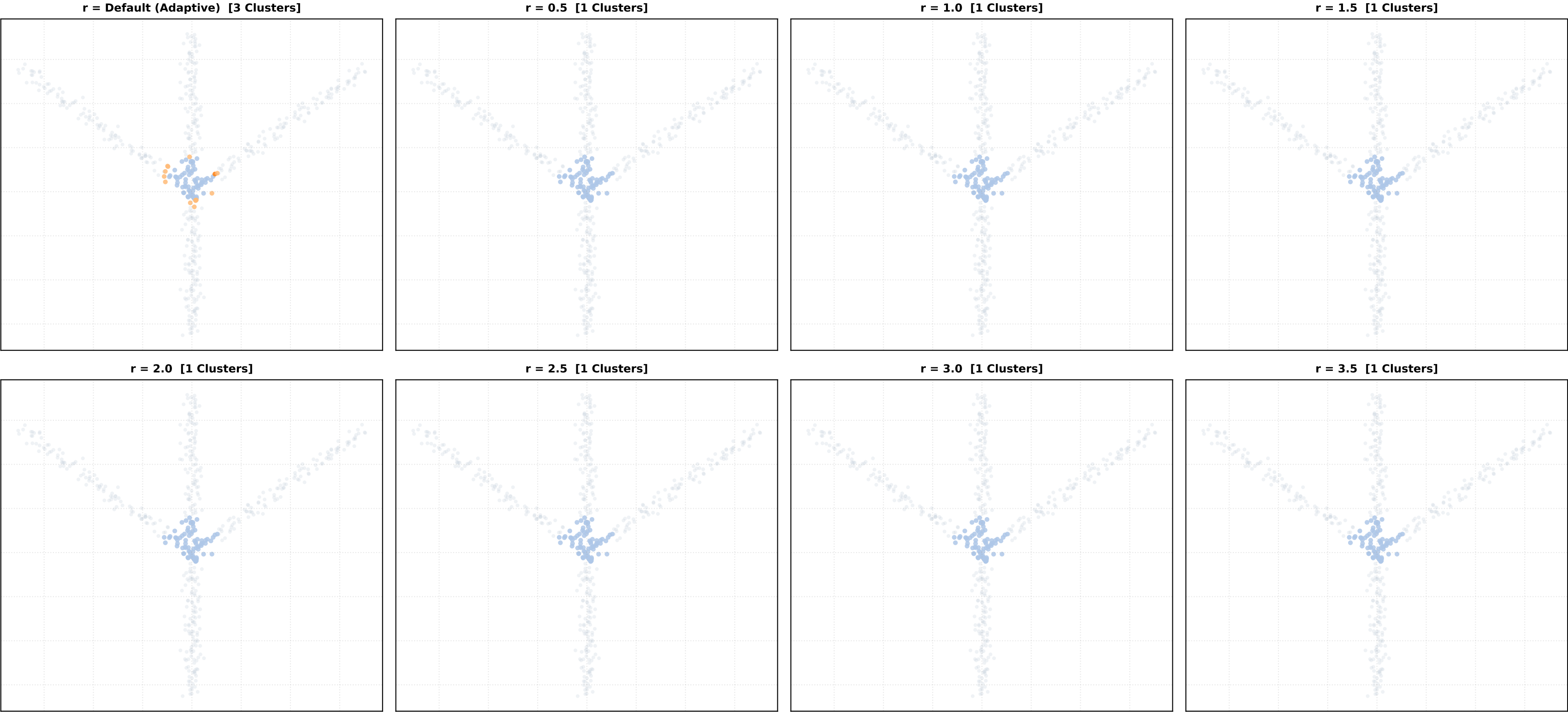
r = 3.0 [1 Clusters]



r = 3.5 [1 Clusters]

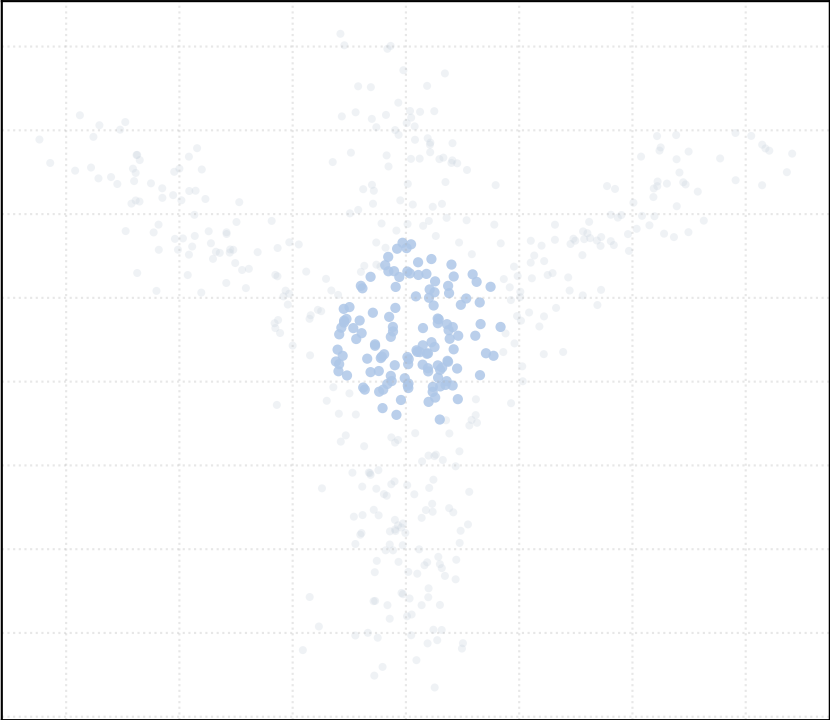


Dataset: bifurcation_3way_noise_low | Search Radius Sweep (r in grow_ensemble)

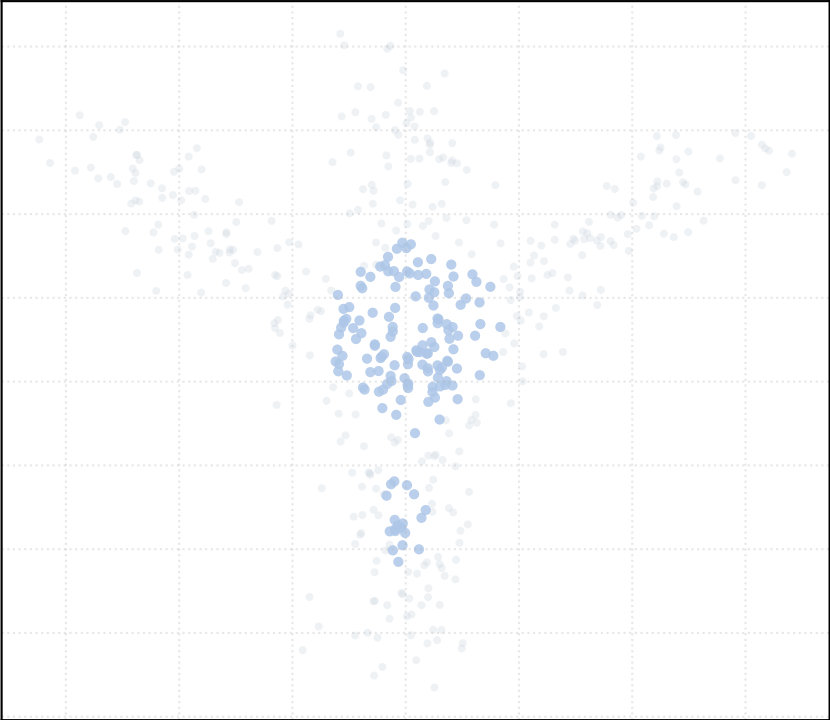


Dataset: bifurcation_3way_noise_medium | Search Radius Sweep (r in grow_ensemble)

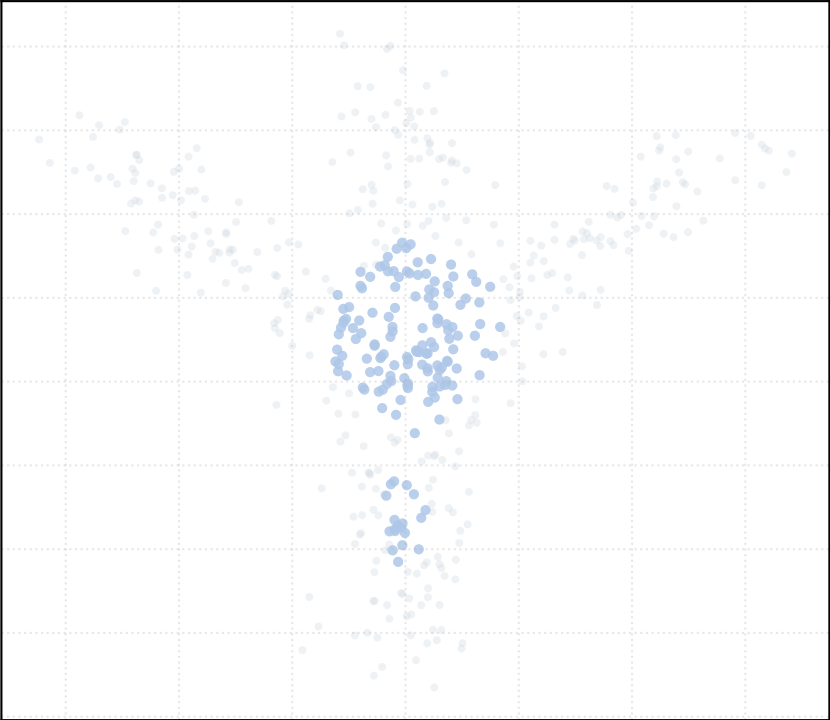
r = Default (Adaptive) [1 Clusters]



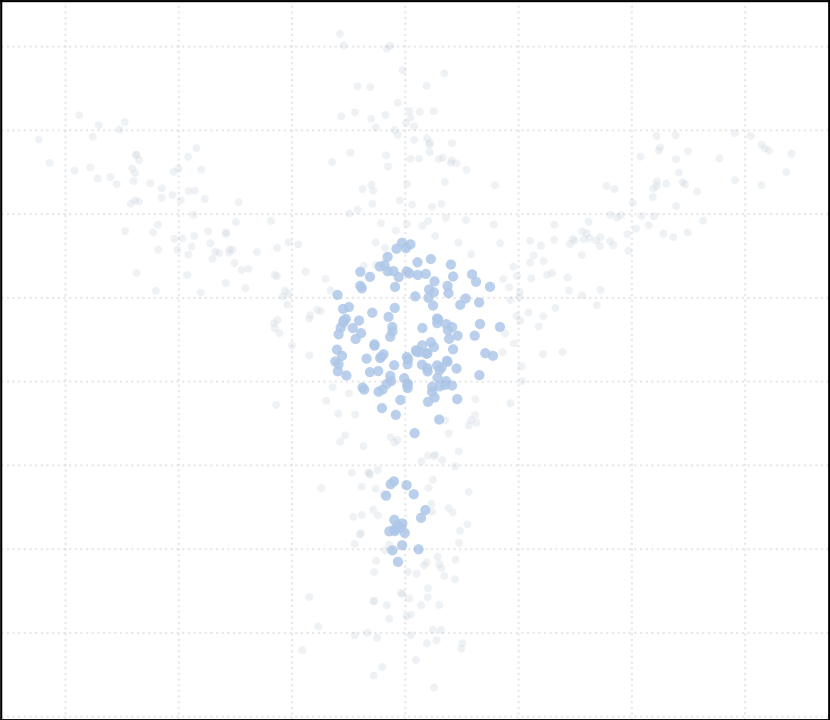
r = 0.5 [1 Clusters]



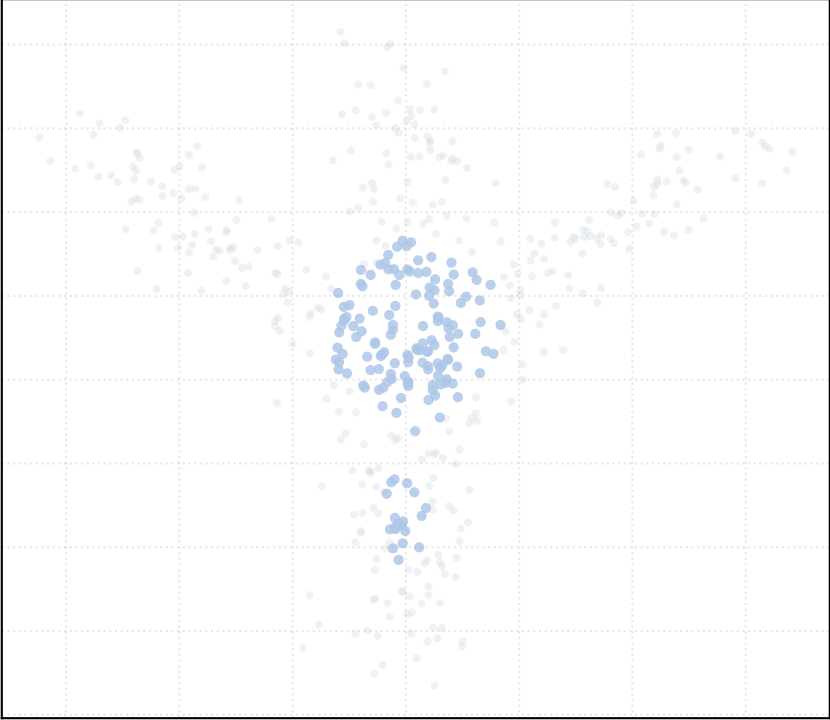
r = 1.0 [1 Clusters]



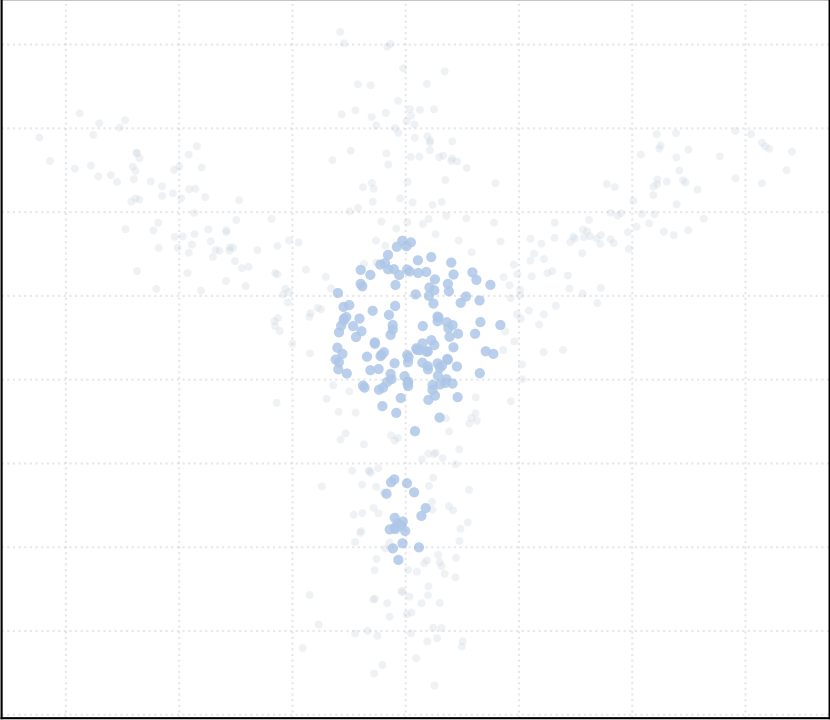
r = 1.5 [1 Clusters]



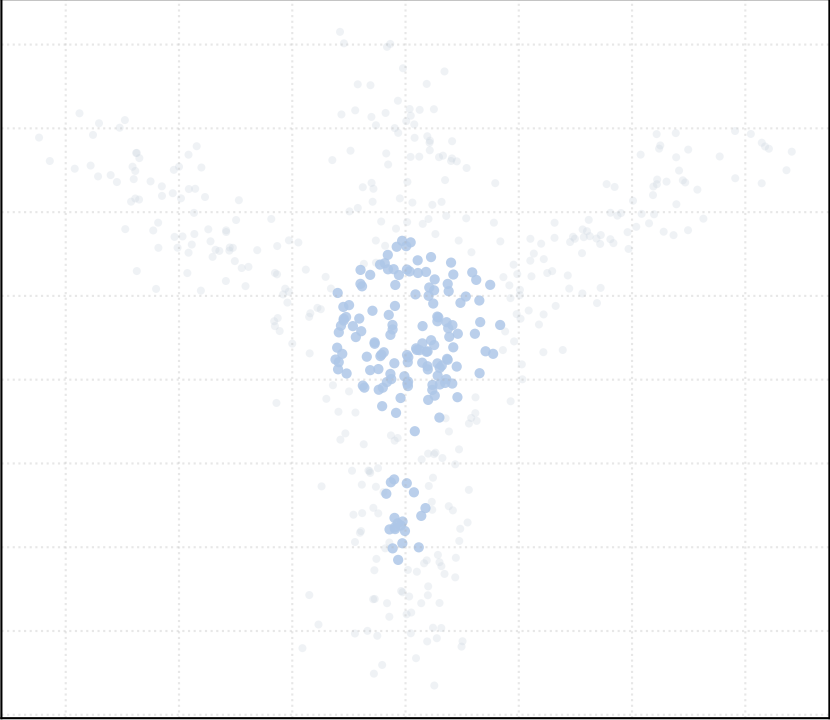
r = 2.0 [1 Clusters]



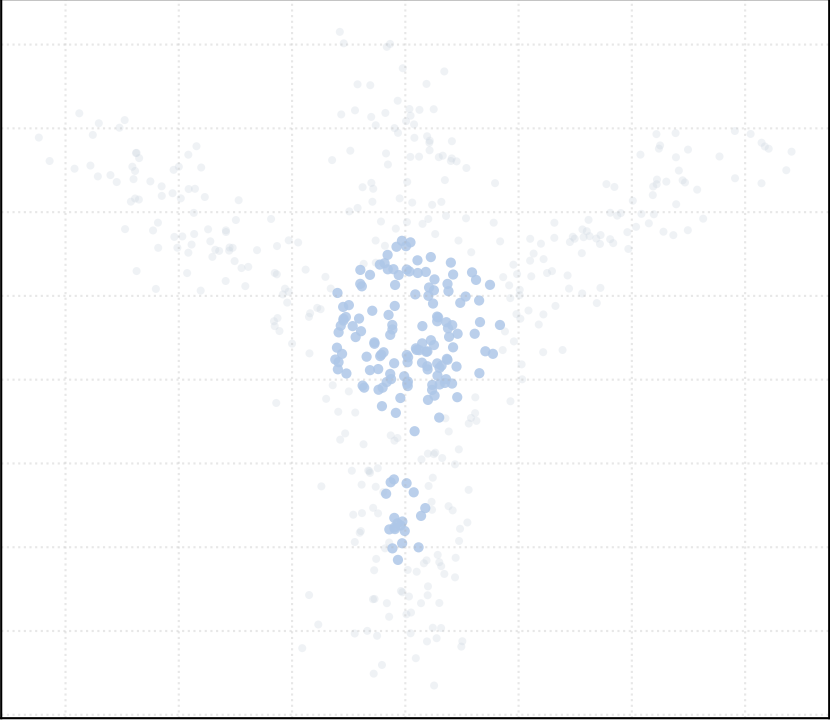
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

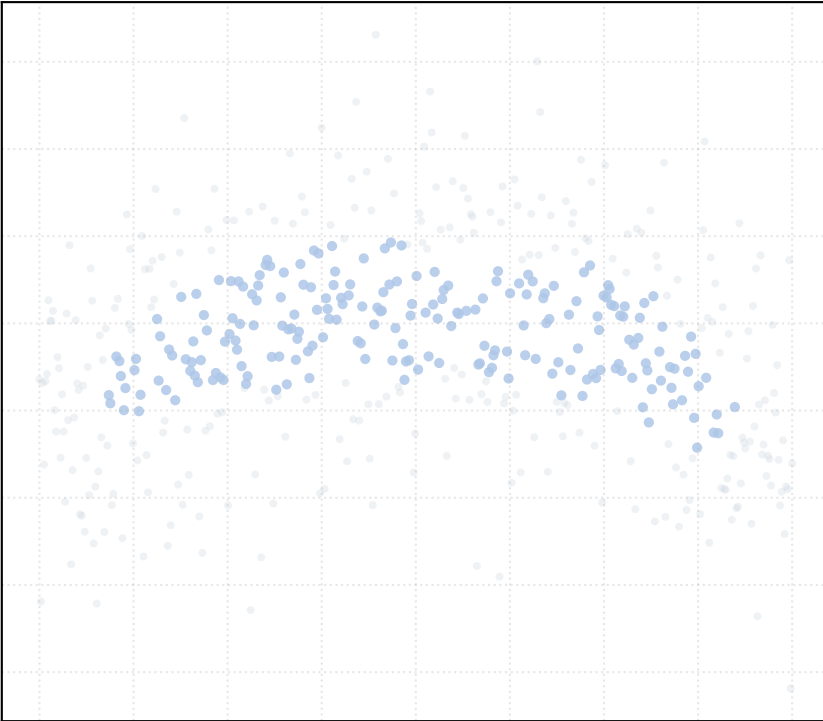


r = 3.5 [1 Clusters]

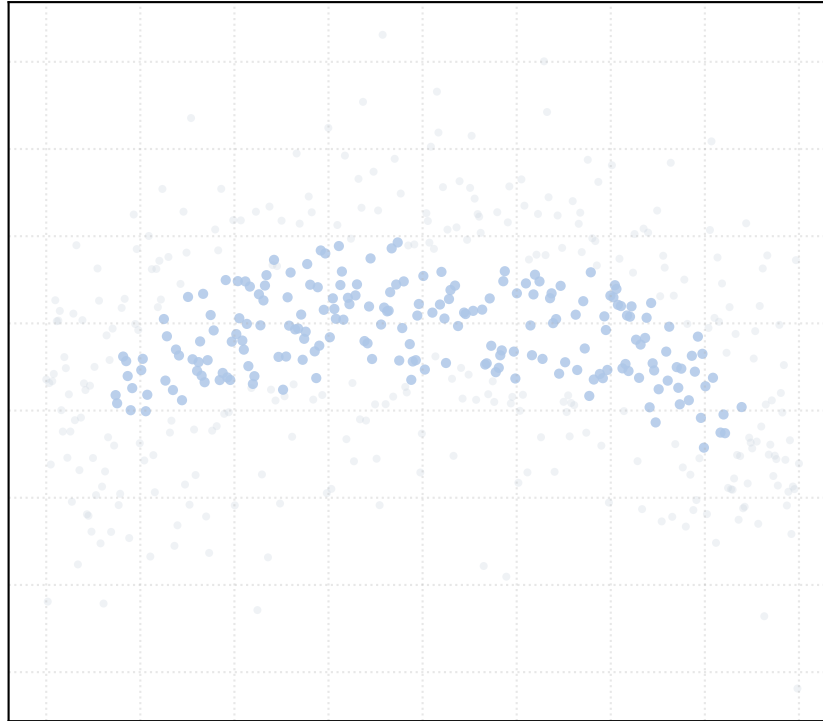


Dataset: circular_arc_noise_high | Search Radius Sweep (r in grow_ensemble)

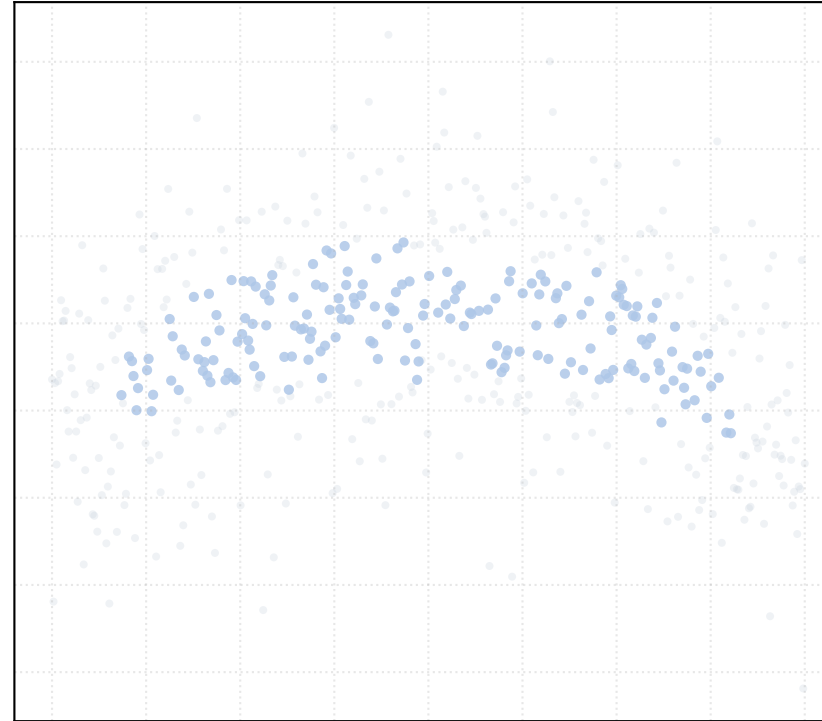
r = Default (Adaptive) [1 Clusters]



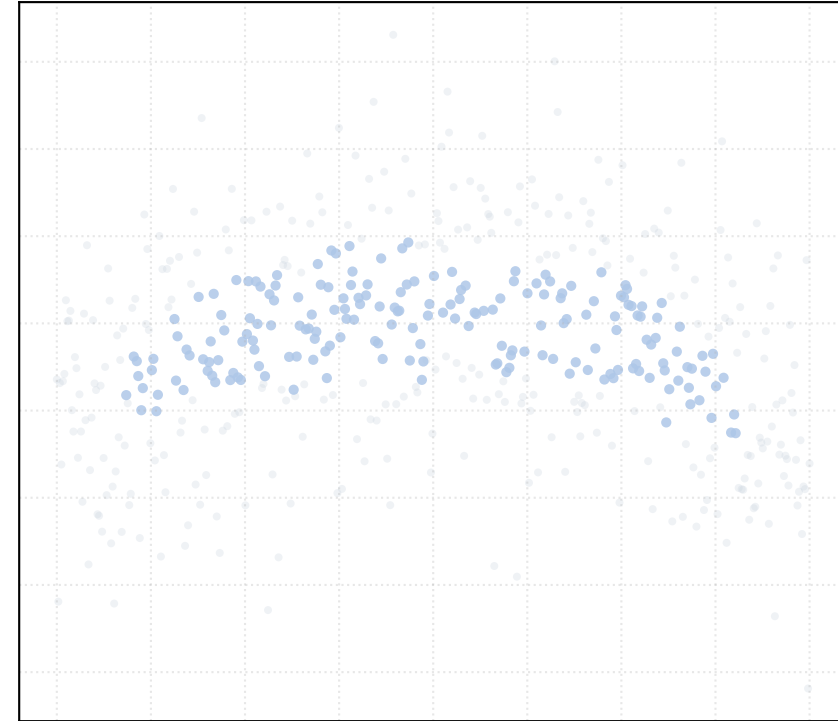
r = 0.5 [1 Clusters]



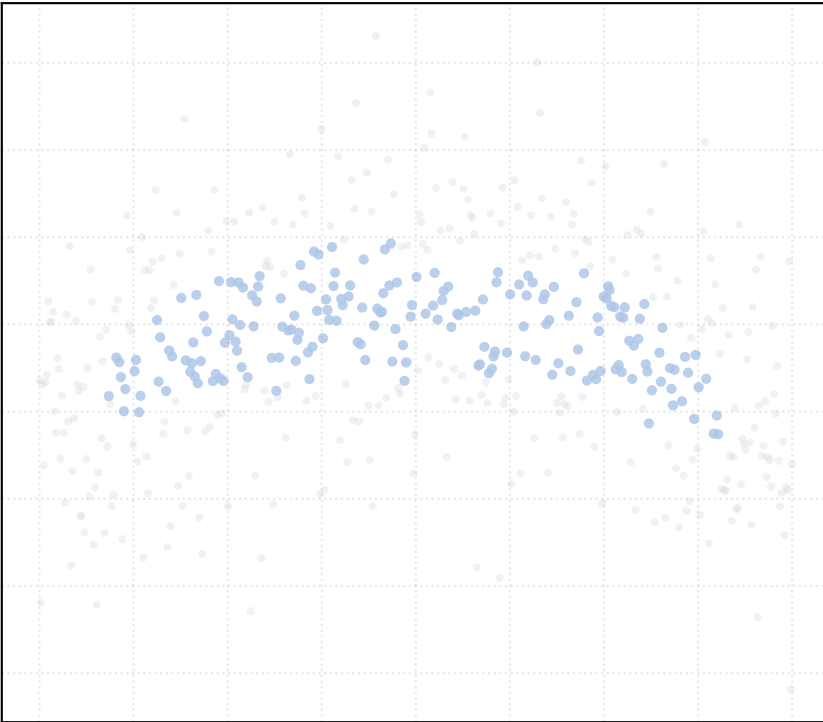
r = 1.0 [1 Clusters]



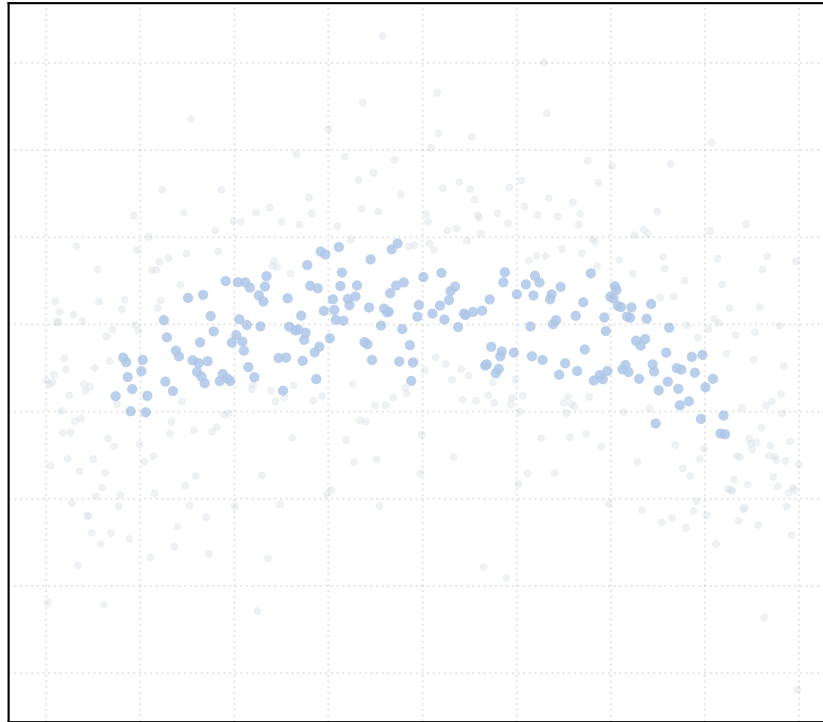
r = 1.5 [1 Clusters]



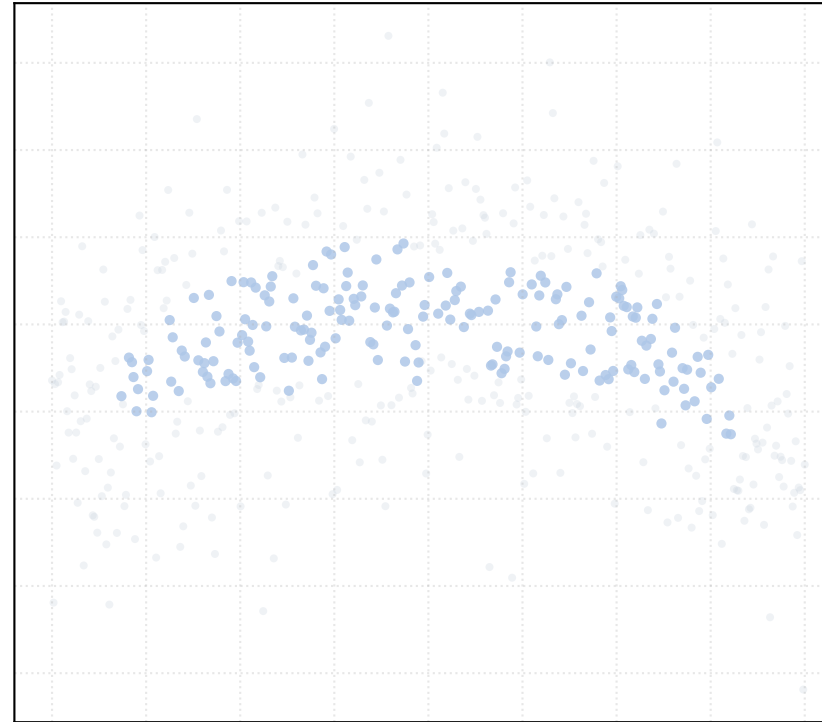
r = 2.0 [1 Clusters]



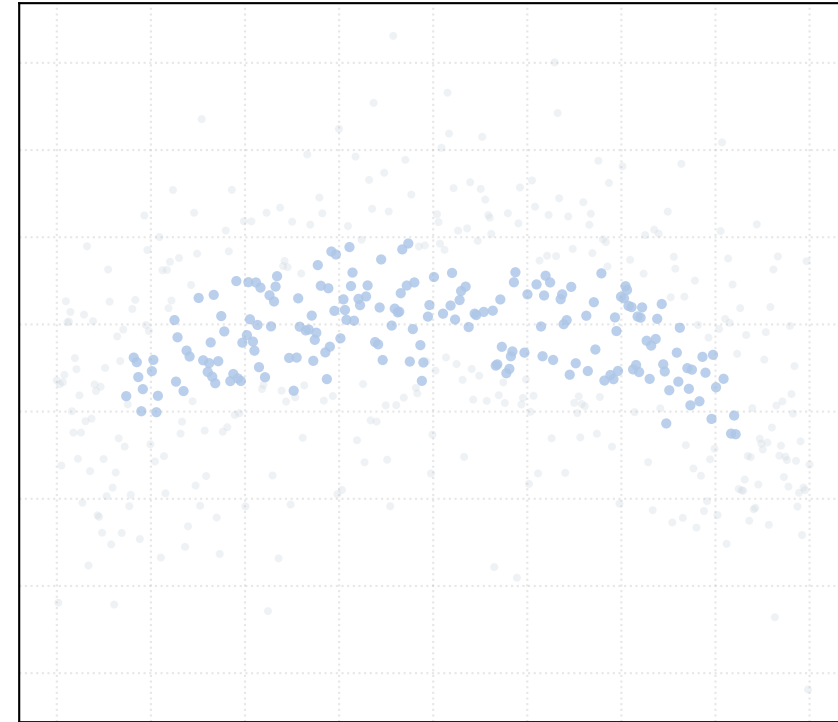
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

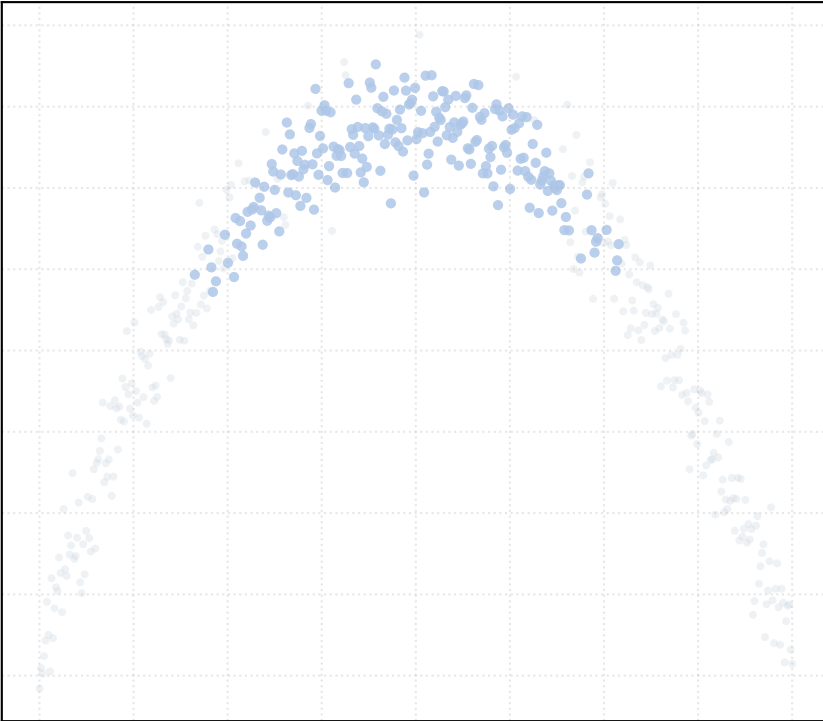


r = 3.5 [1 Clusters]

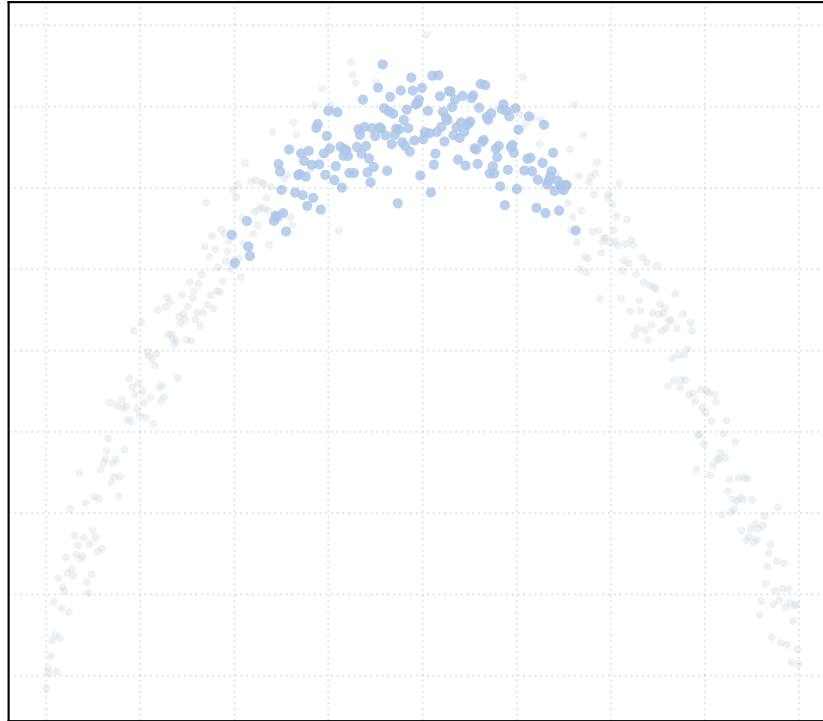


Dataset: circular_arc_noise_low | Search Radius Sweep (r in grow_ensemble)

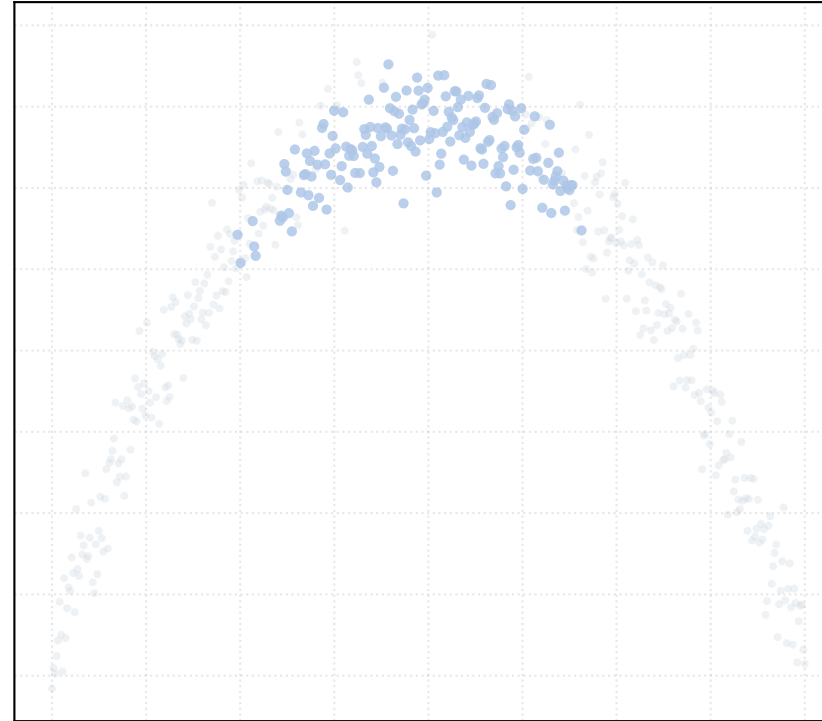
r = Default (Adaptive) [1 Clusters]



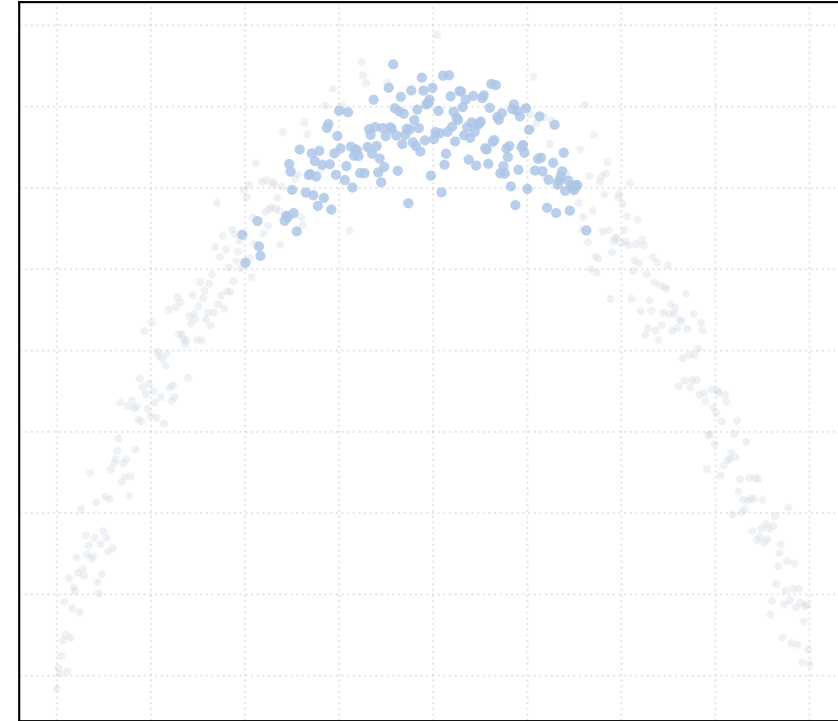
r = 0.5 [1 Clusters]



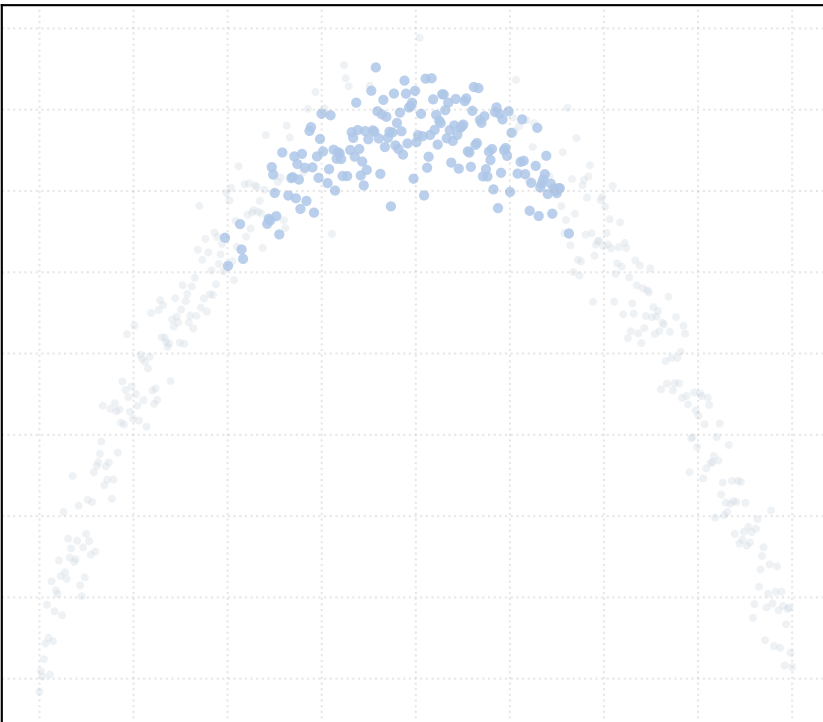
r = 1.0 [1 Clusters]



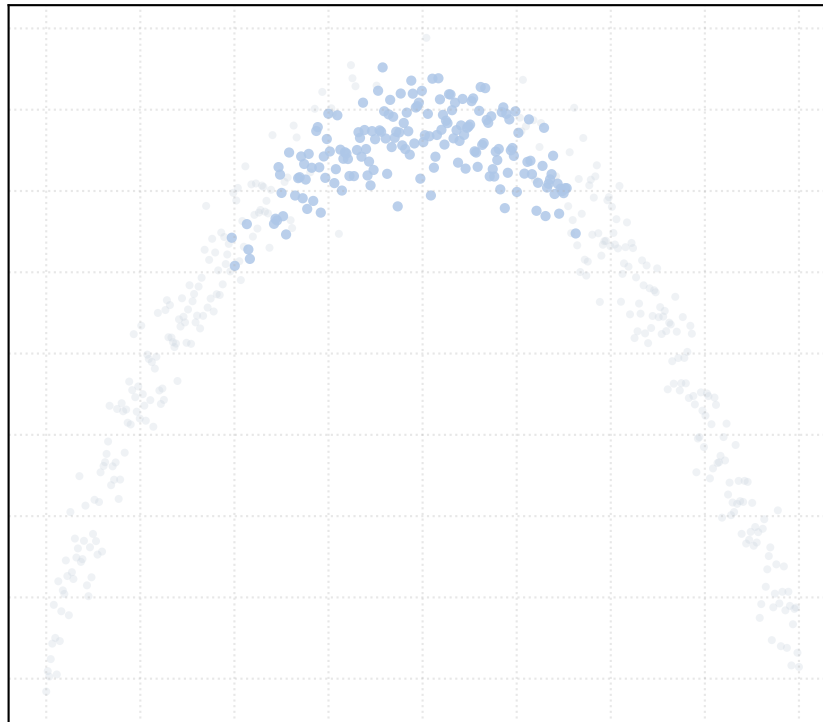
r = 1.5 [1 Clusters]



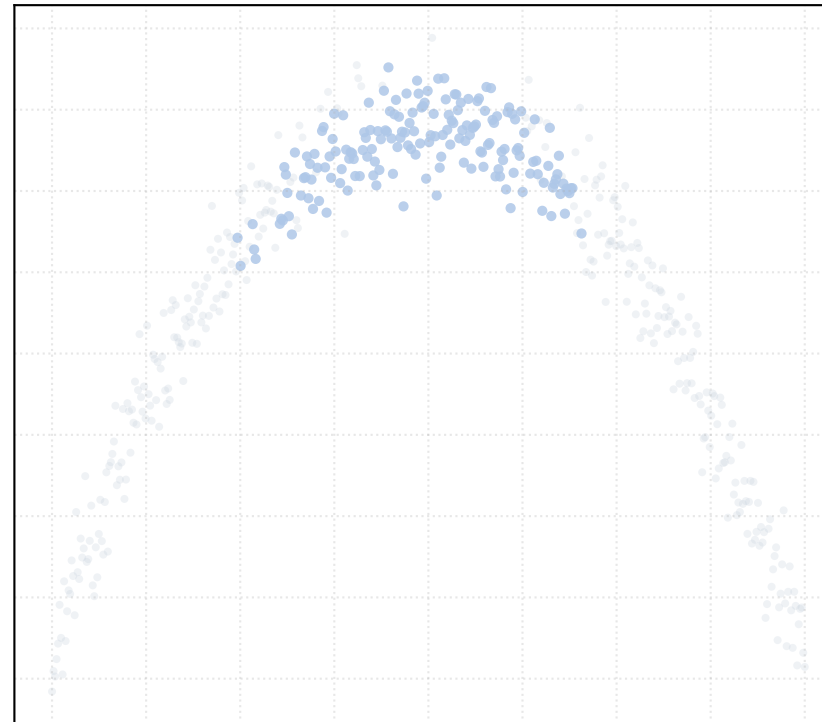
r = 2.0 [1 Clusters]



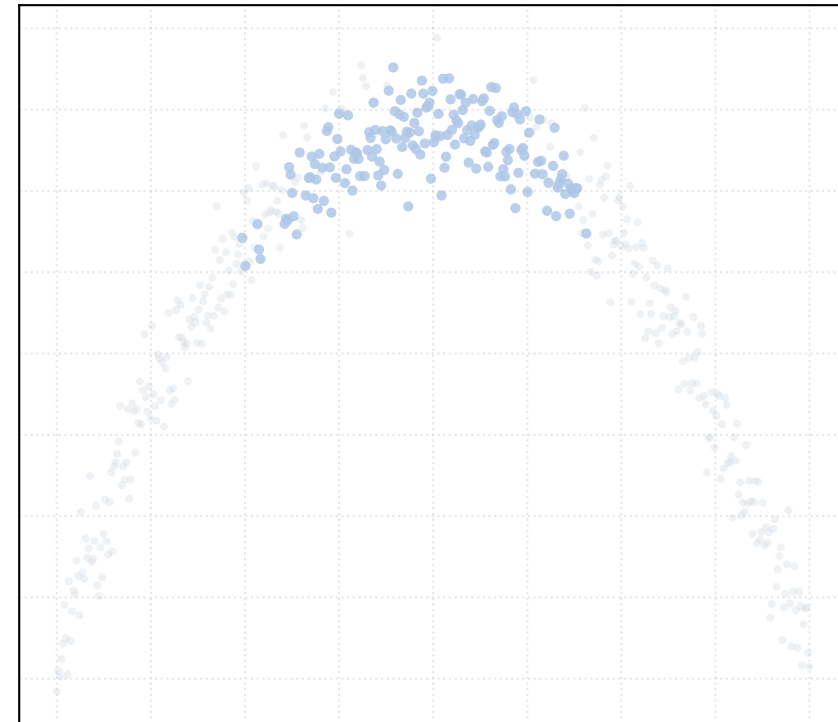
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

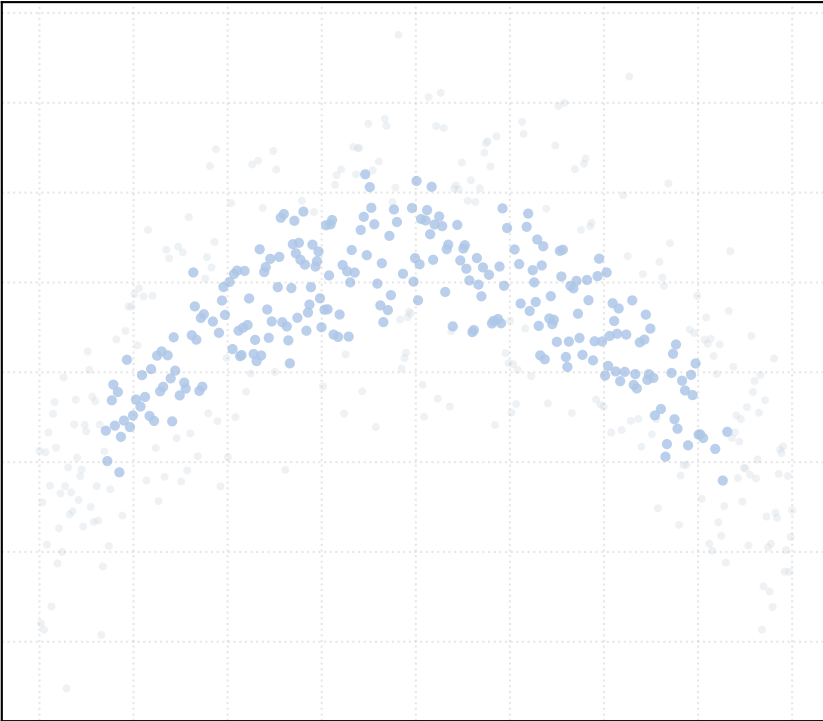


r = 3.5 [1 Clusters]

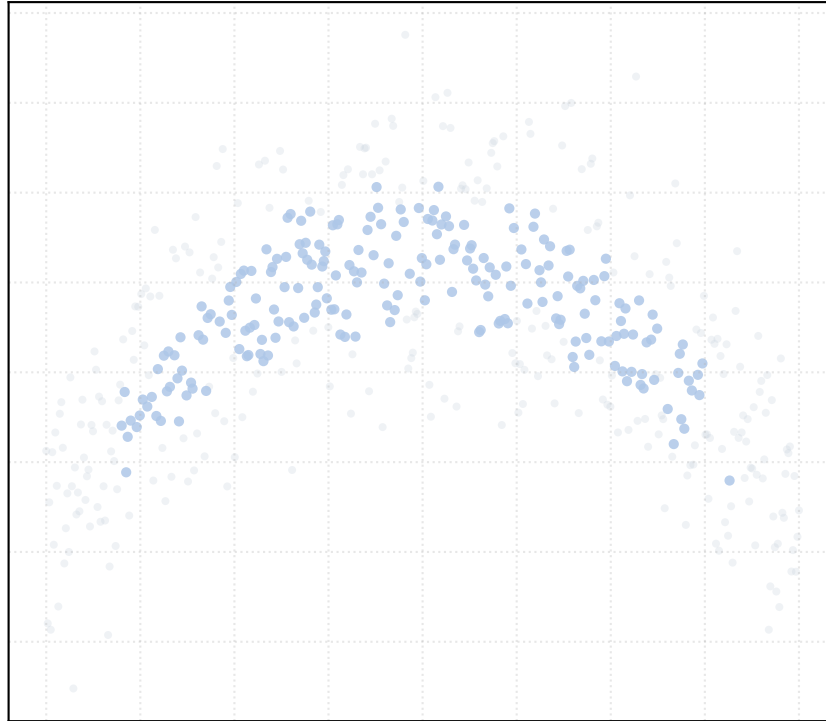


Dataset: circular_arc_noise_medium | Search Radius Sweep (r in grow_ensemble)

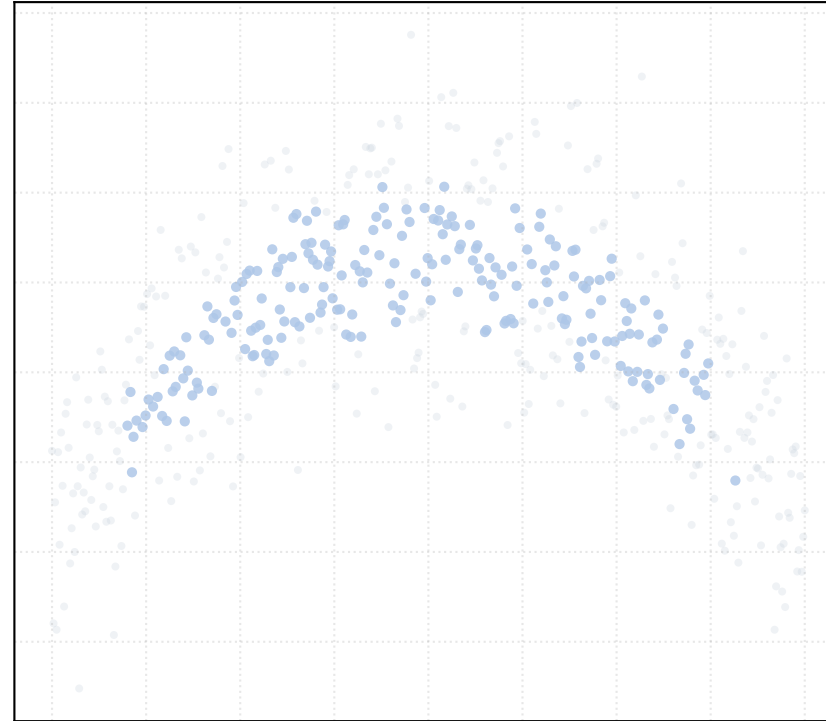
r = Default (Adaptive) [1 Clusters]



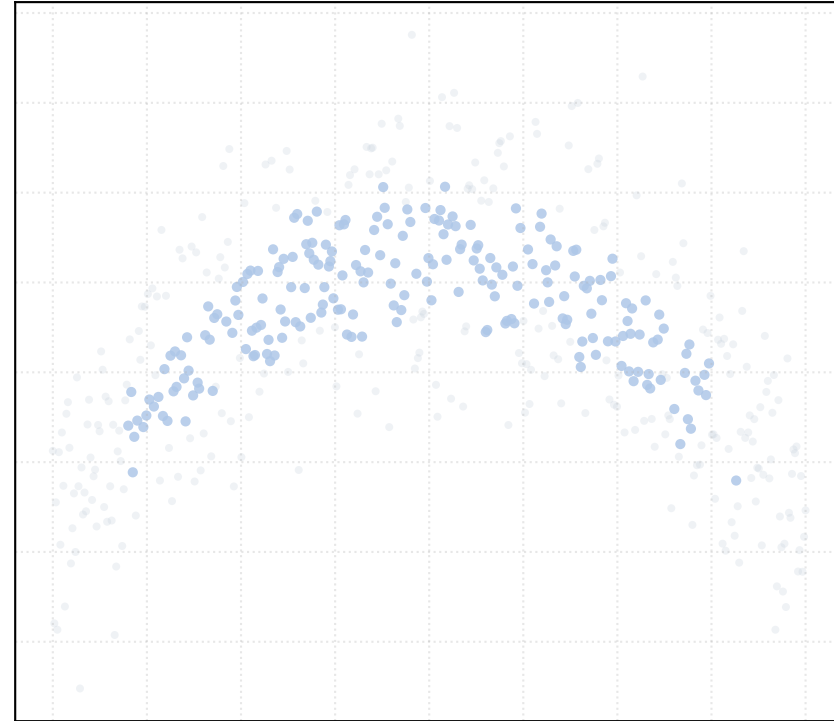
r = 0.5 [1 Clusters]



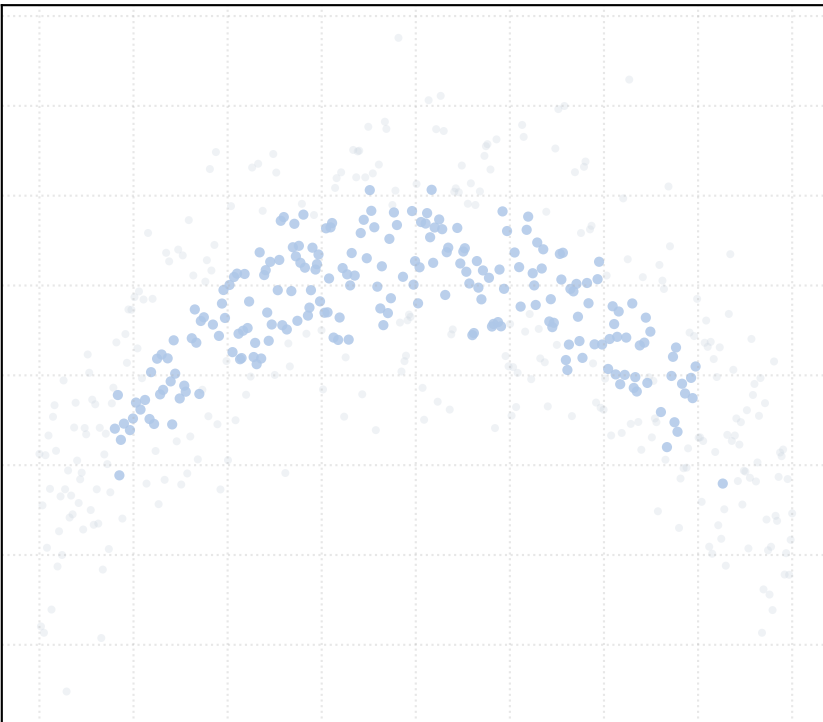
r = 1.0 [1 Clusters]



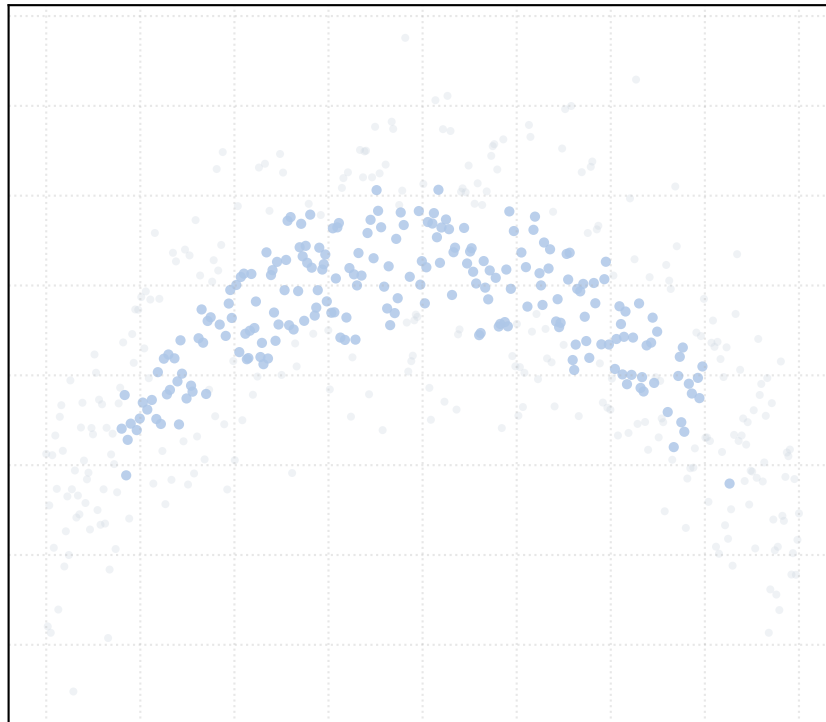
r = 1.5 [1 Clusters]



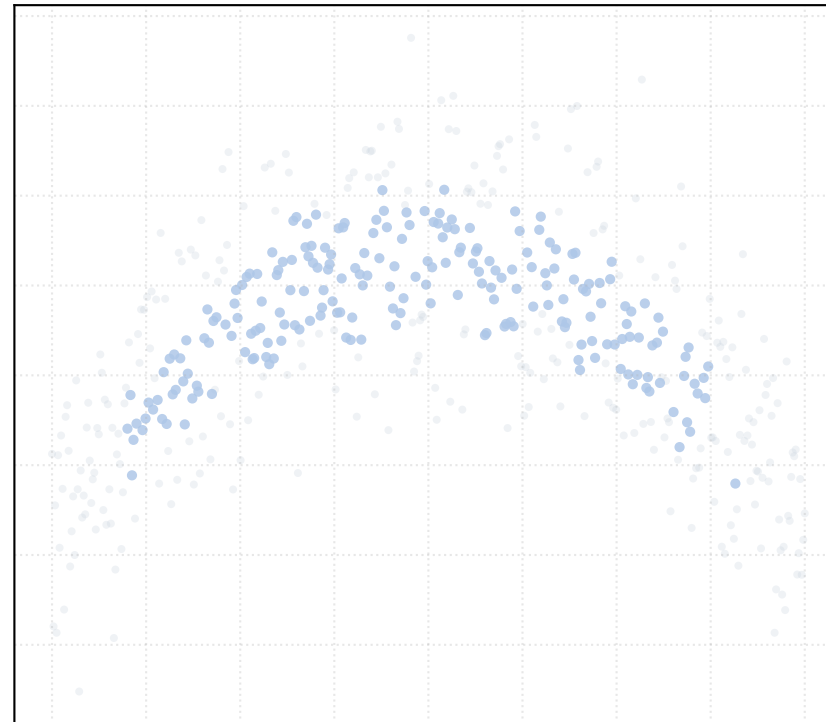
r = 2.0 [1 Clusters]



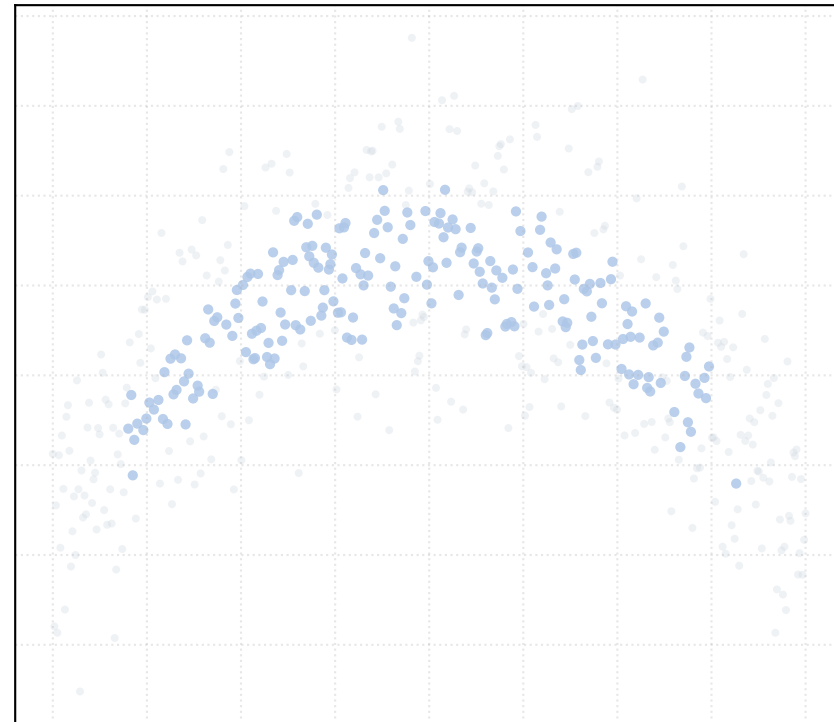
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

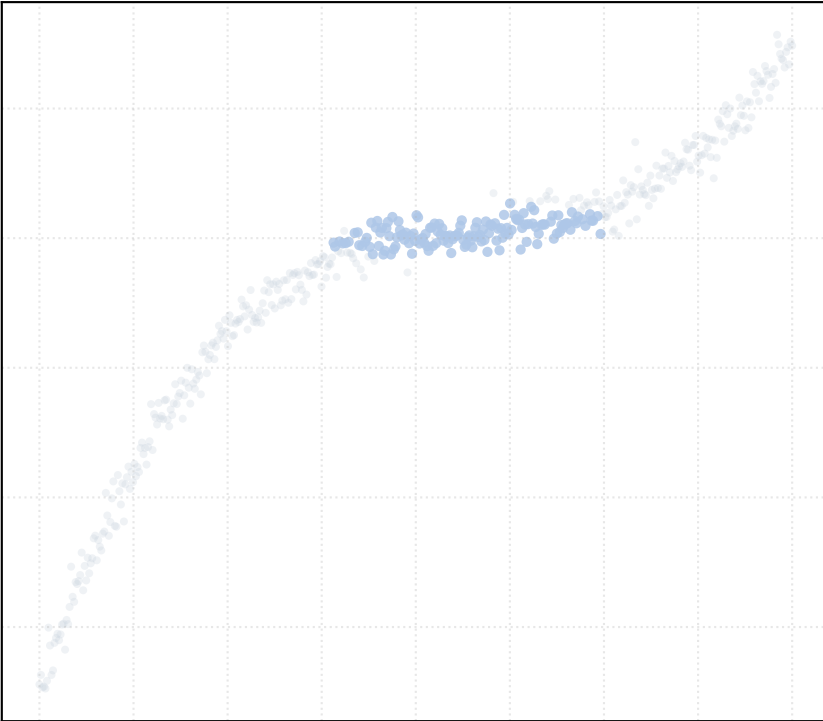


r = 3.5 [1 Clusters]

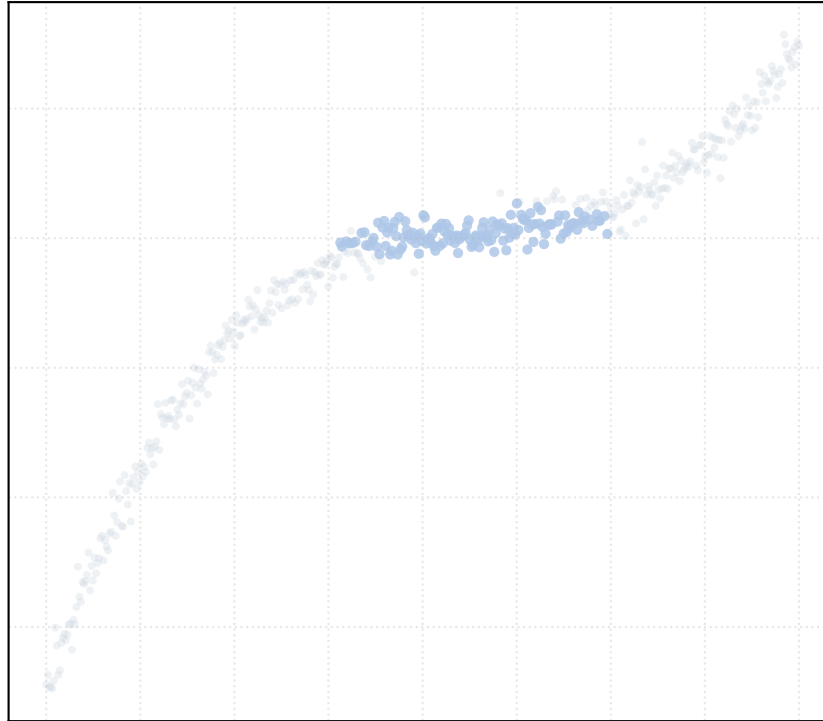


Dataset: cubic_noise_high | Search Radius Sweep (r in grow_ensemble)

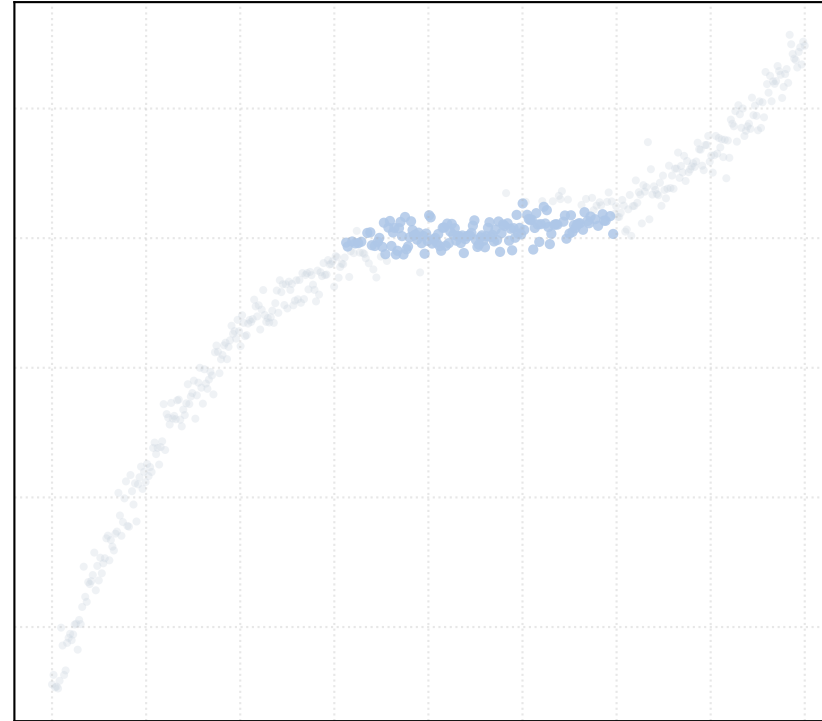
r = Default (Adaptive) [1 Clusters]



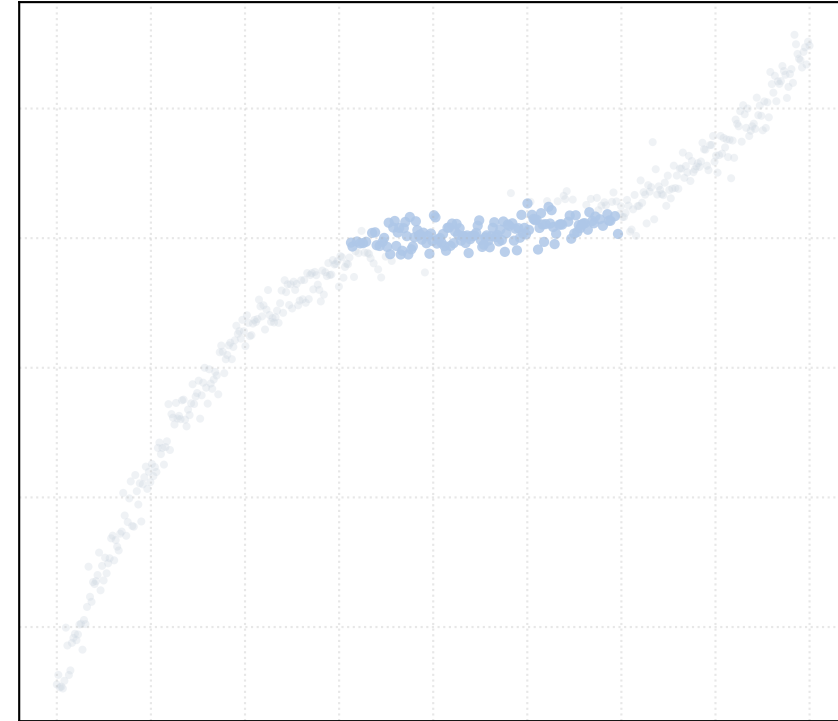
r = 0.5 [1 Clusters]



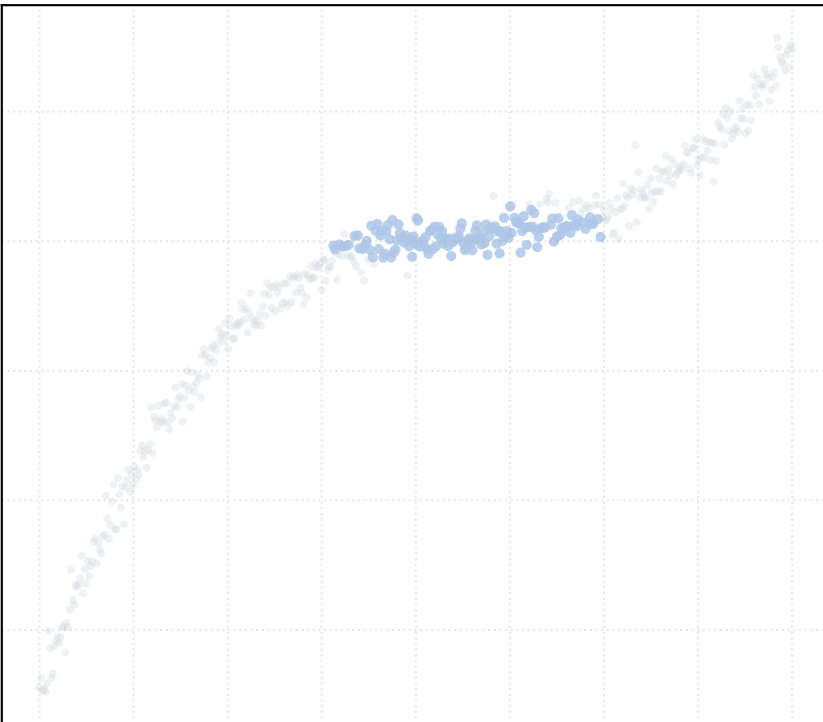
r = 1.0 [1 Clusters]



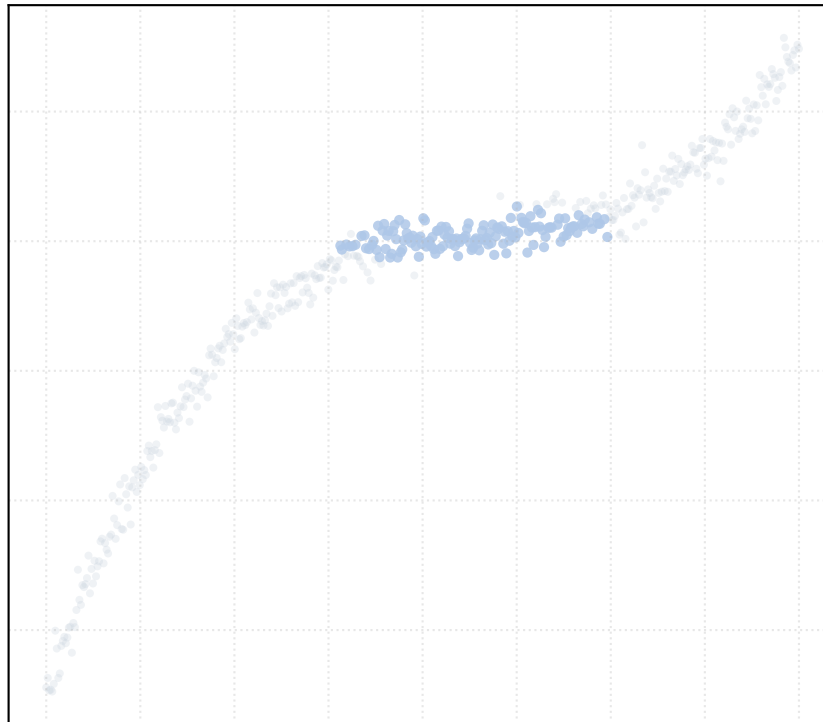
r = 1.5 [1 Clusters]



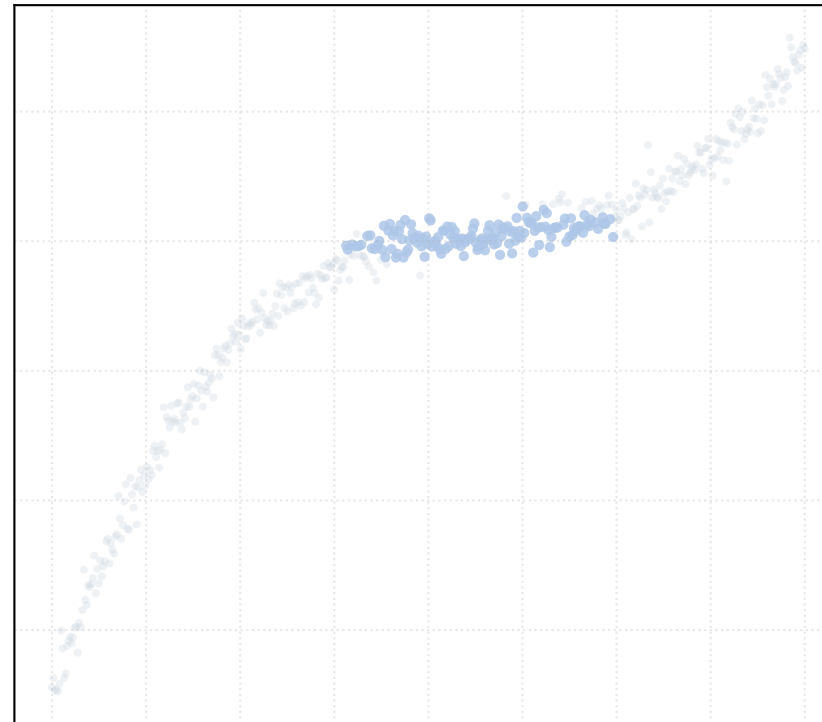
r = 2.0 [1 Clusters]



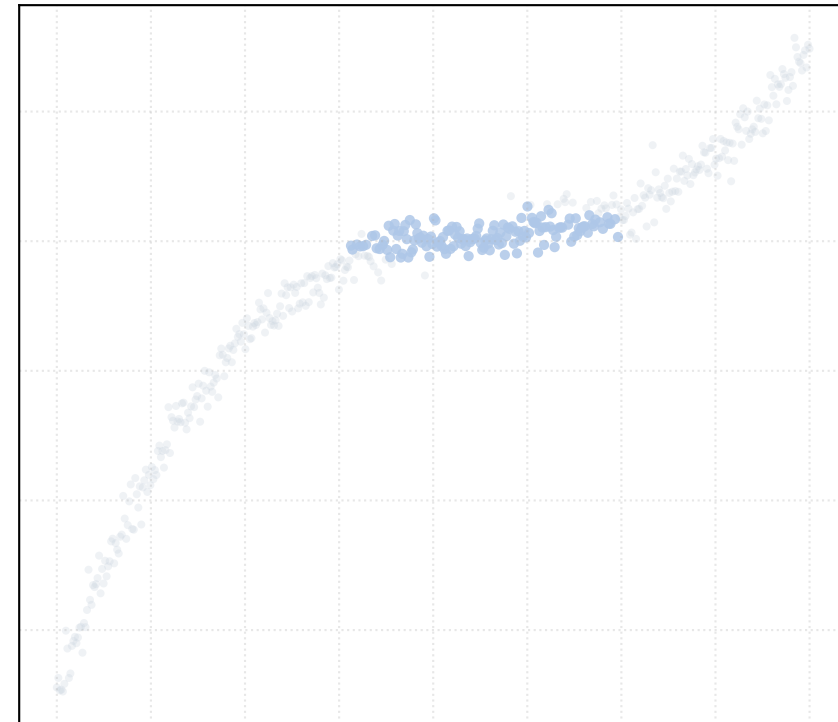
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

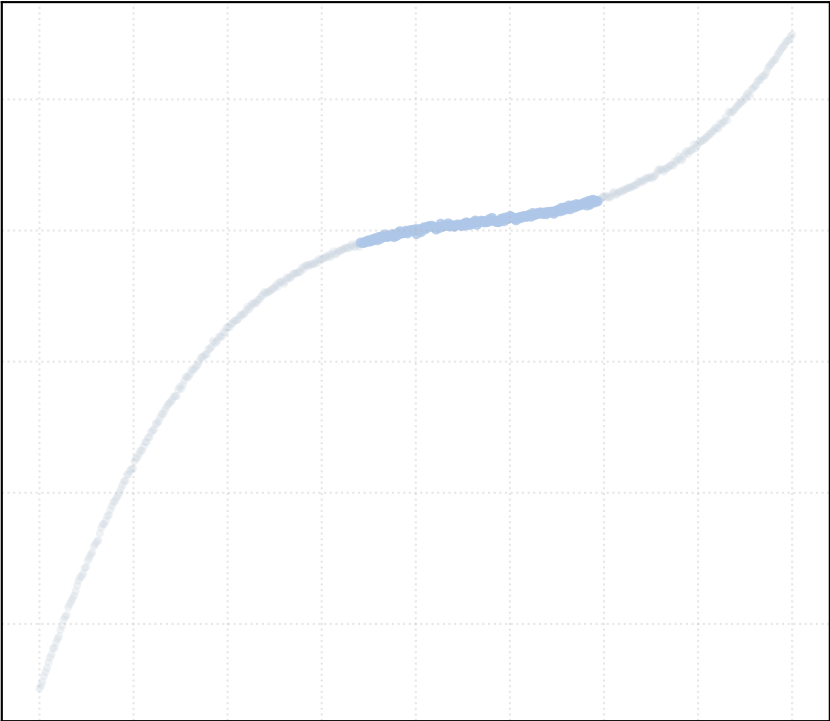


r = 3.5 [1 Clusters]

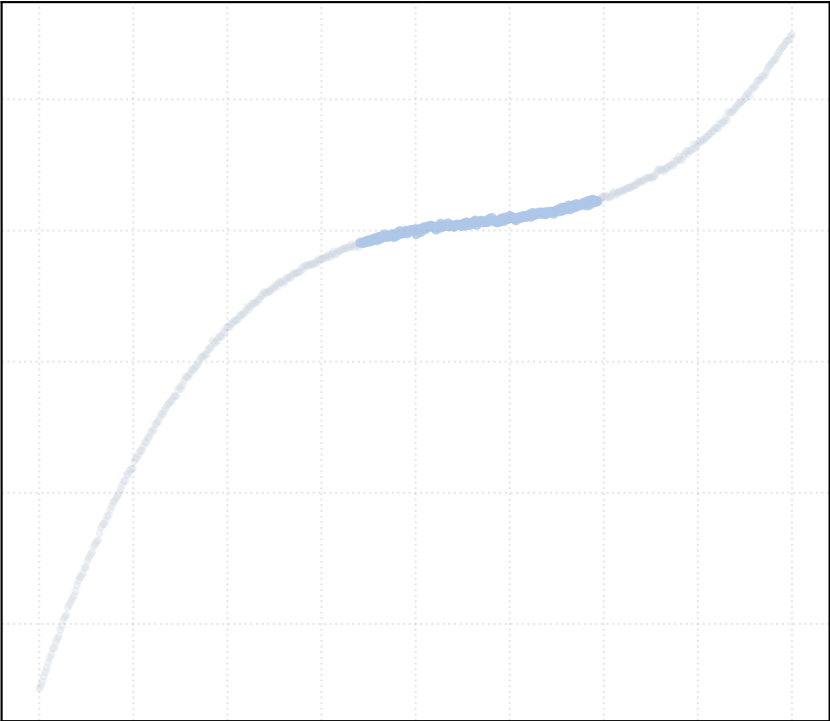


Dataset: cubic_noise_low | Search Radius Sweep (r in grow_ensemble)

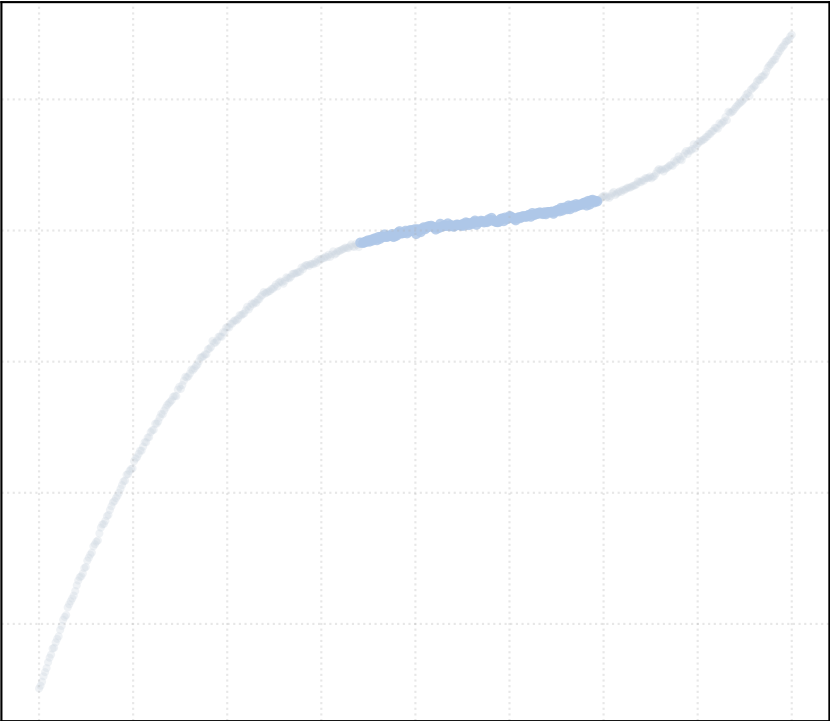
r = Default (Adaptive) [1 Clusters]



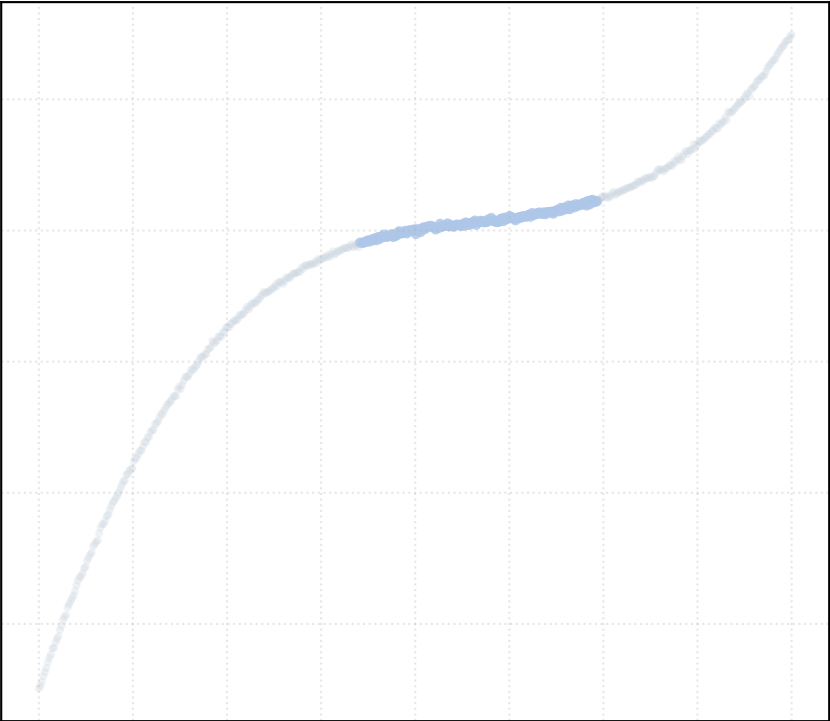
r = 0.5 [1 Clusters]



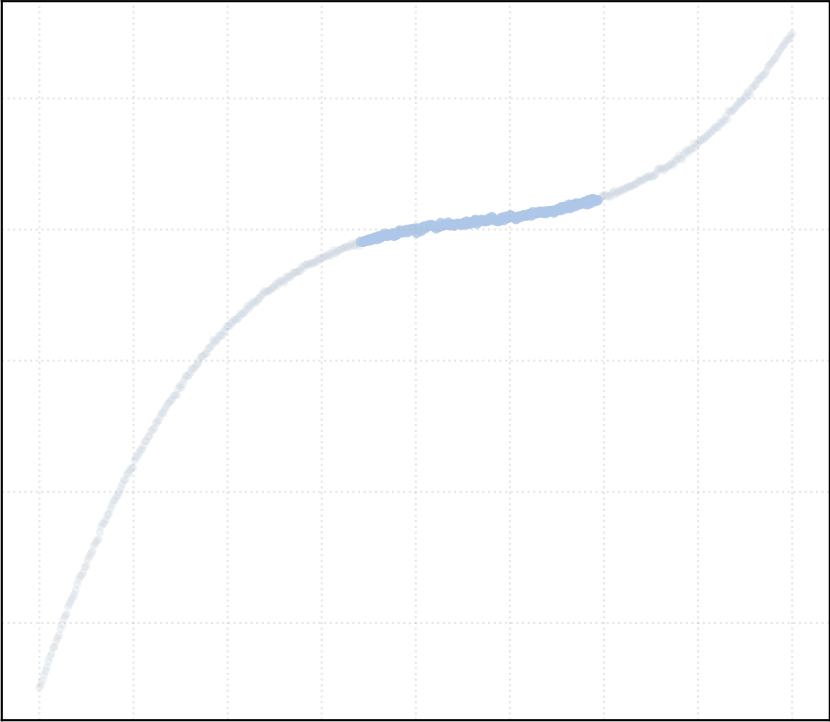
r = 1.0 [1 Clusters]



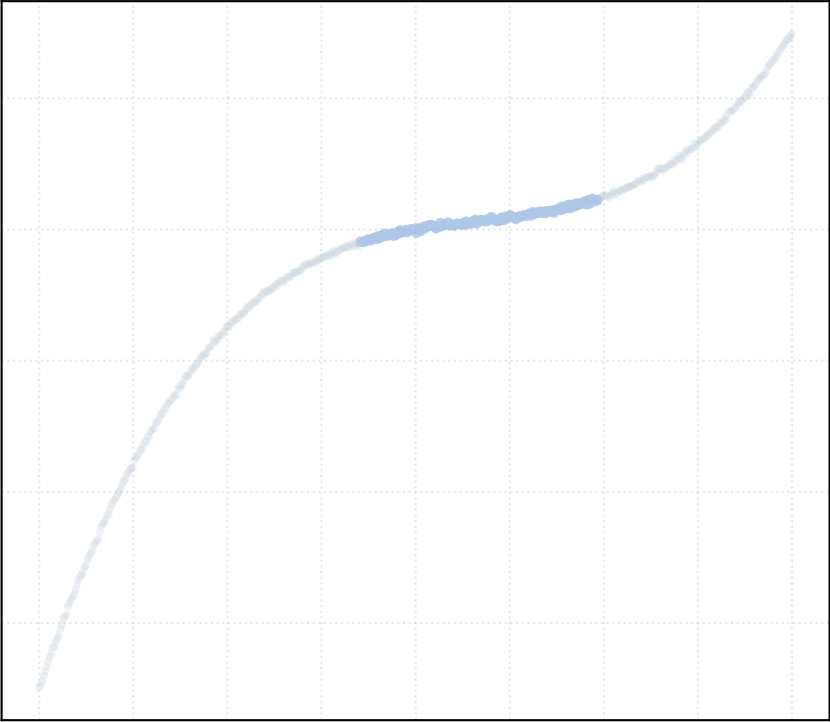
r = 1.5 [1 Clusters]



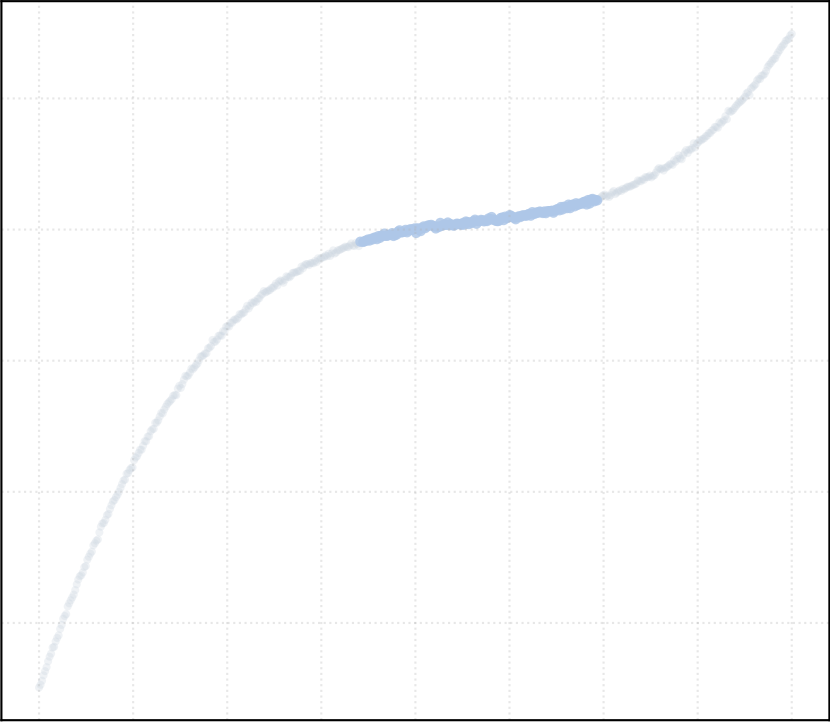
r = 2.0 [1 Clusters]



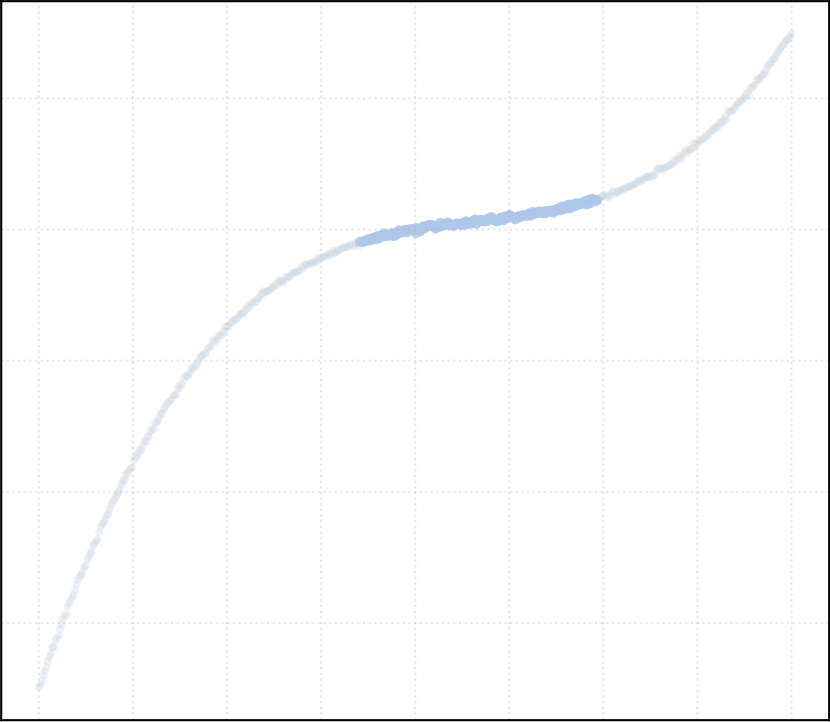
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

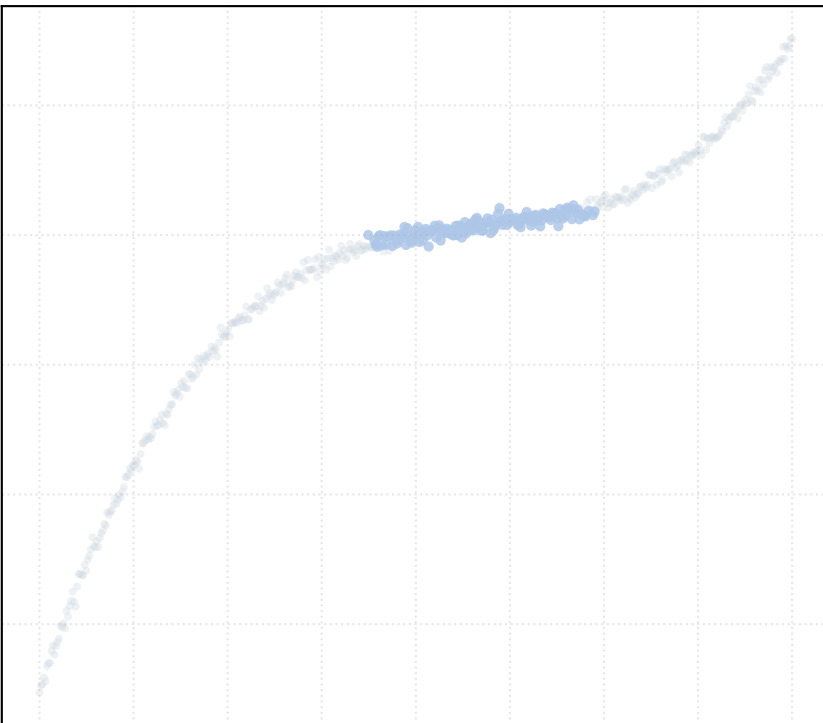


r = 3.5 [1 Clusters]

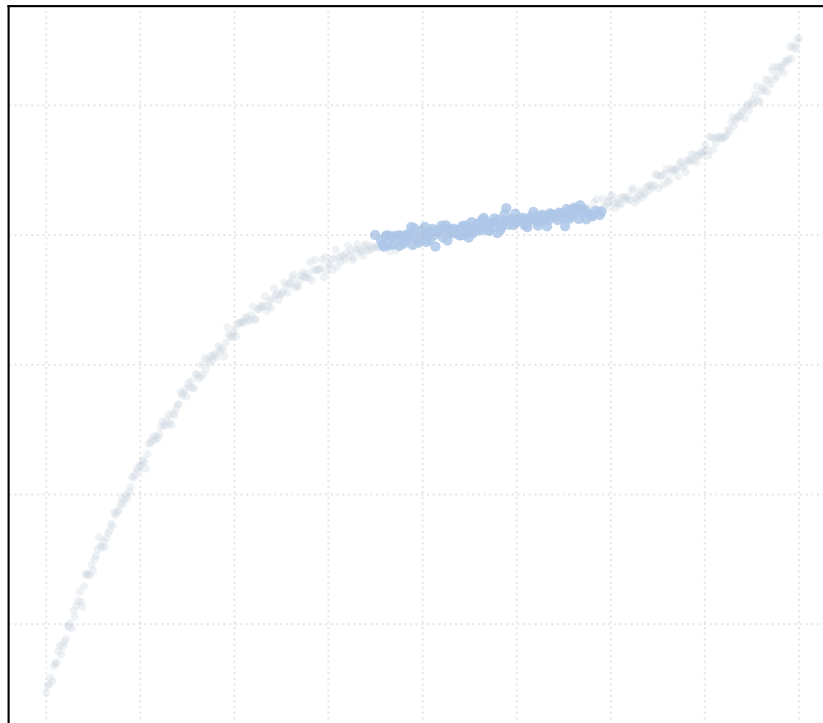


Dataset: cubic_noise_medium | Search Radius Sweep (r in grow_ensemble)

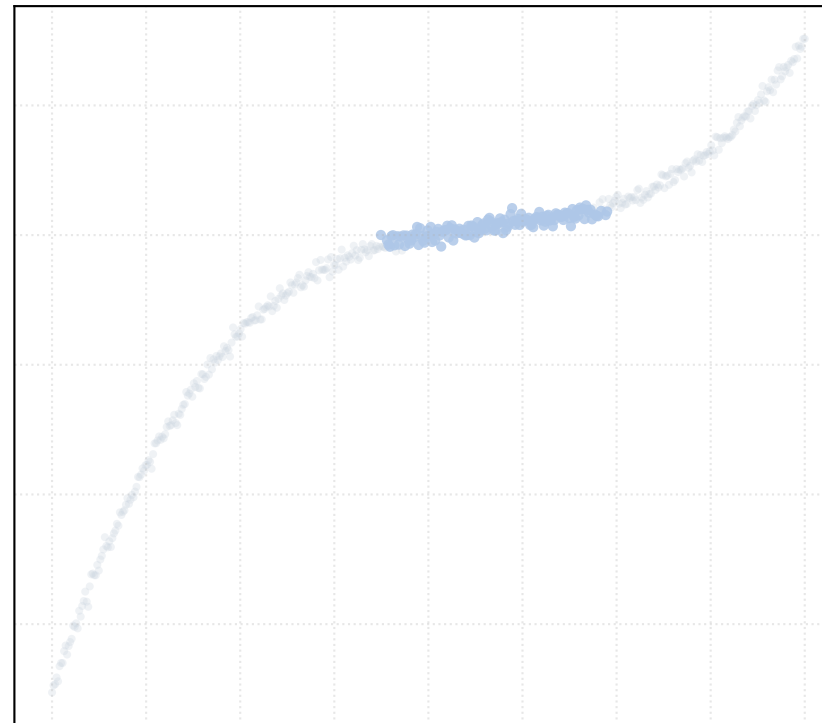
r = Default (Adaptive) [1 Clusters]



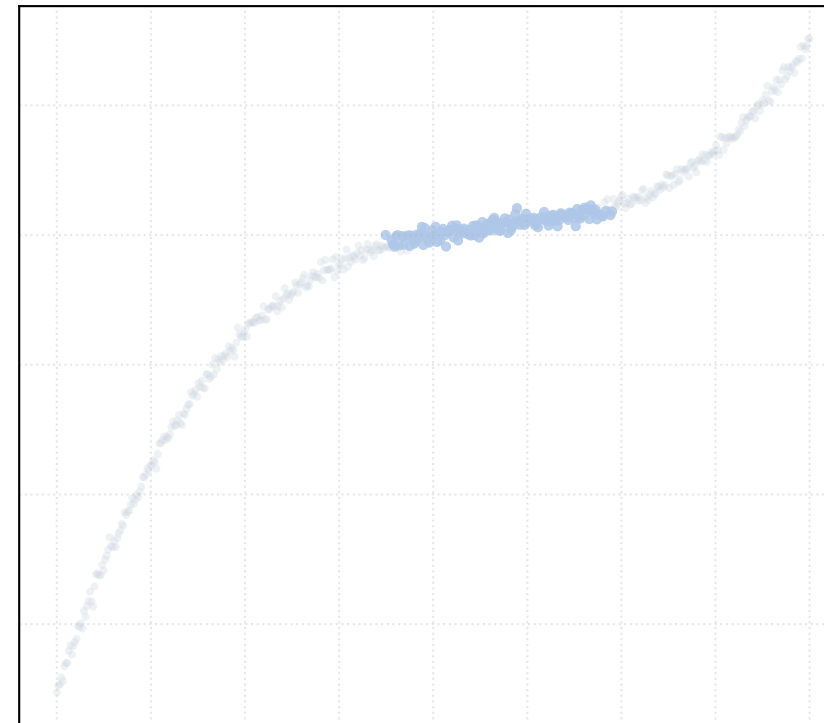
r = 0.5 [1 Clusters]



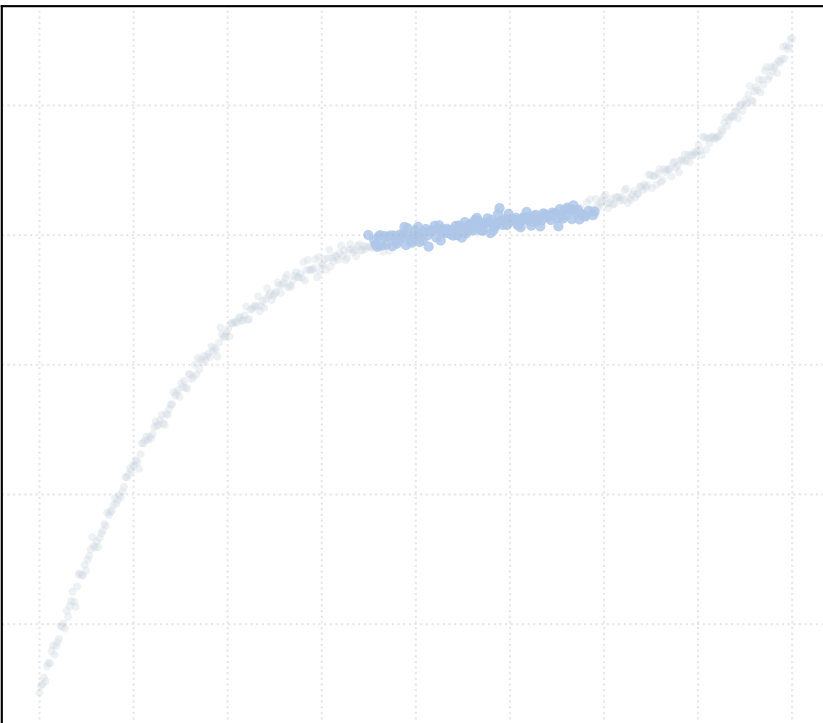
r = 1.0 [1 Clusters]



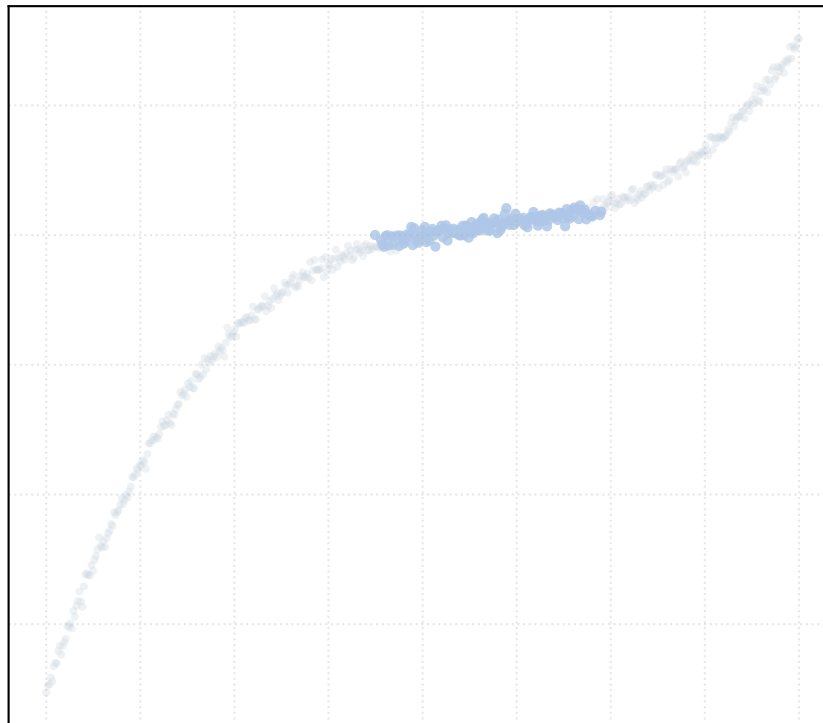
r = 1.5 [1 Clusters]



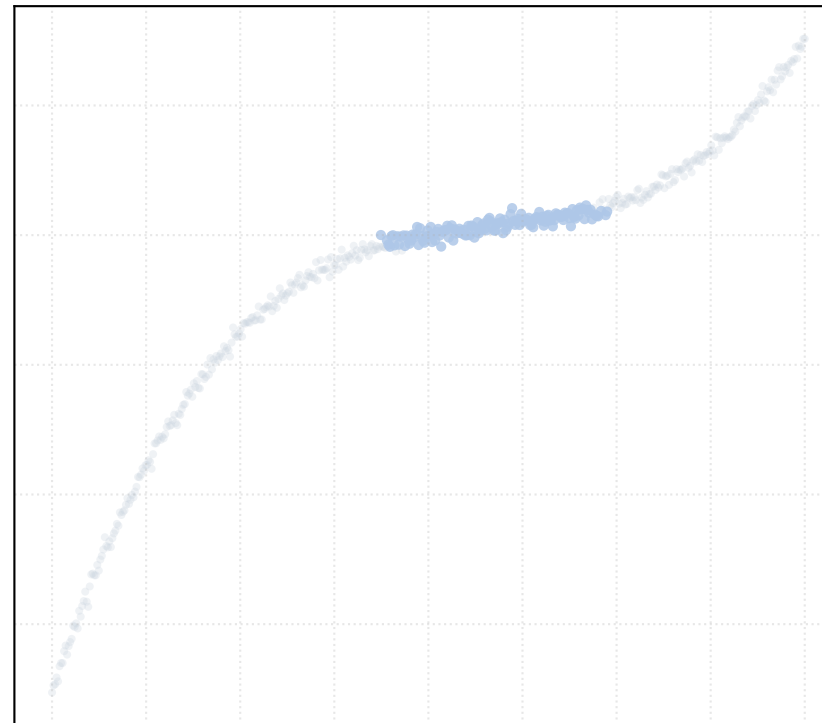
r = 2.0 [1 Clusters]



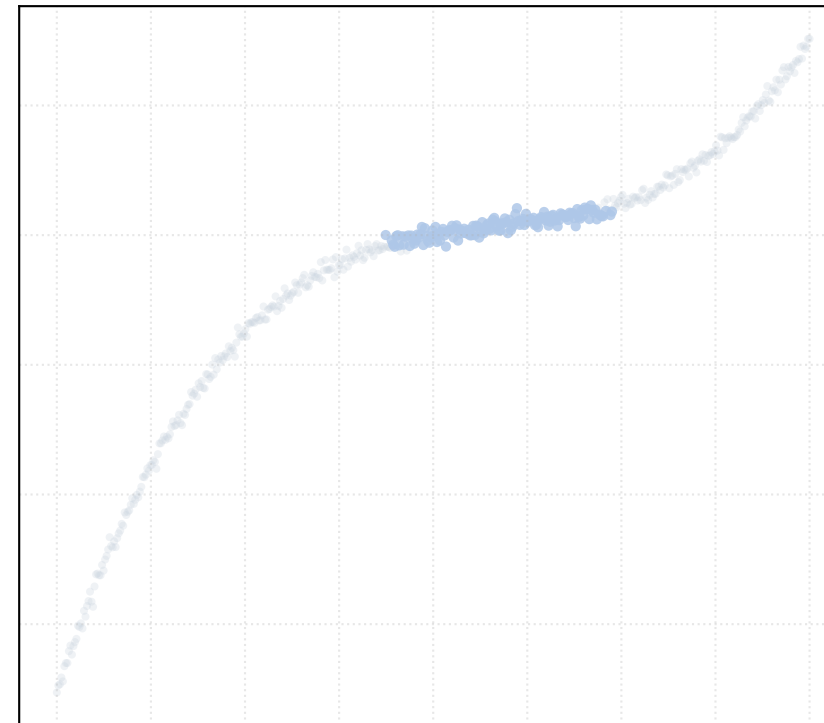
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

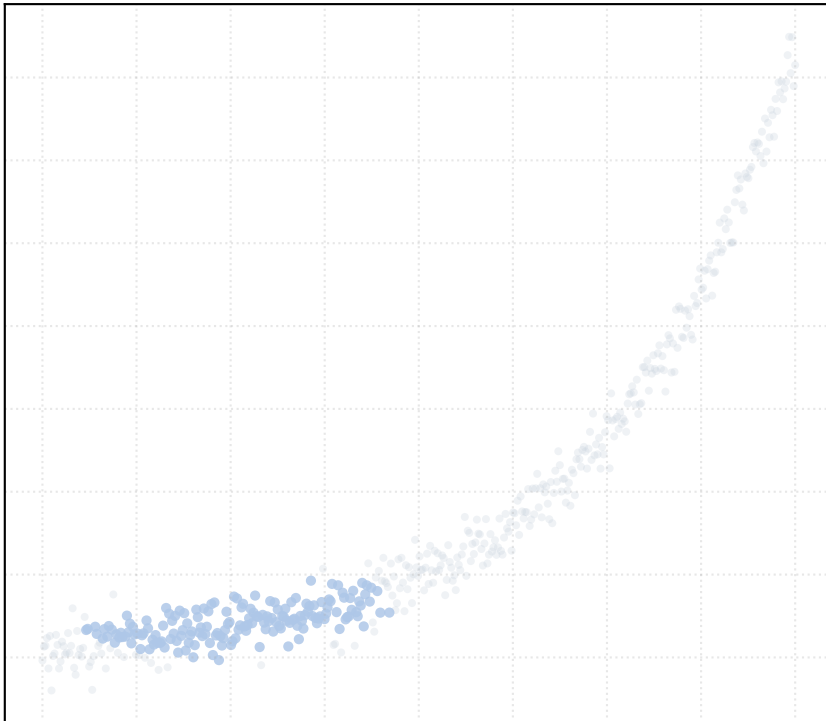


r = 3.5 [1 Clusters]

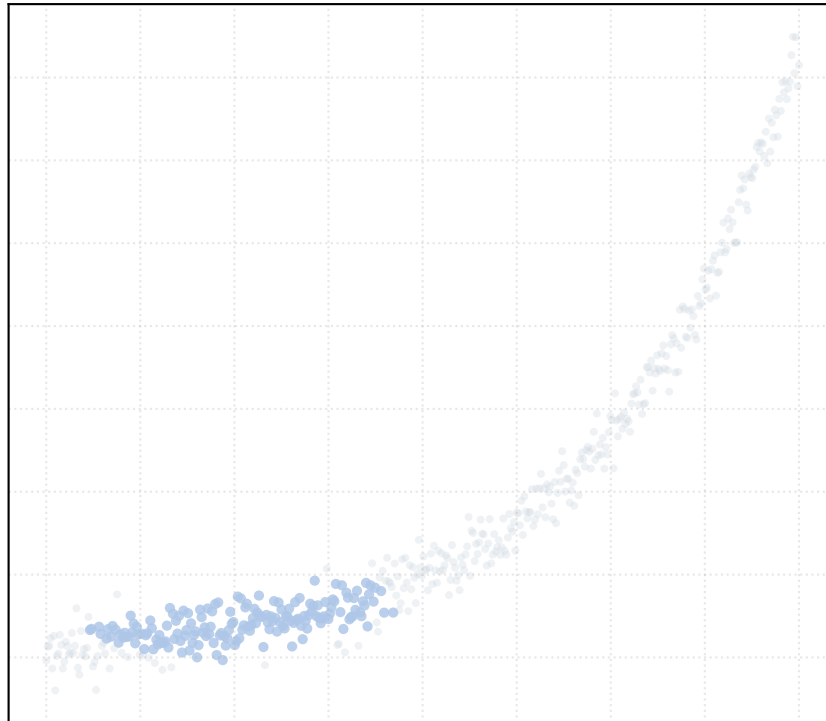


Dataset: exponential_noise_high | Search Radius Sweep (r in grow_ensemble)

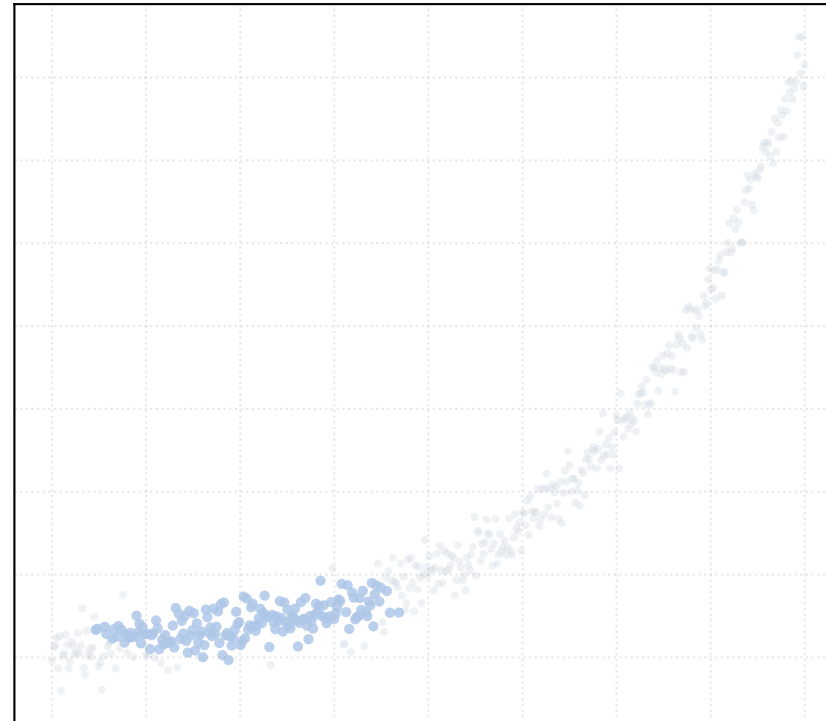
r = Default (Adaptive) [1 Clusters]



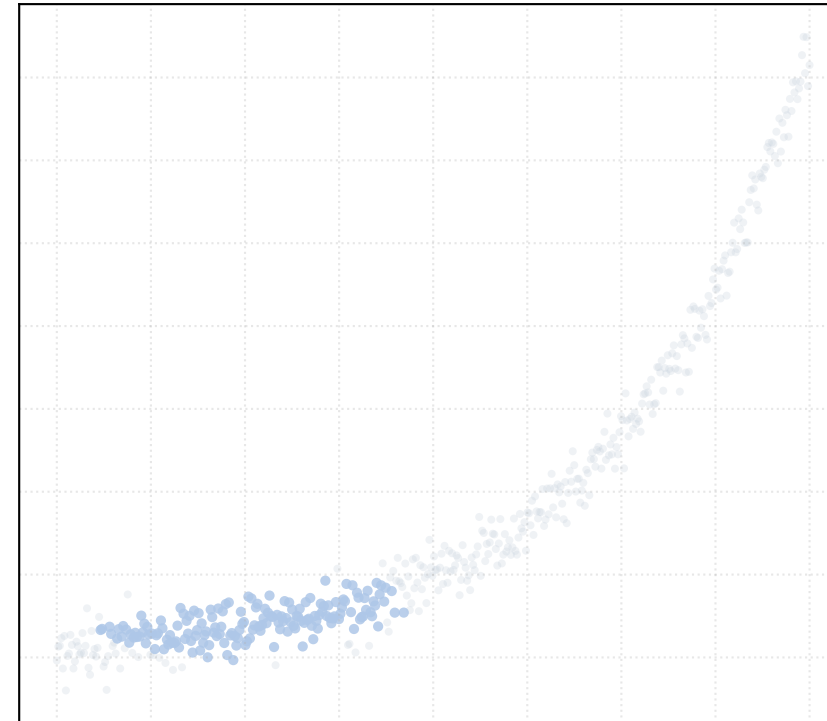
r = 0.5 [1 Clusters]



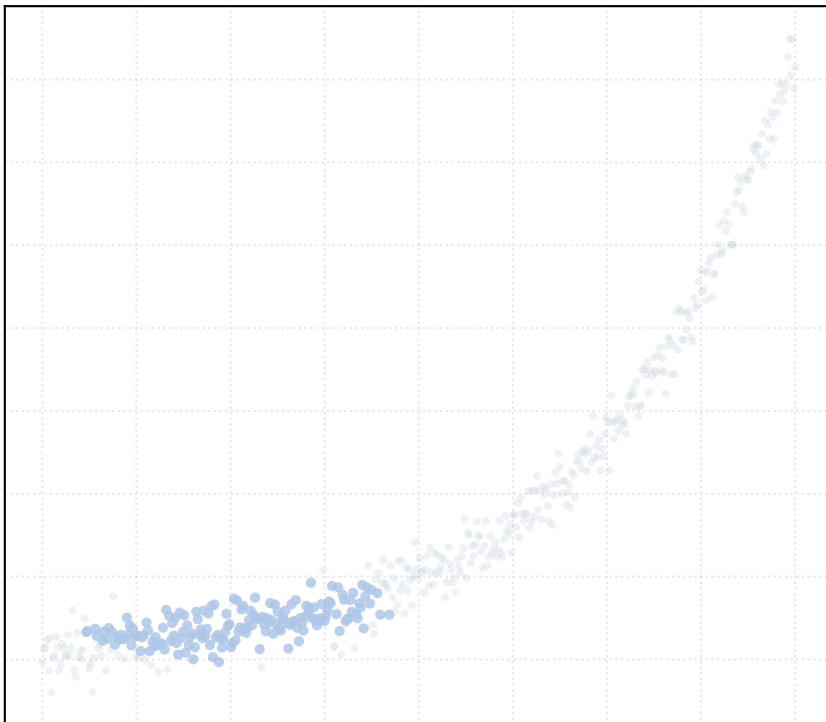
r = 1.0 [1 Clusters]



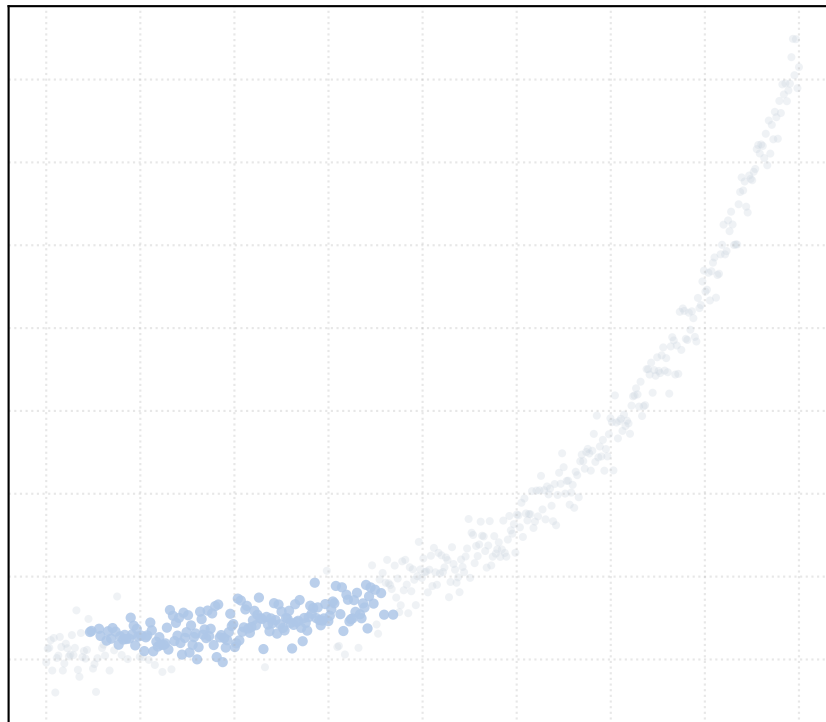
r = 1.5 [1 Clusters]



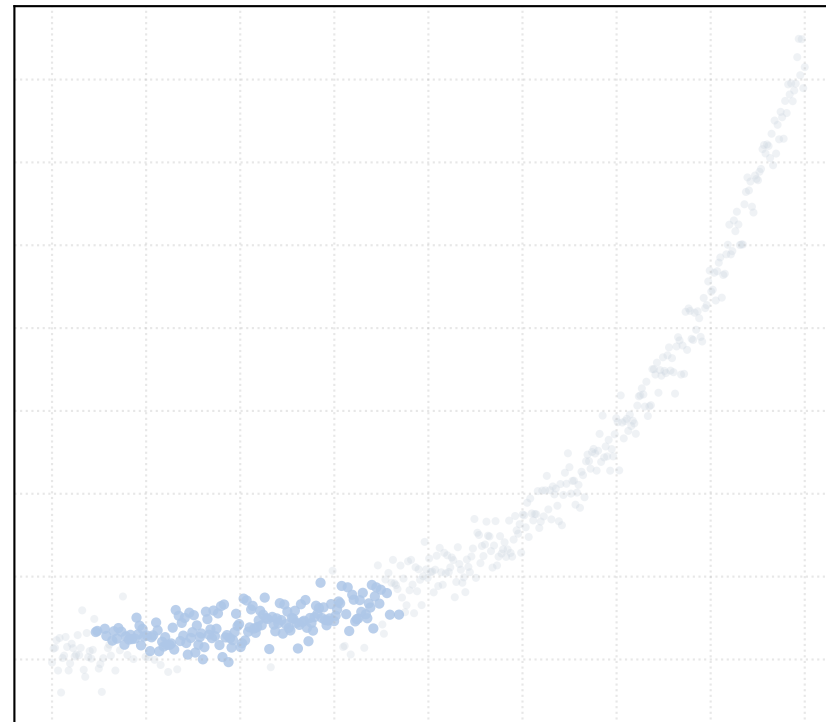
r = 2.0 [1 Clusters]



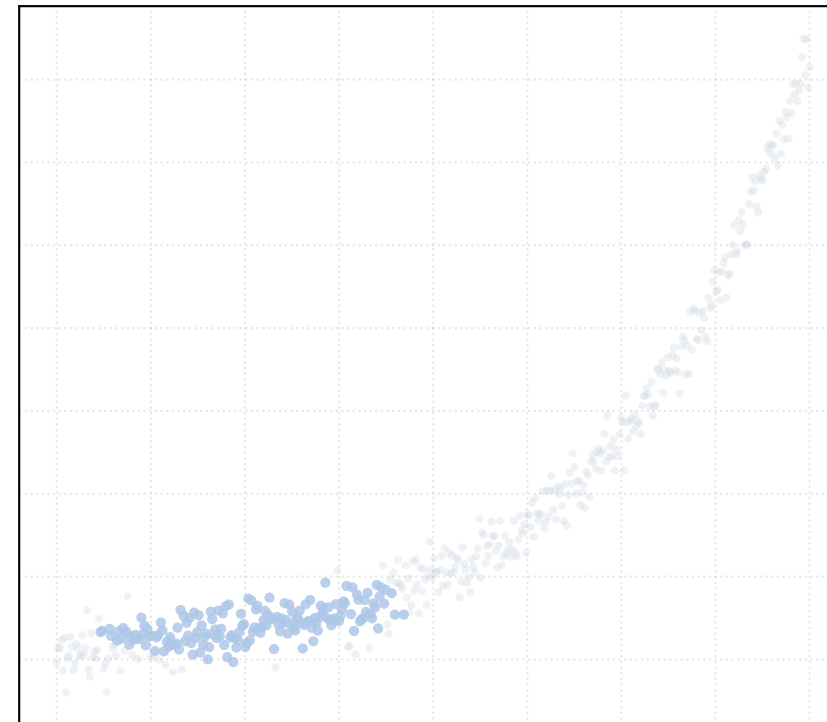
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

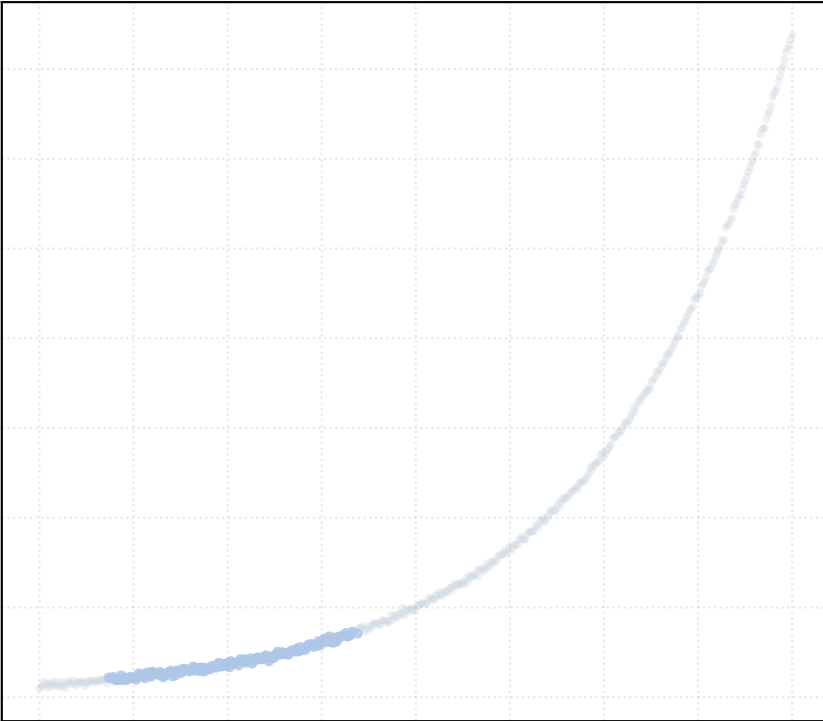


r = 3.5 [1 Clusters]

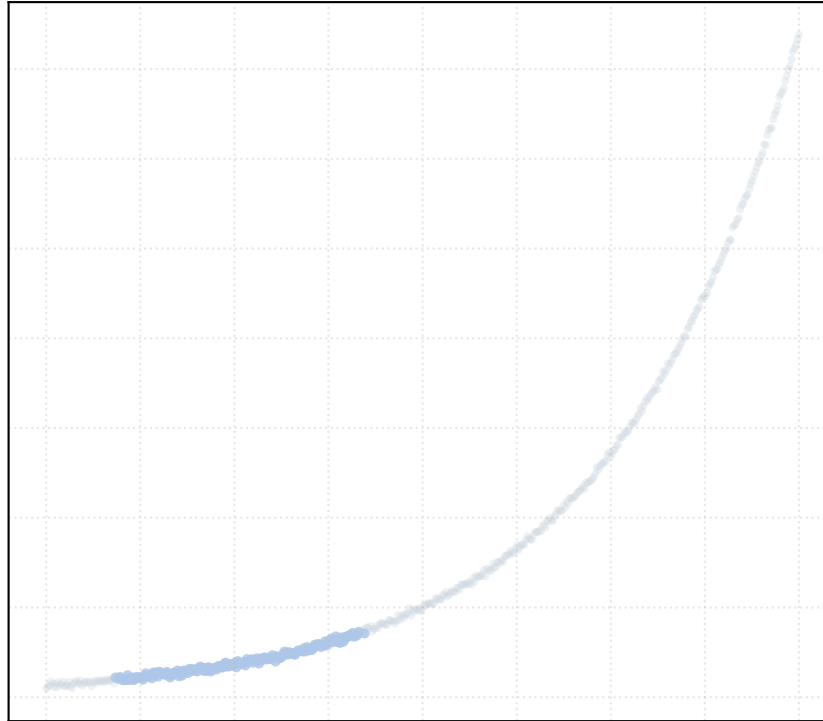


Dataset: exponential_noise_low | Search Radius Sweep (r in grow_ensemble)

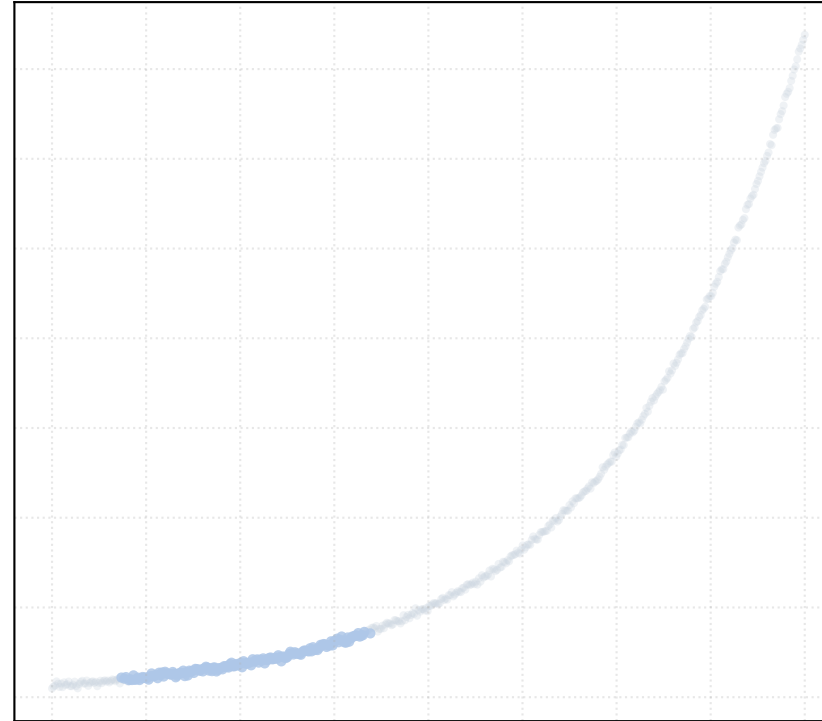
r = Default (Adaptive) [1 Clusters]



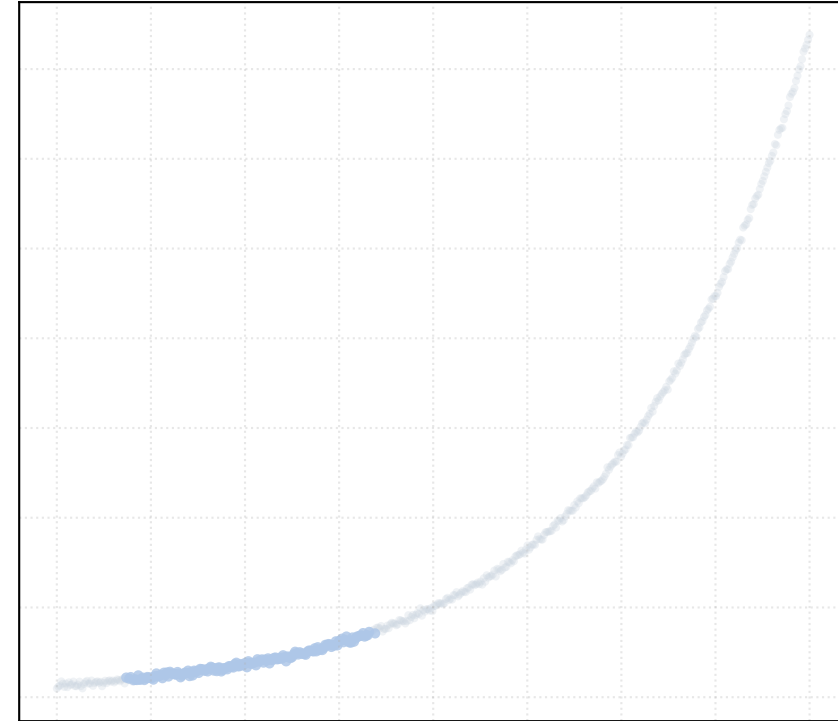
r = 0.5 [1 Clusters]



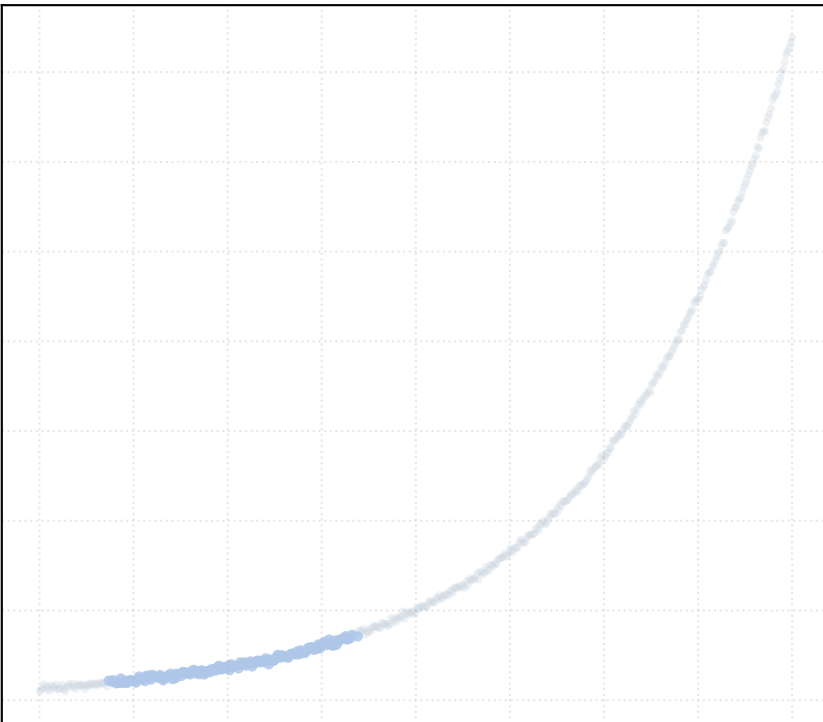
r = 1.0 [1 Clusters]



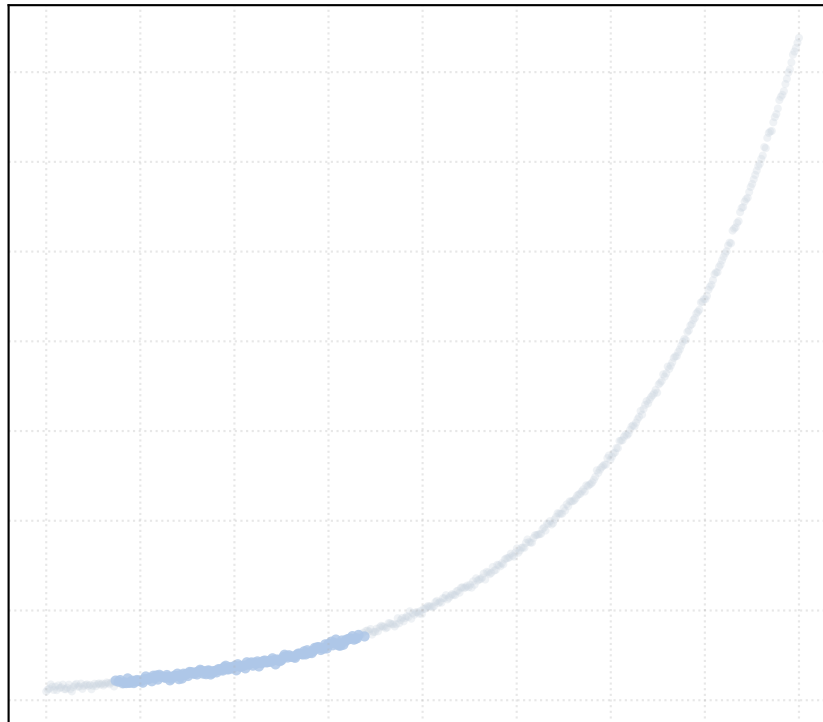
r = 1.5 [1 Clusters]



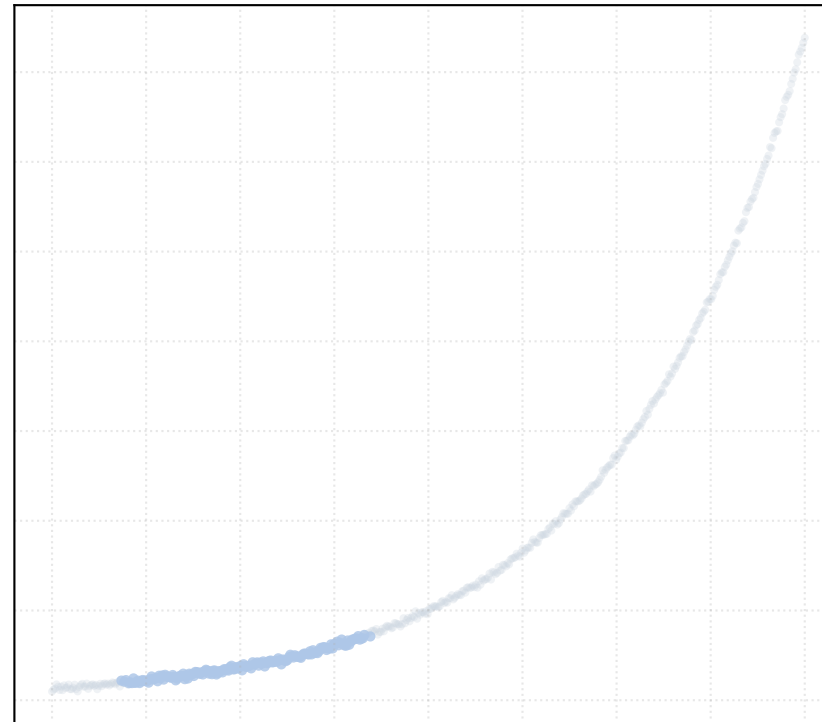
r = 2.0 [1 Clusters]



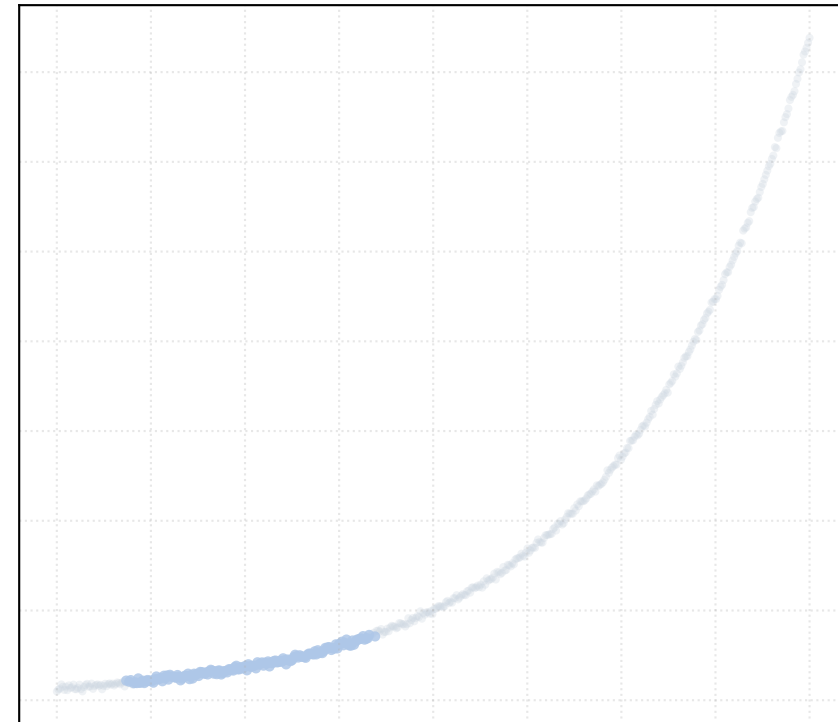
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

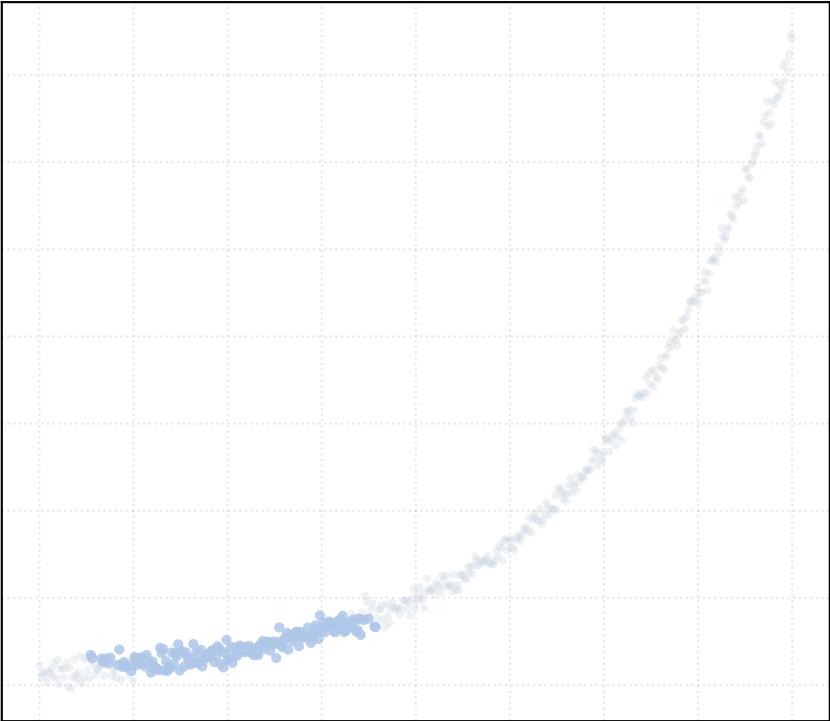


r = 3.5 [1 Clusters]

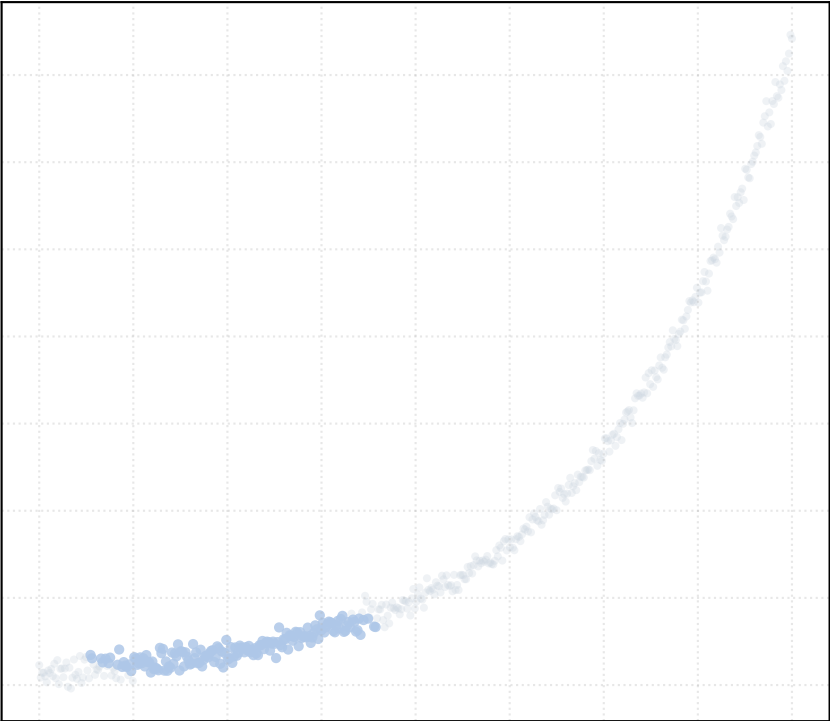


Dataset: exponential_noise_medium | Search Radius Sweep (r in grow_ensemble)

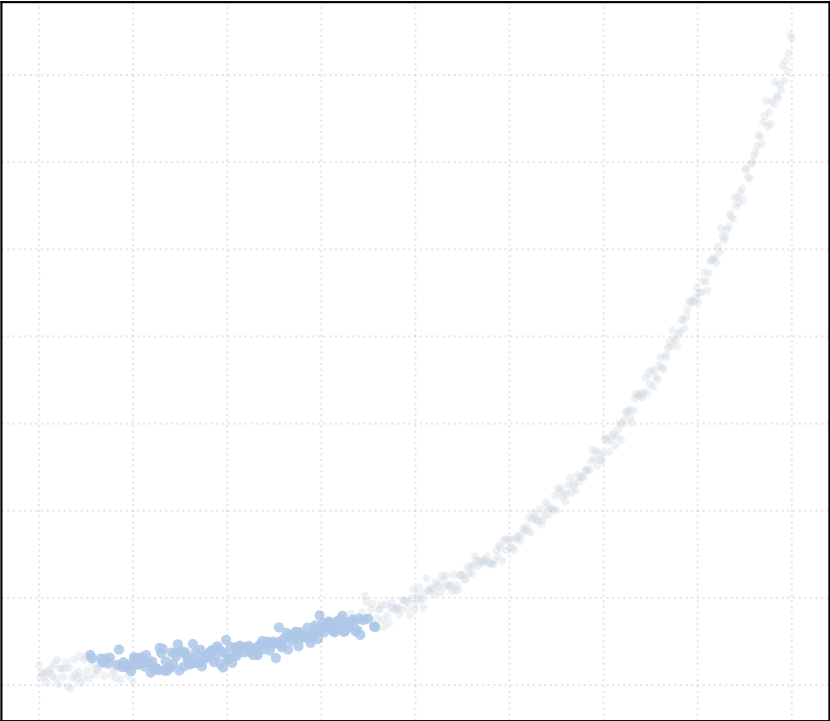
r = Default (Adaptive) [1 Clusters]



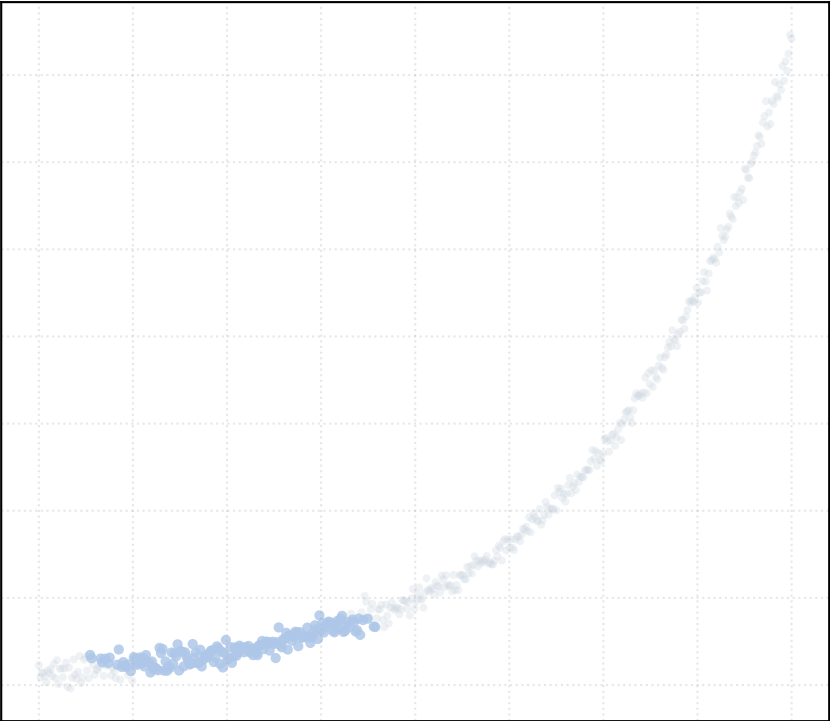
r = 0.5 [1 Clusters]



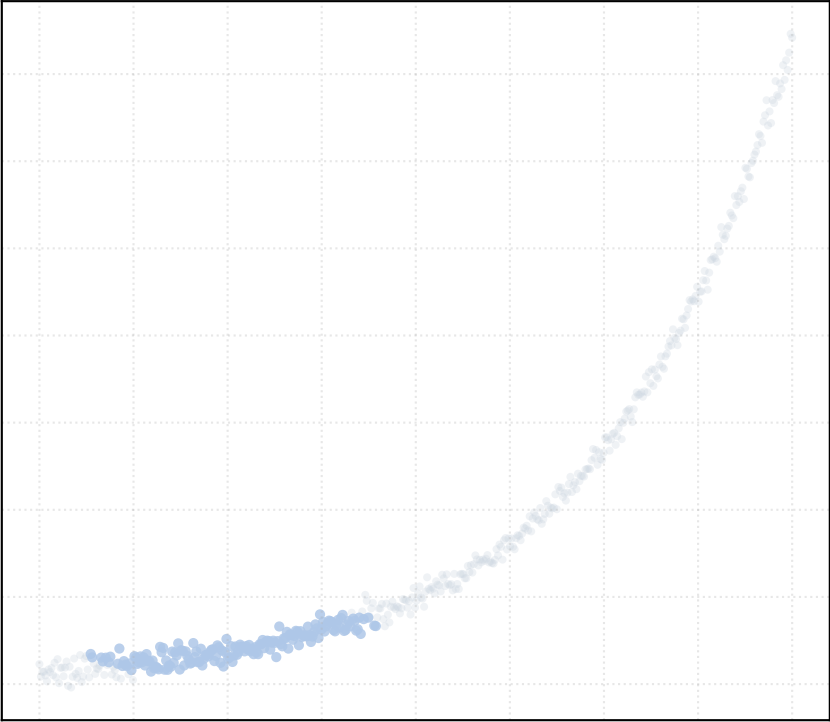
r = 1.0 [1 Clusters]



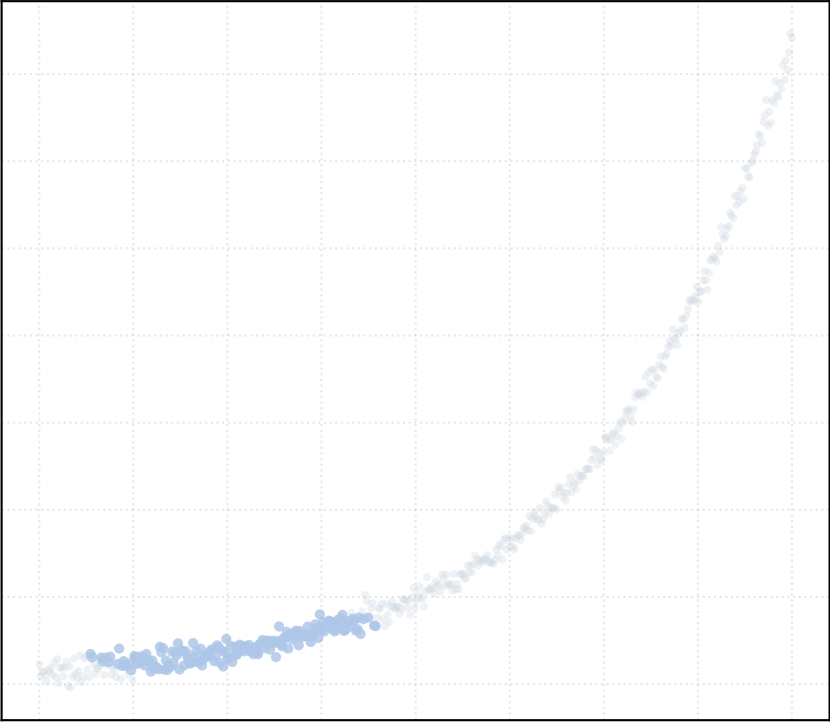
r = 1.5 [1 Clusters]



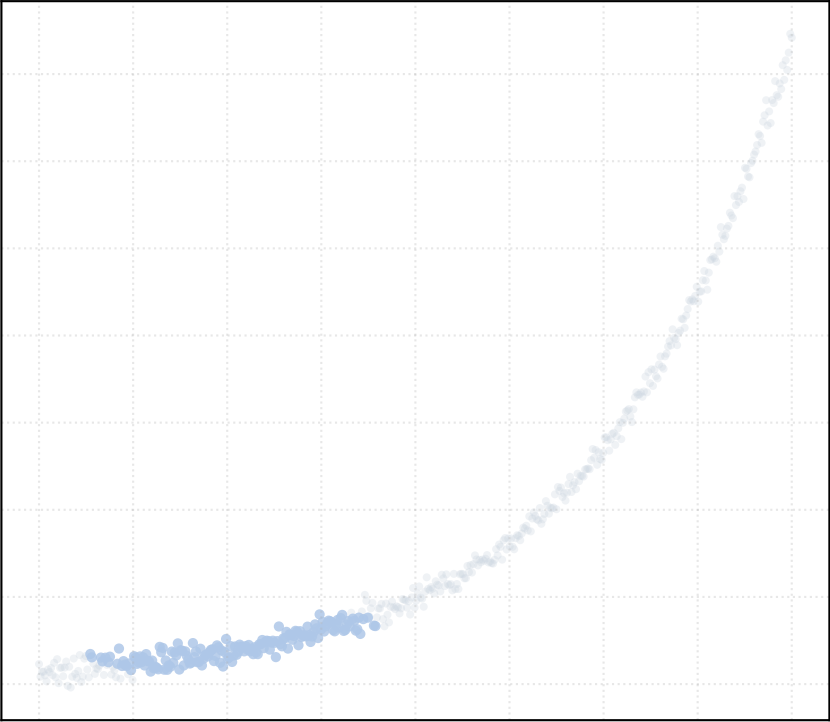
r = 2.0 [1 Clusters]



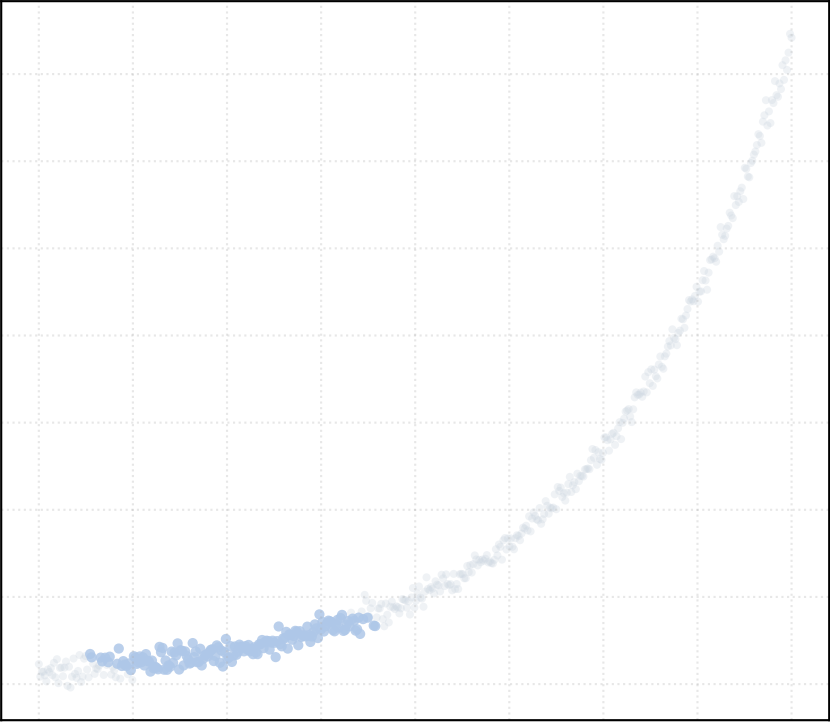
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

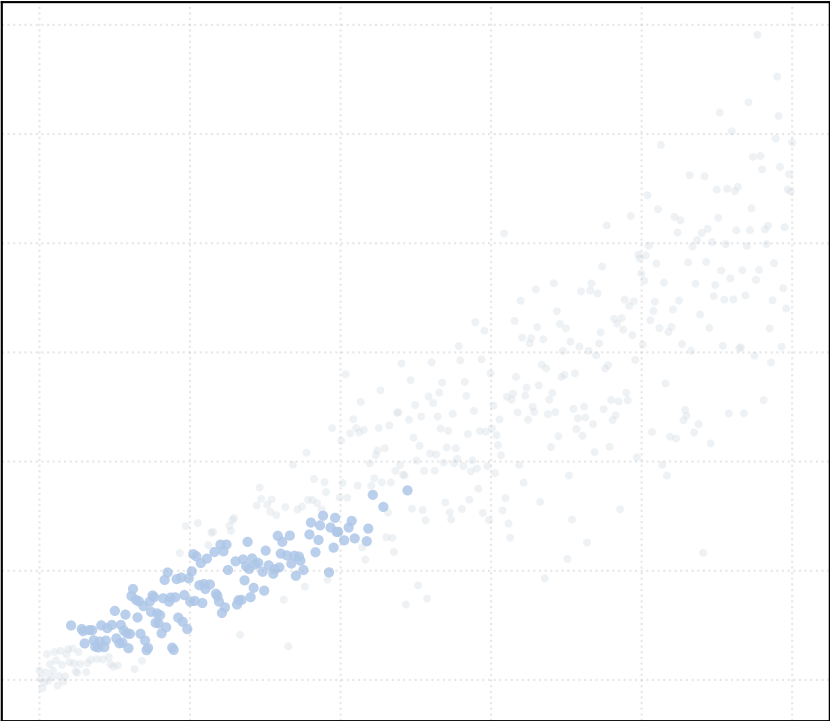


r = 3.5 [1 Clusters]



Dataset: heteroscedastic_noise_high | Search Radius Sweep (r in grow_ensemble)

r = Default (Adaptive) [1 Clusters]



r = 0.5 [1 Clusters]



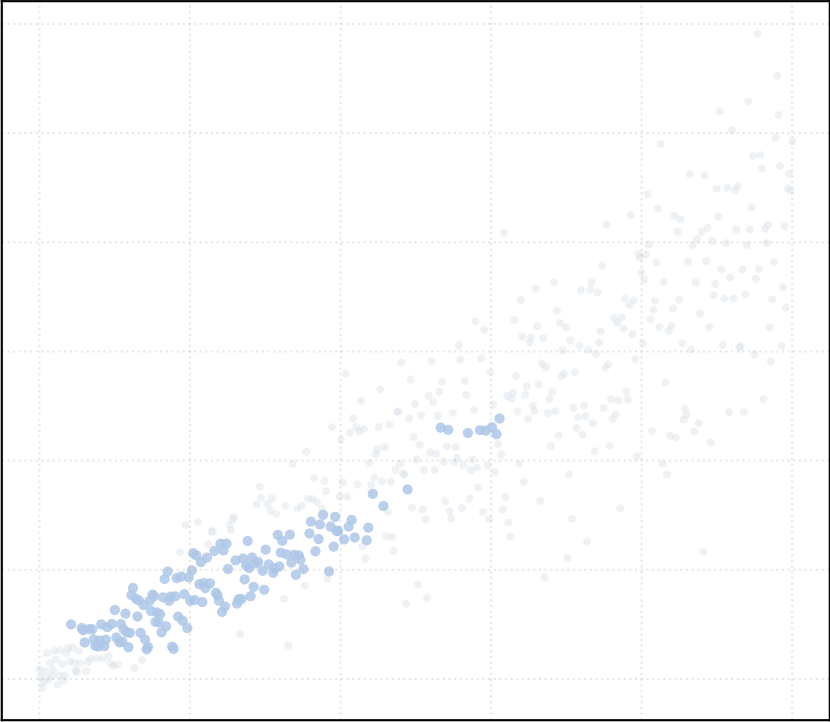
r = 1.0 [1 Clusters]



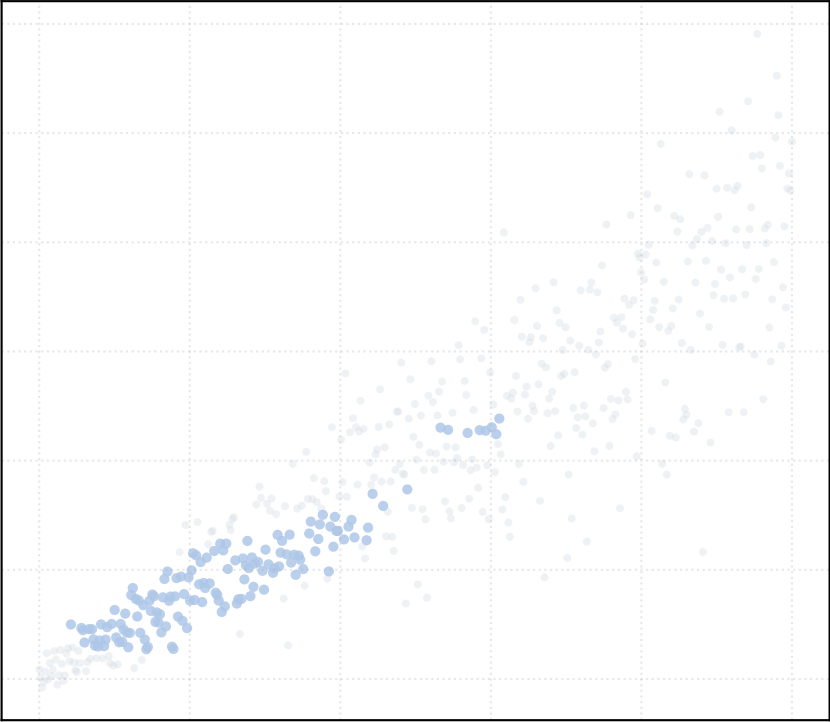
r = 1.5 [1 Clusters]



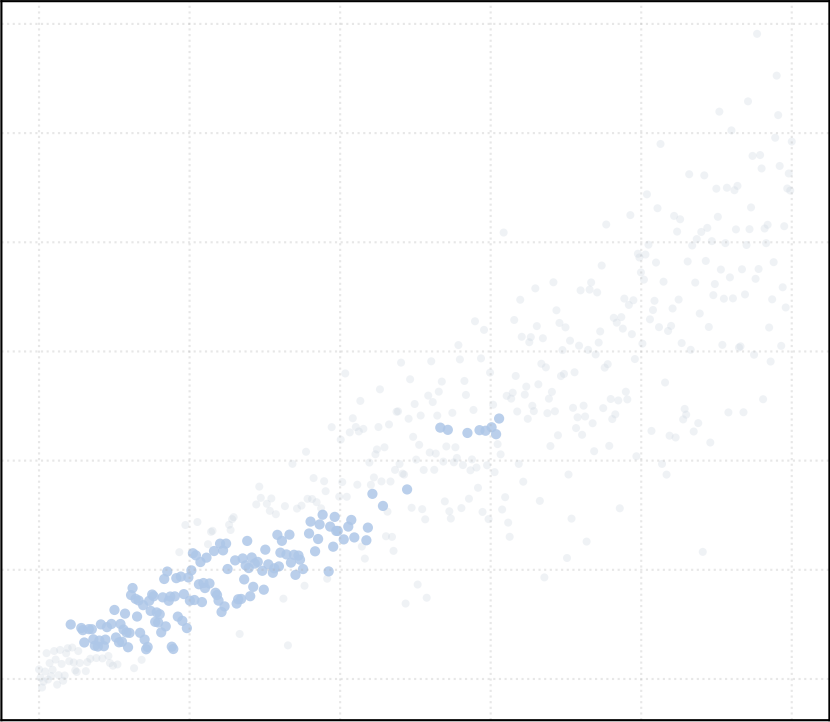
r = 2.0 [1 Clusters]



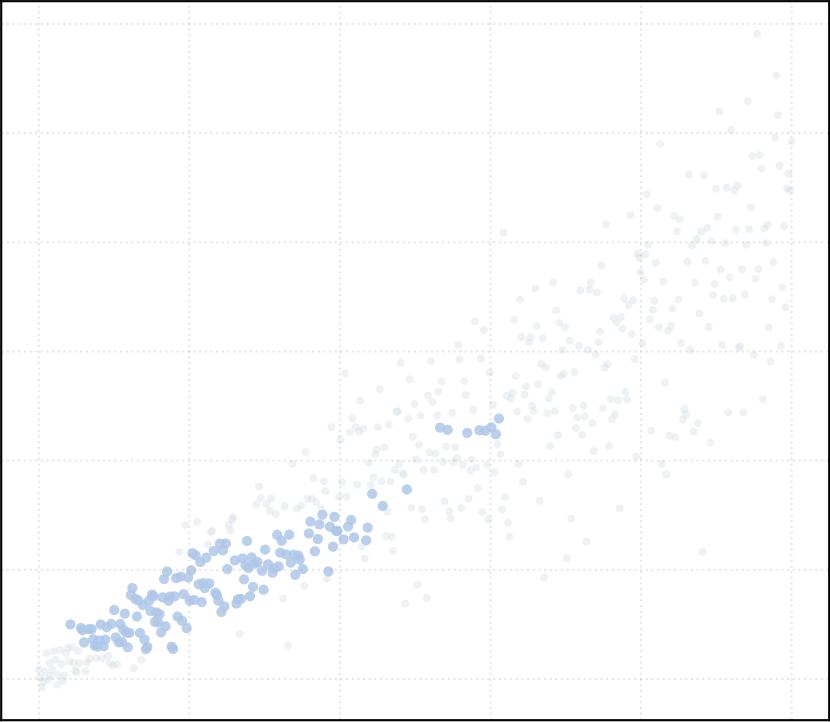
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

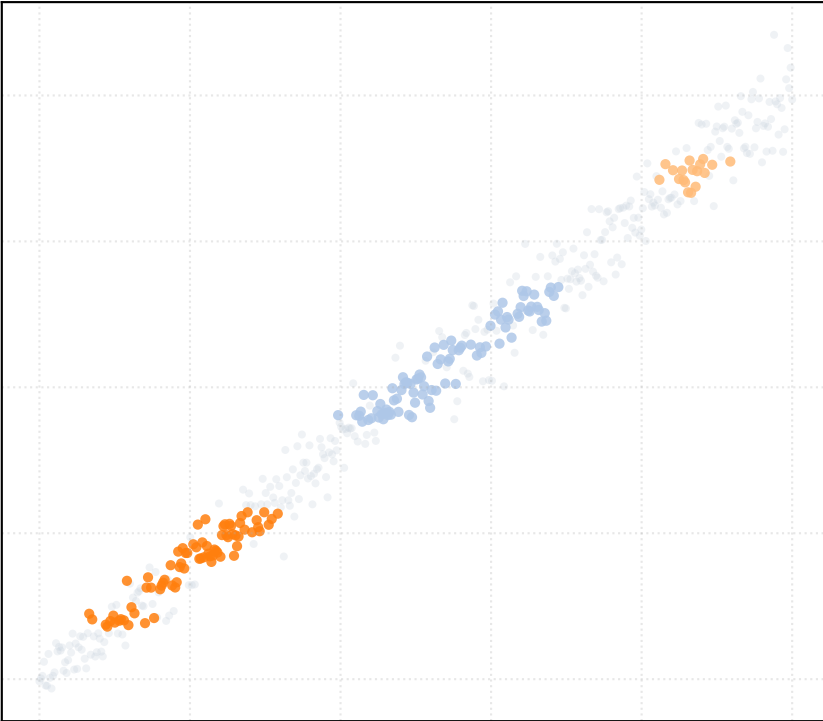


r = 3.5 [1 Clusters]

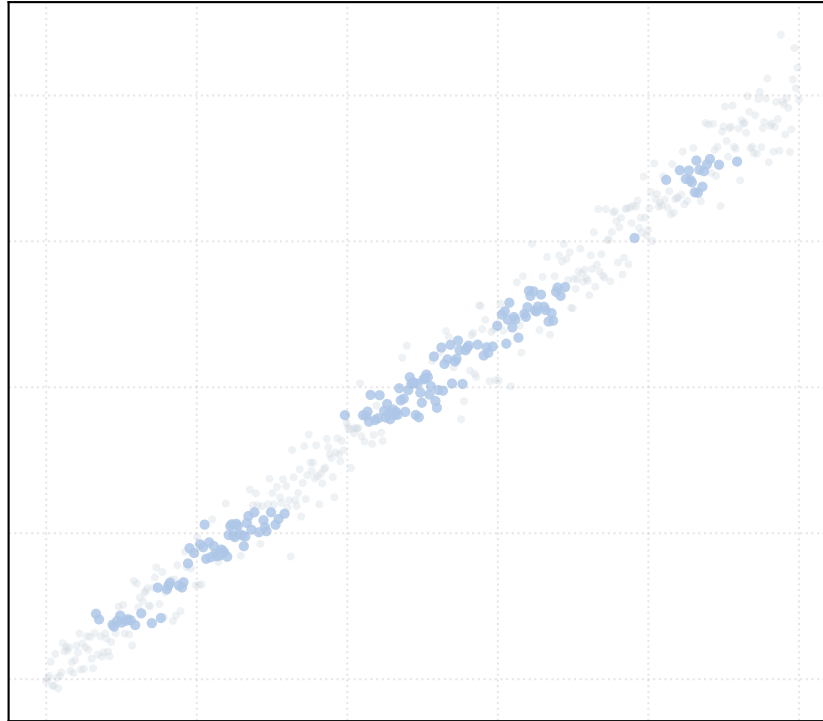


Dataset: heteroscedastic_noise_low | Search Radius Sweep (r in grow_ensemble)

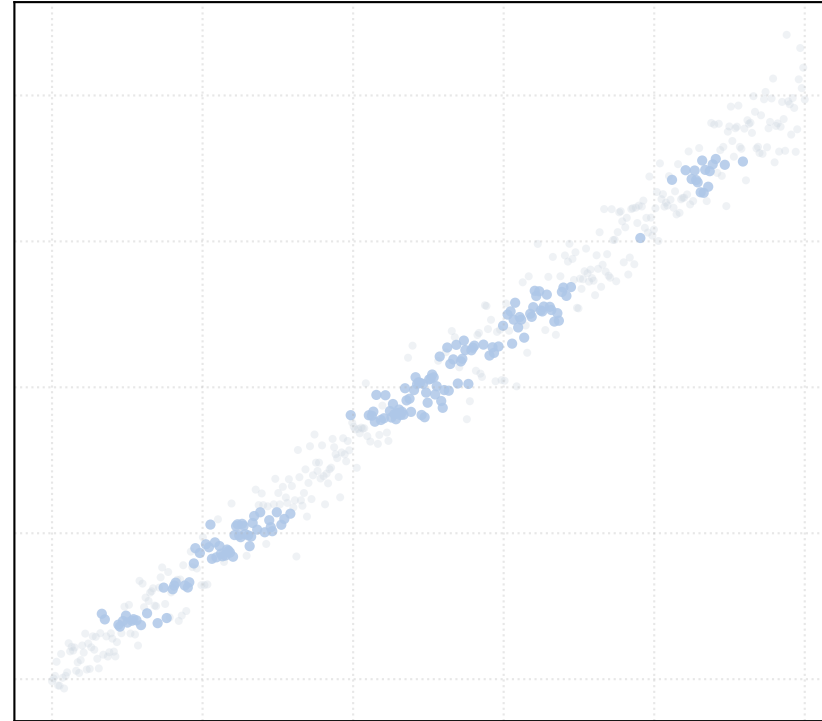
r = Default (Adaptive) [3 Clusters]



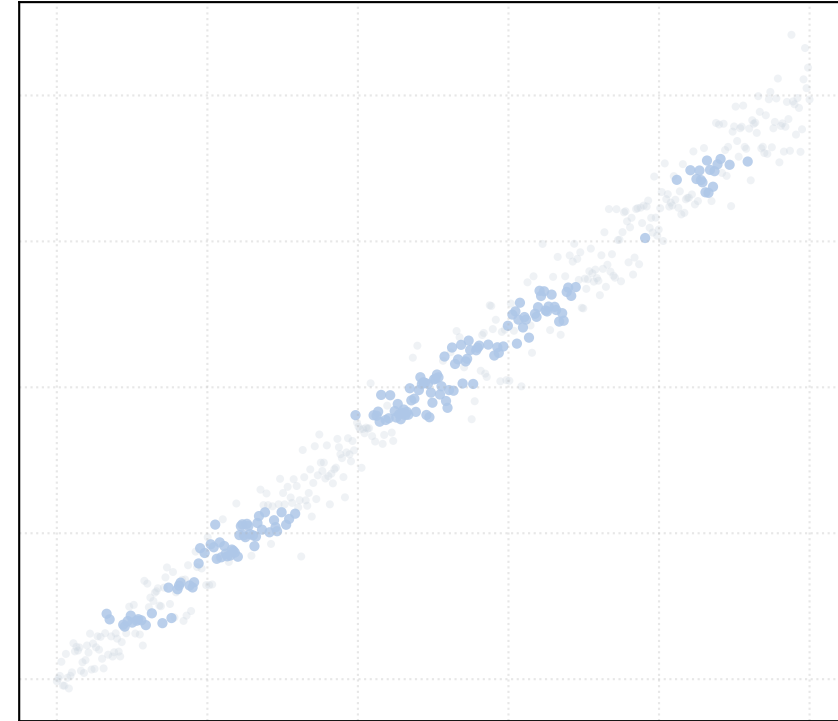
r = 0.5 [1 Clusters]



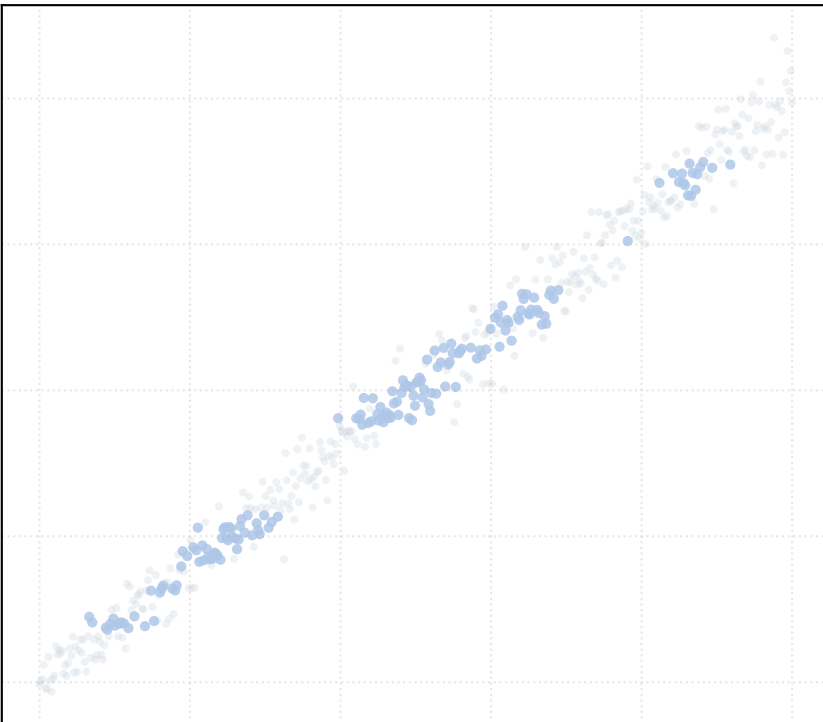
r = 1.0 [1 Clusters]



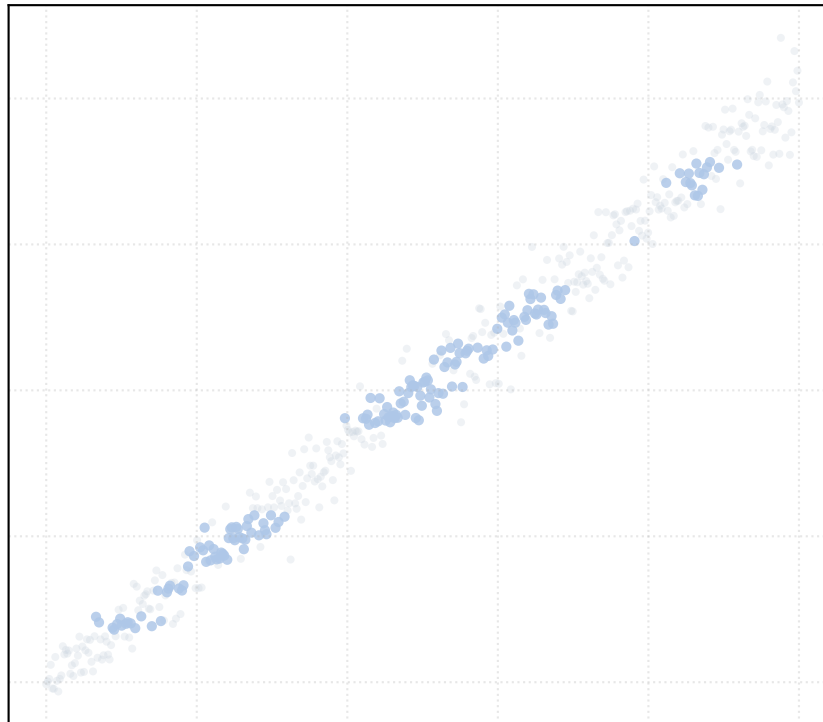
r = 1.5 [1 Clusters]



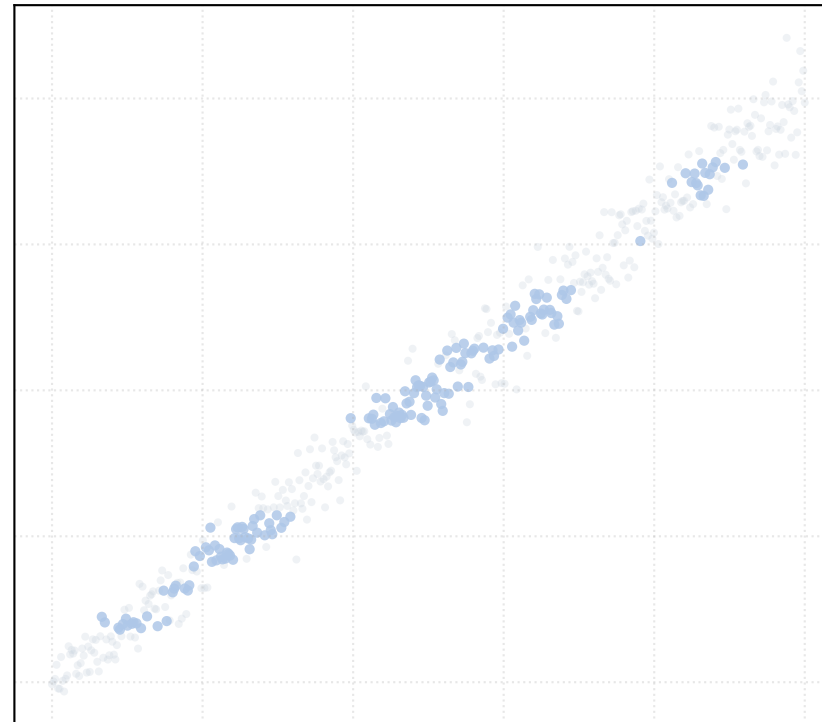
r = 2.0 [1 Clusters]



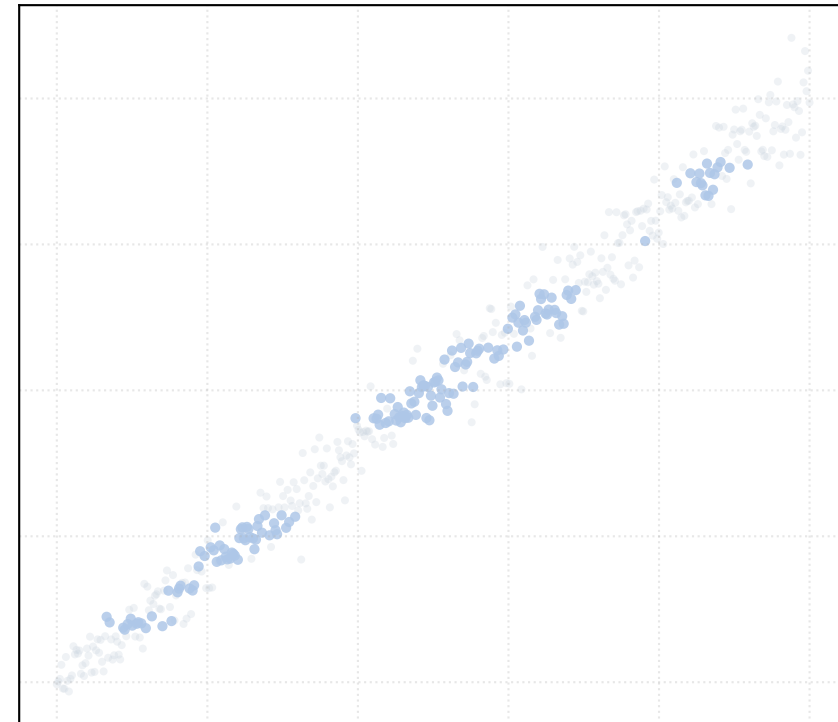
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

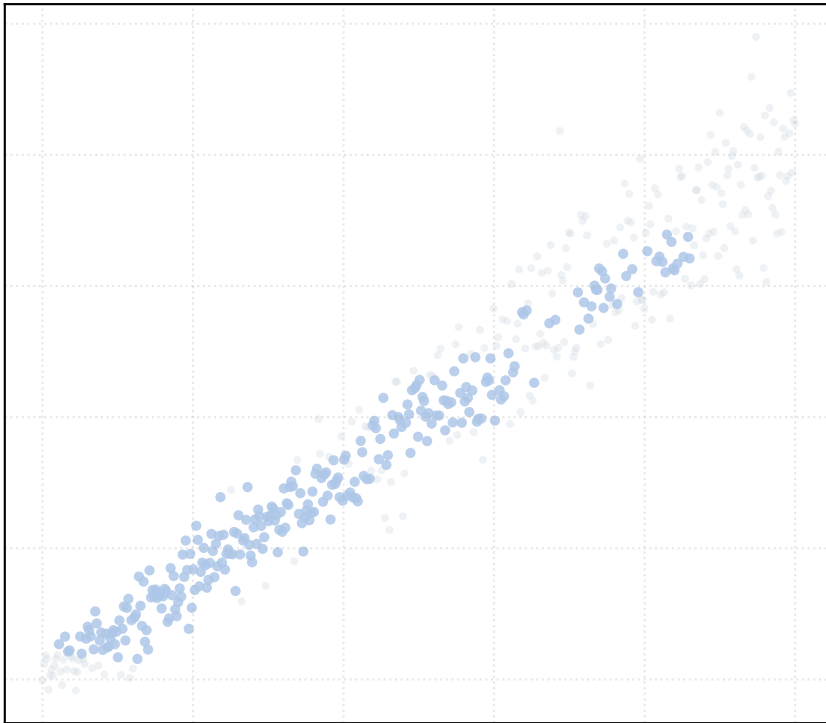


r = 3.5 [1 Clusters]

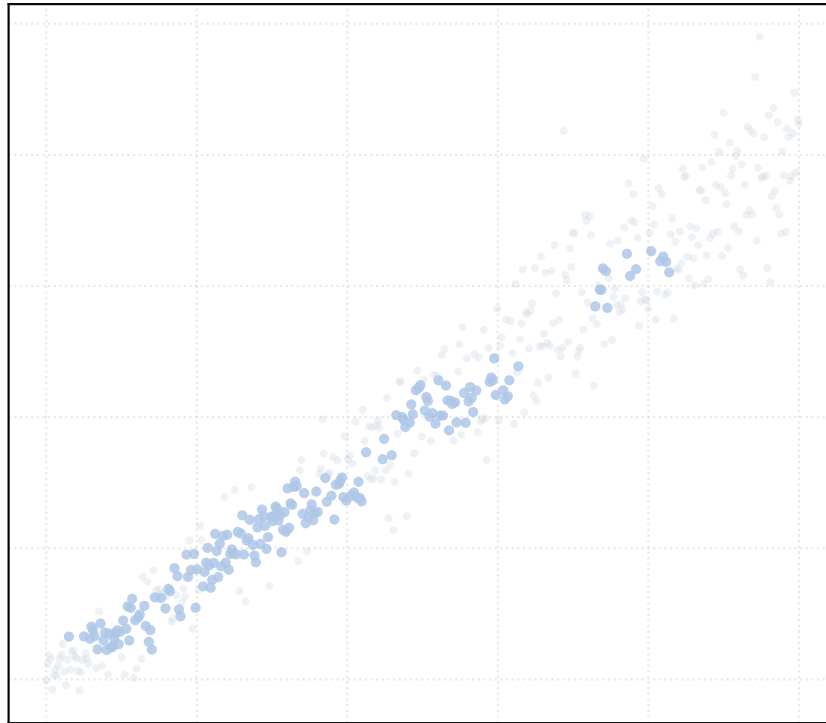


Dataset: heteroscedastic_noise_medium | Search Radius Sweep (r in grow_ensemble)

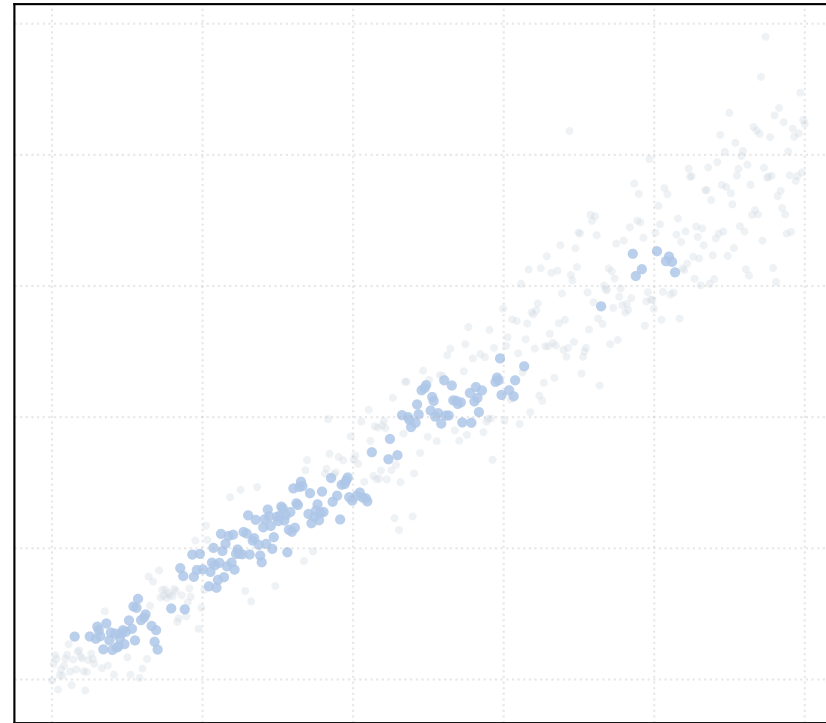
r = Default (Adaptive) [1 Clusters]



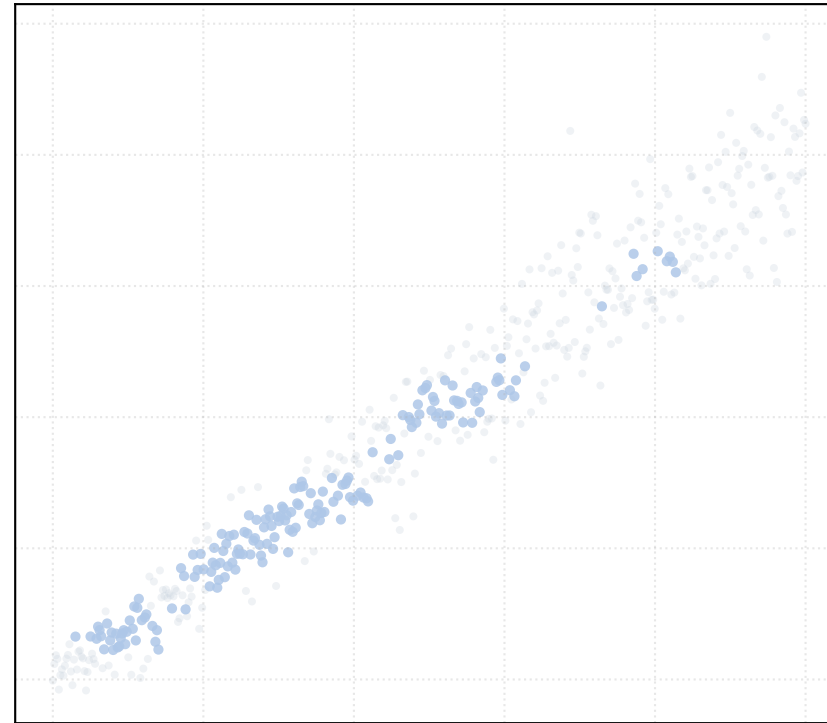
r = 0.5 [1 Clusters]



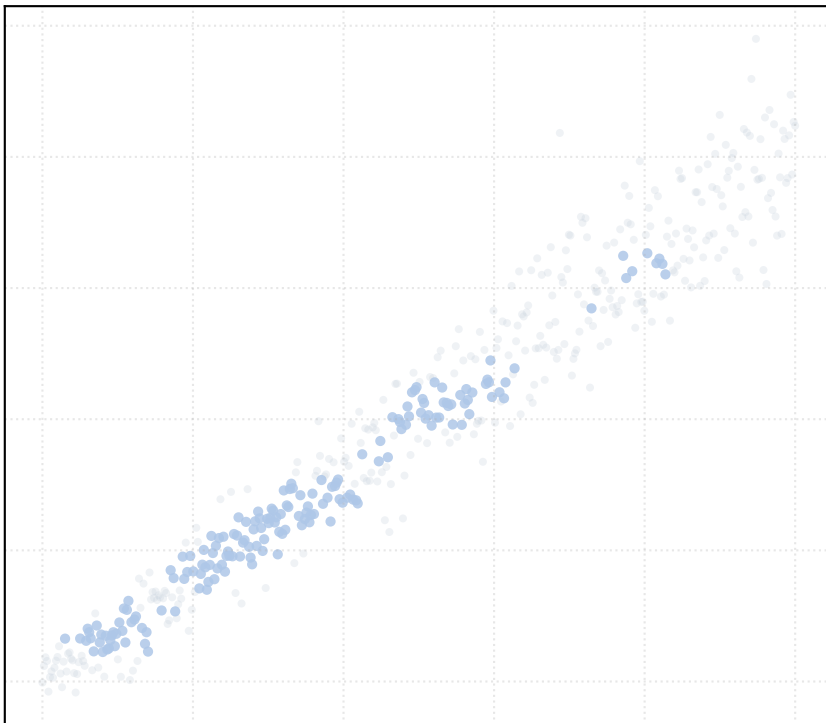
r = 1.0 [1 Clusters]



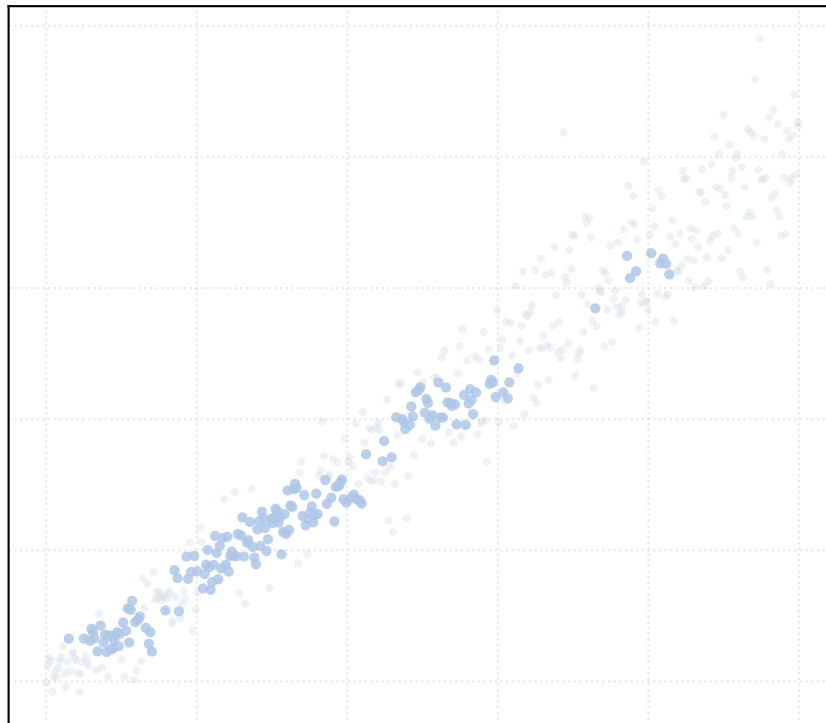
r = 1.5 [1 Clusters]



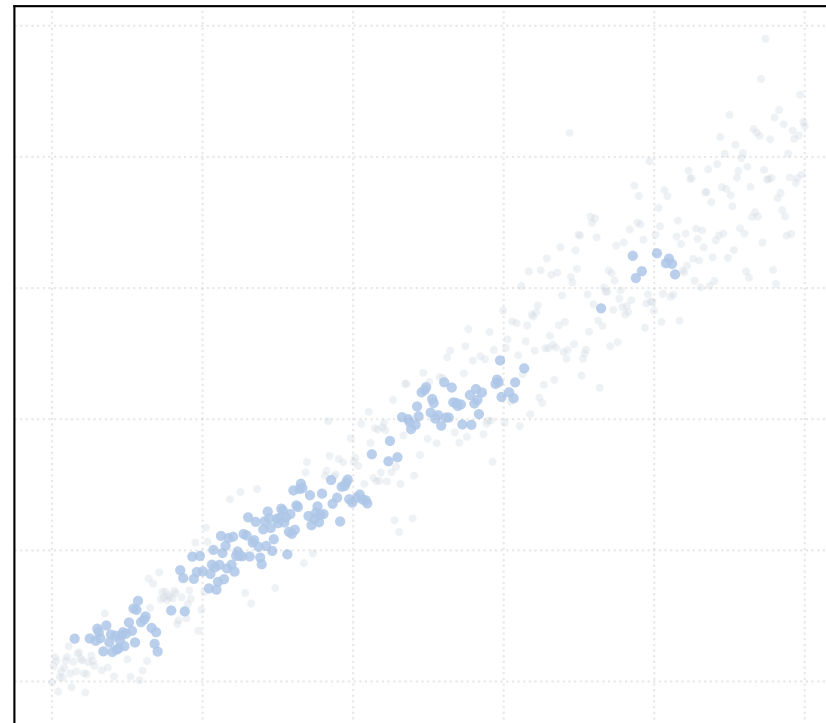
r = 2.0 [1 Clusters]



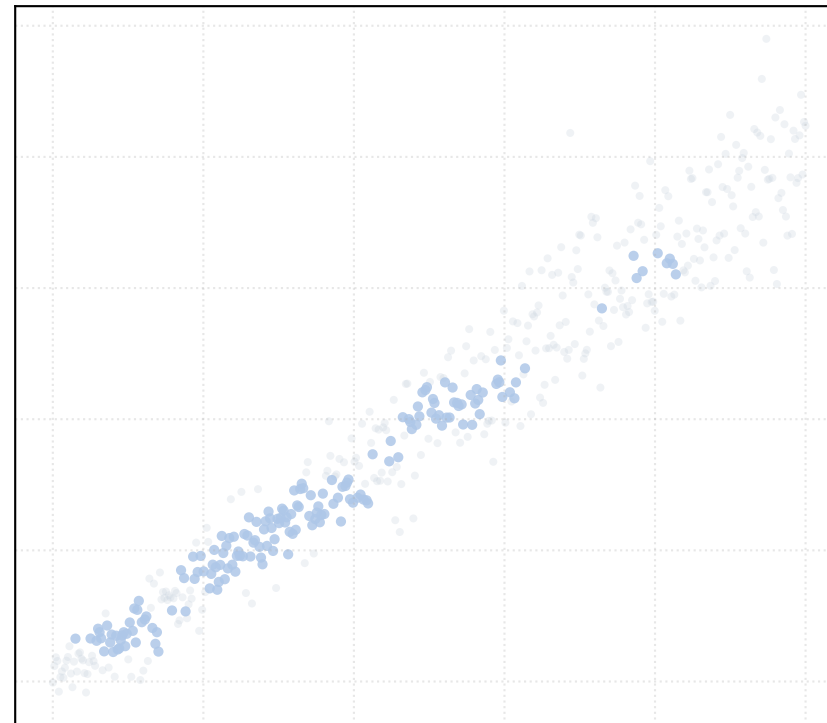
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

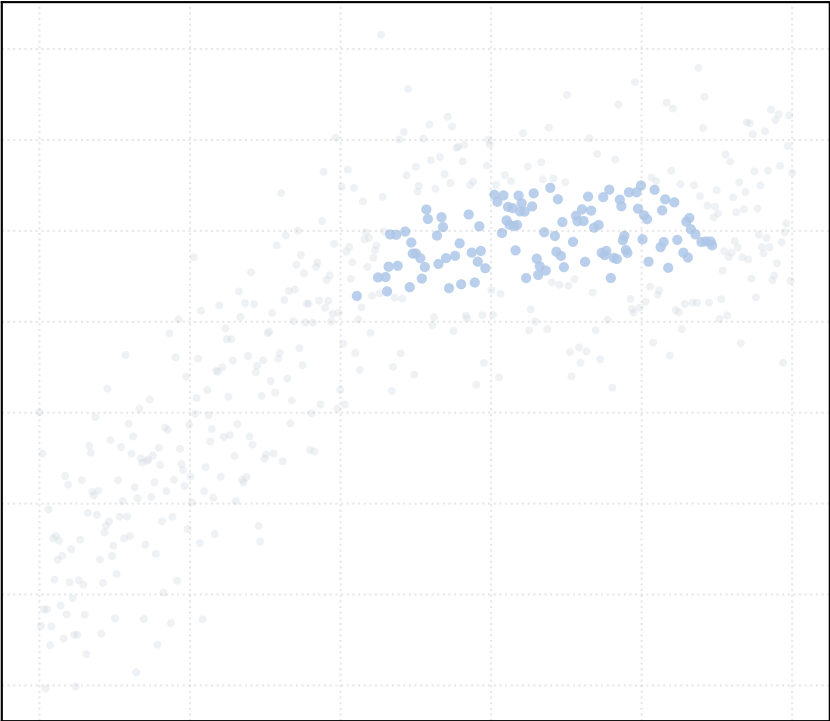


r = 3.5 [1 Clusters]

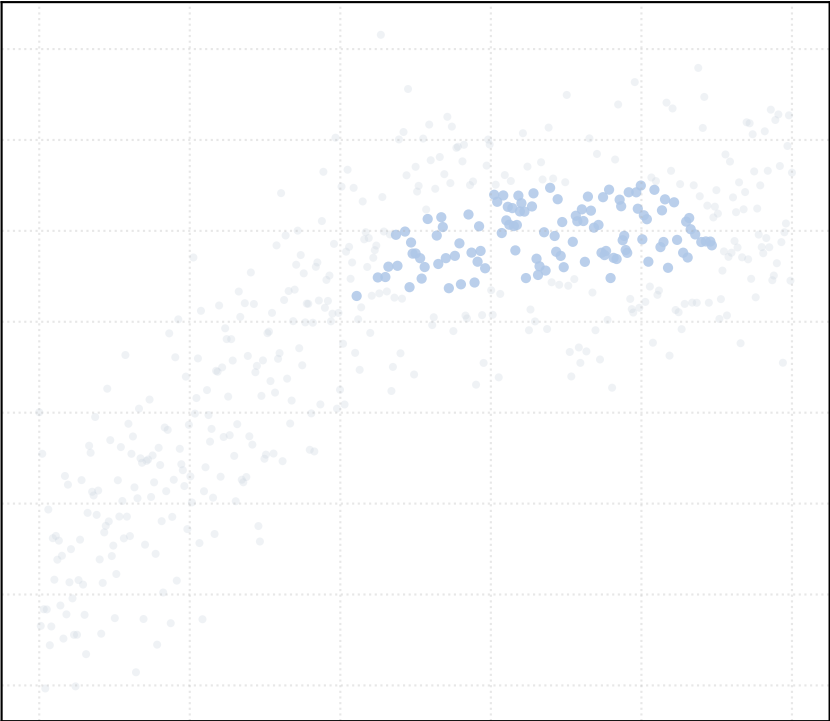


Dataset: kinked_curve_noise_high | Search Radius Sweep (r in grow_ensemble)

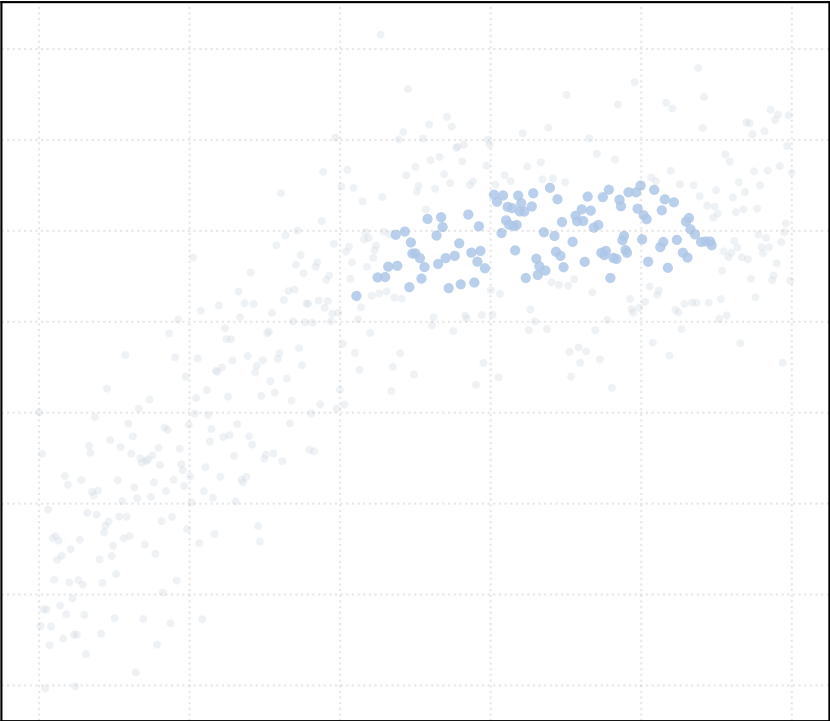
r = Default (Adaptive) [1 Clusters]



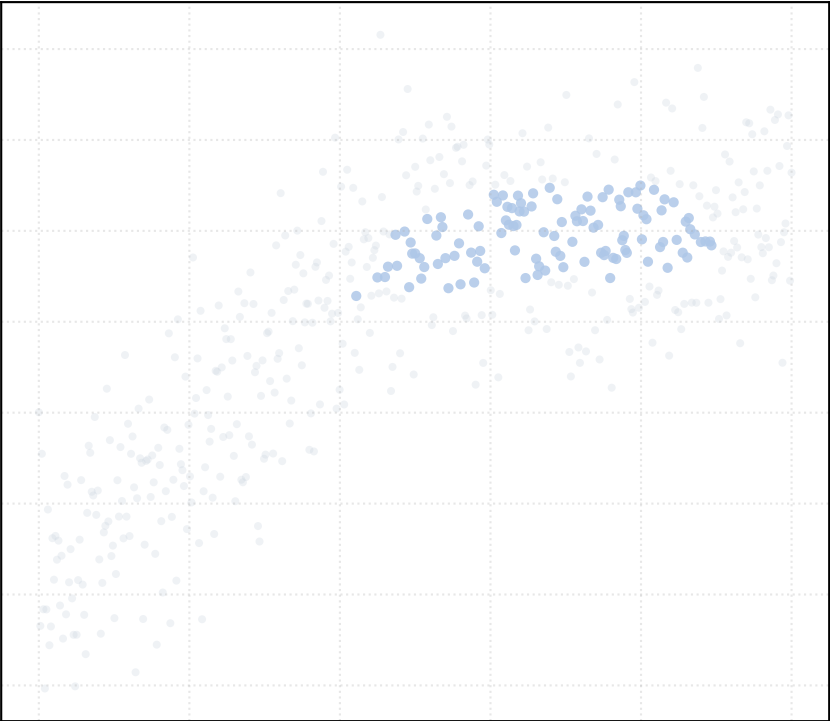
r = 0.5 [1 Clusters]



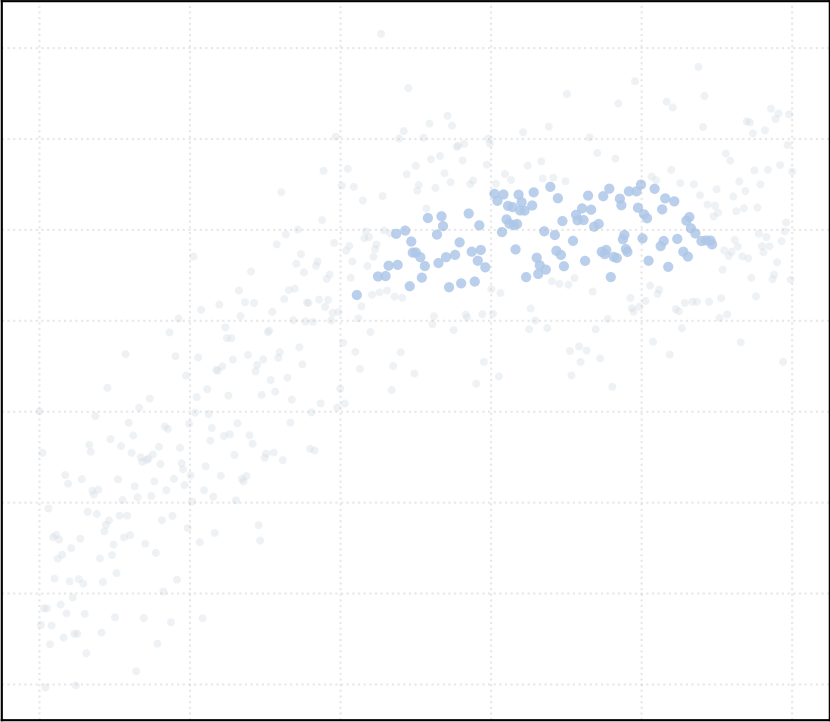
r = 1.0 [1 Clusters]



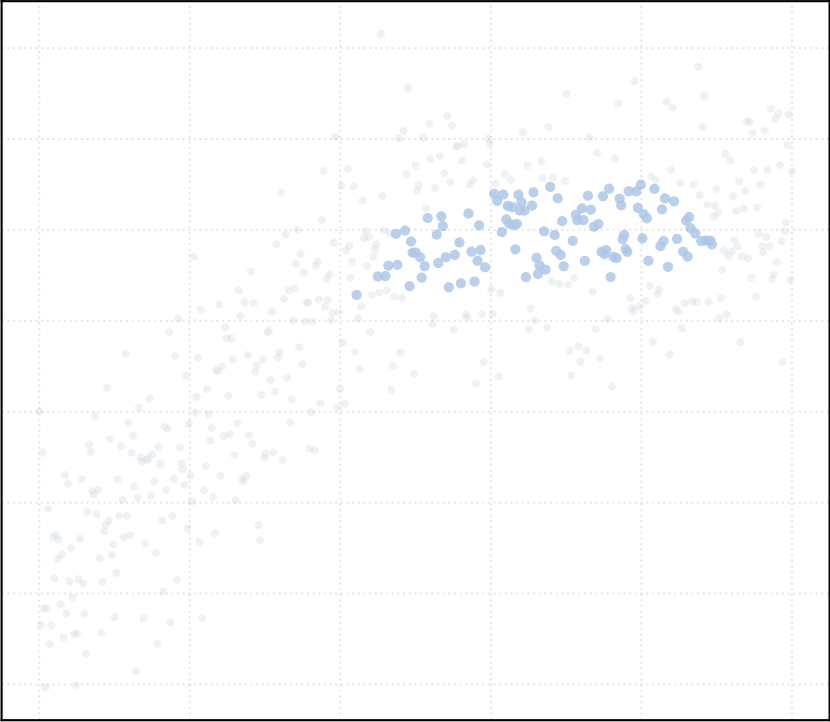
r = 1.5 [1 Clusters]



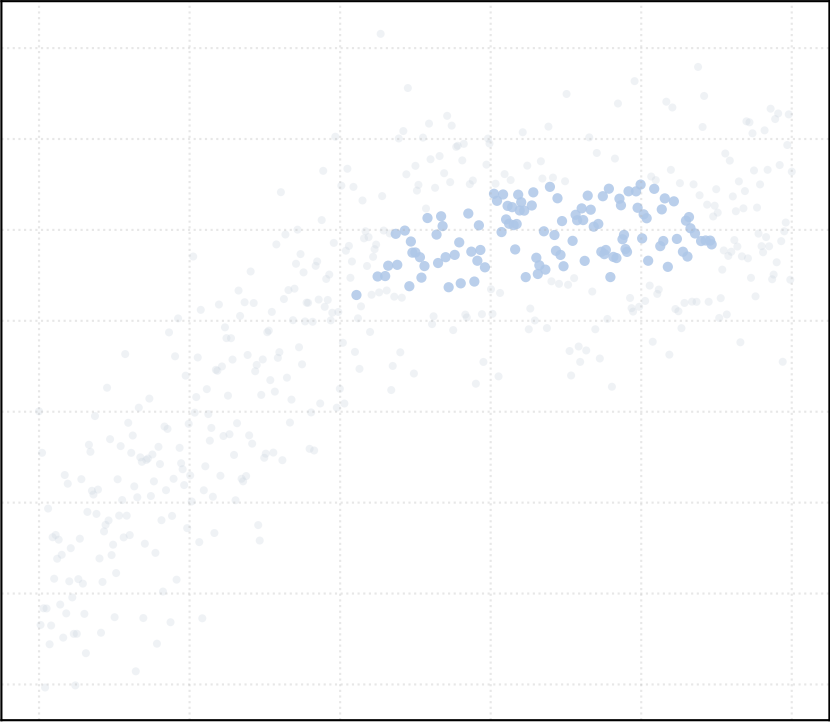
r = 2.0 [1 Clusters]



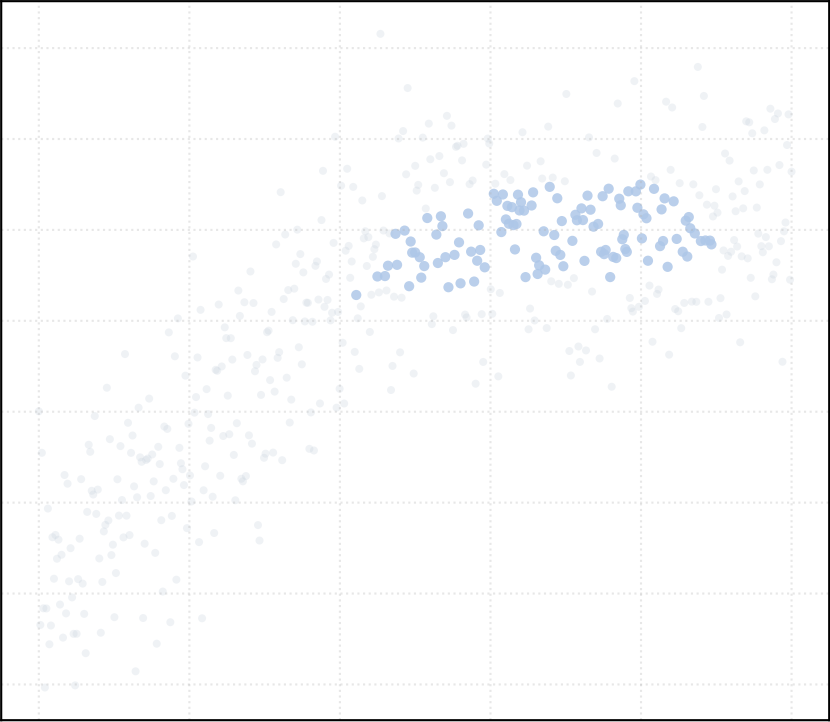
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

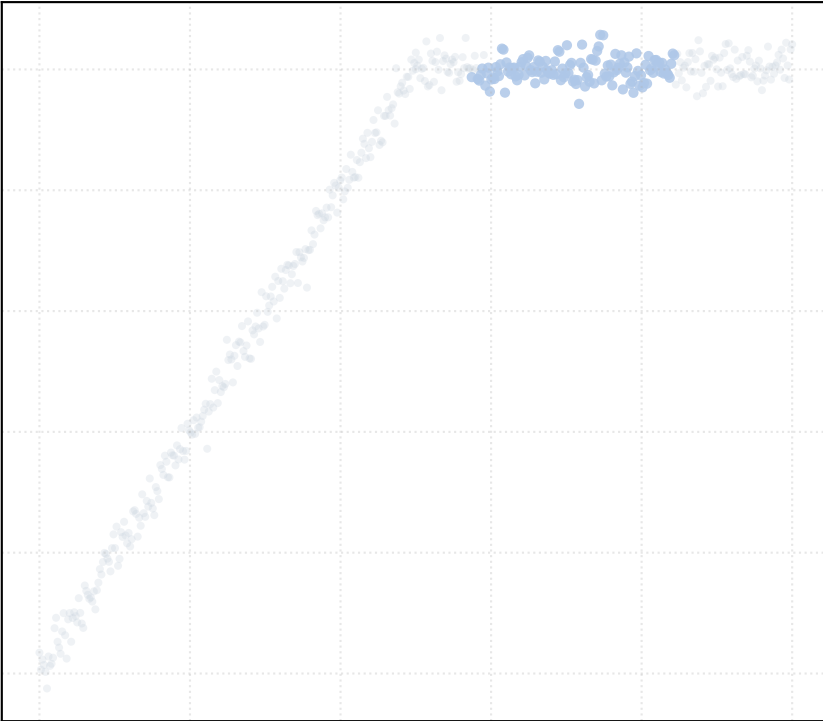


r = 3.5 [1 Clusters]

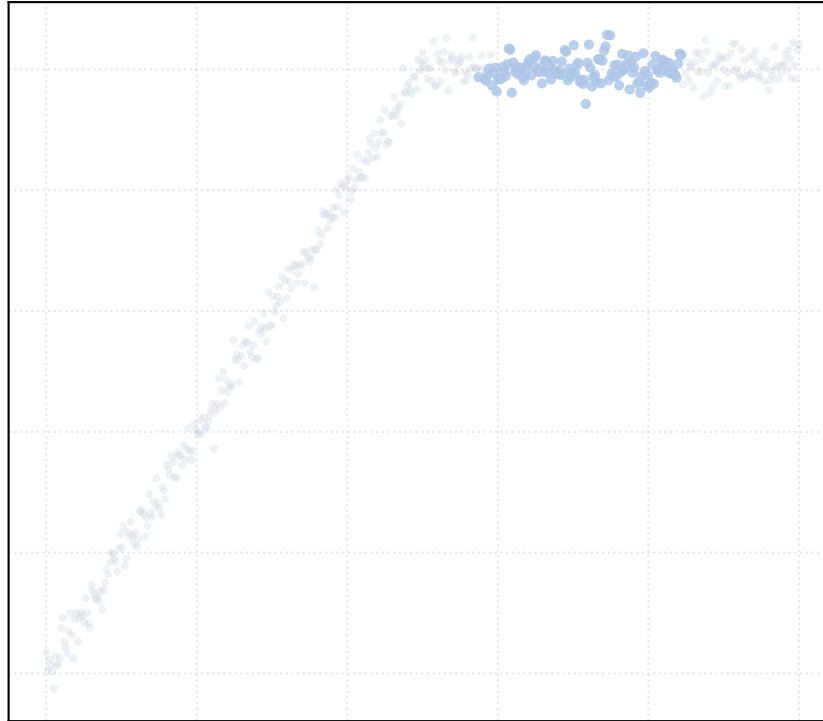


Dataset: kinked_curve_noise_low | Search Radius Sweep (r in grow_ensemble)

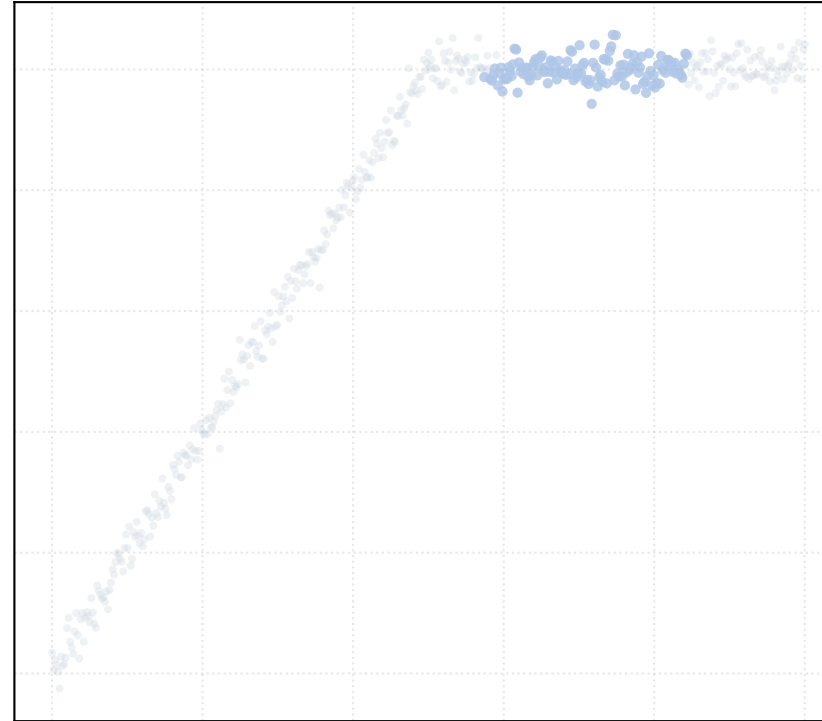
r = Default (Adaptive) [1 Clusters]



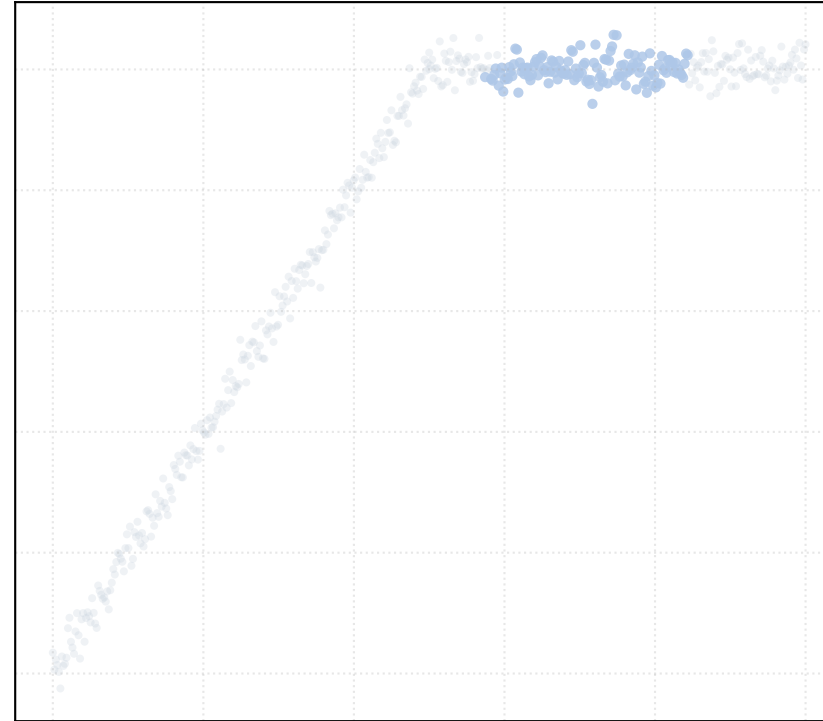
r = 0.5 [1 Clusters]



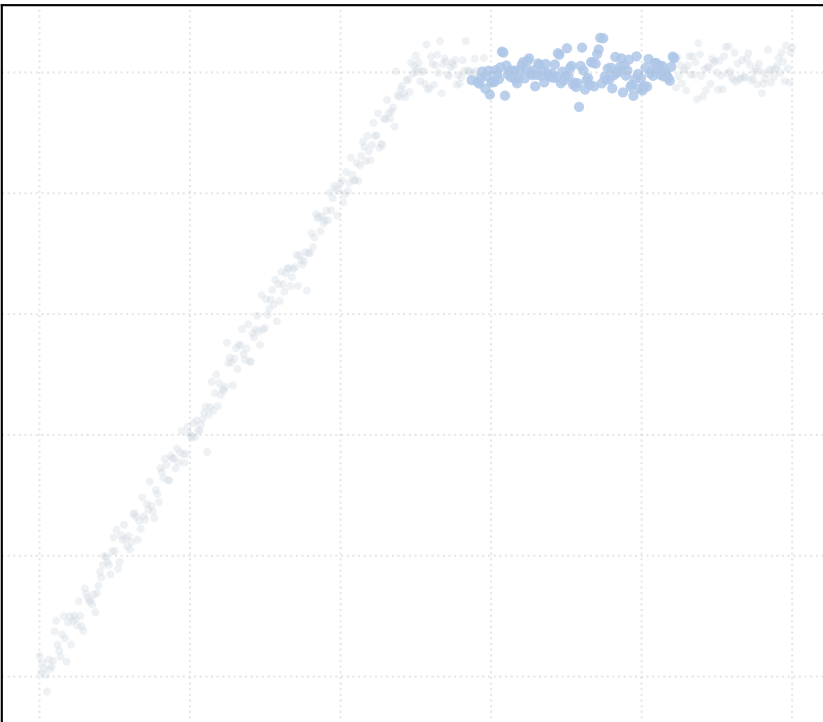
r = 1.0 [1 Clusters]



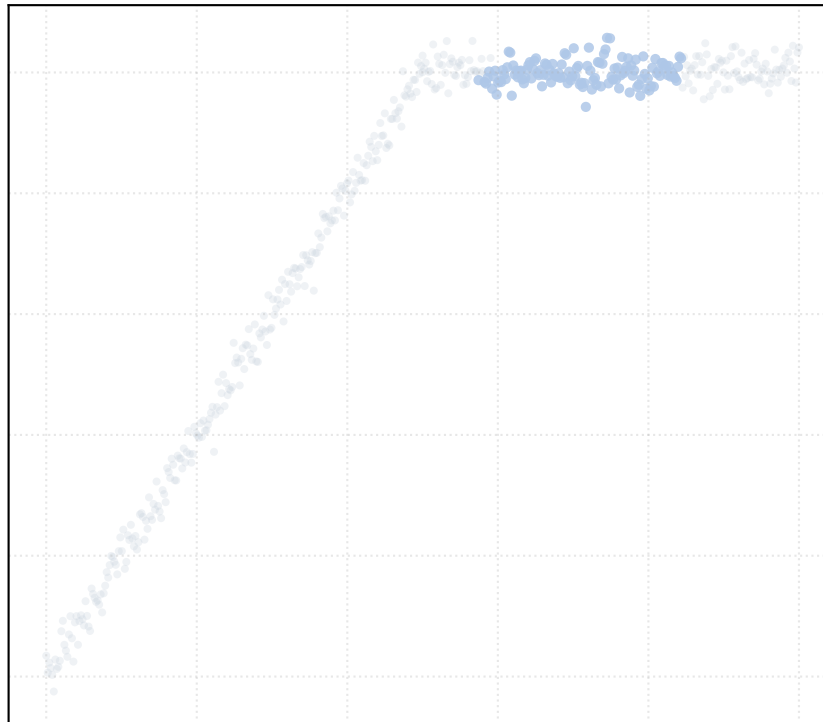
r = 1.5 [1 Clusters]



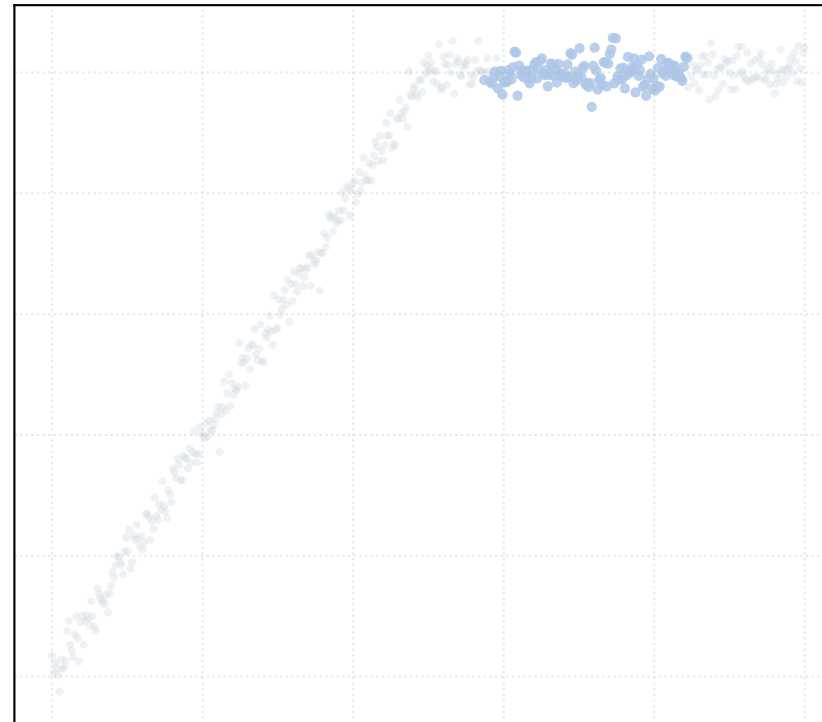
r = 2.0 [1 Clusters]



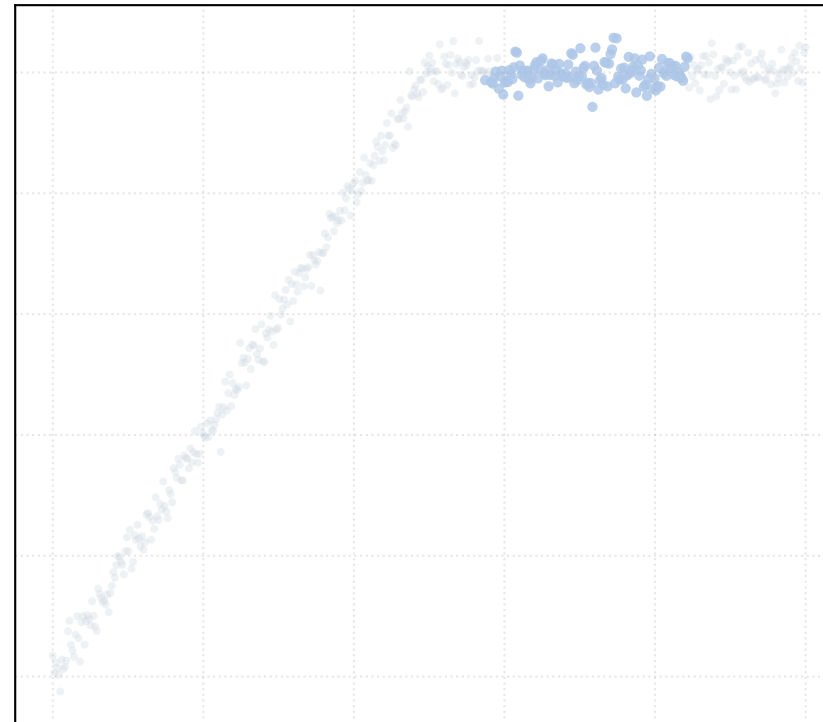
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]



r = 3.5 [1 Clusters]

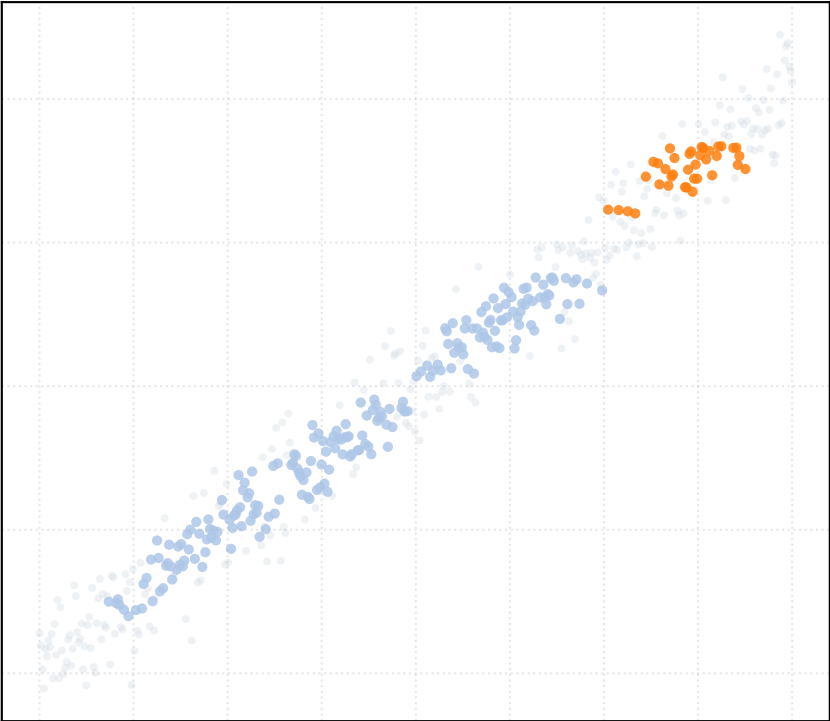


Dataset: kinked_curve_noise_medium | Search Radius Sweep (r in grow_ensemble)

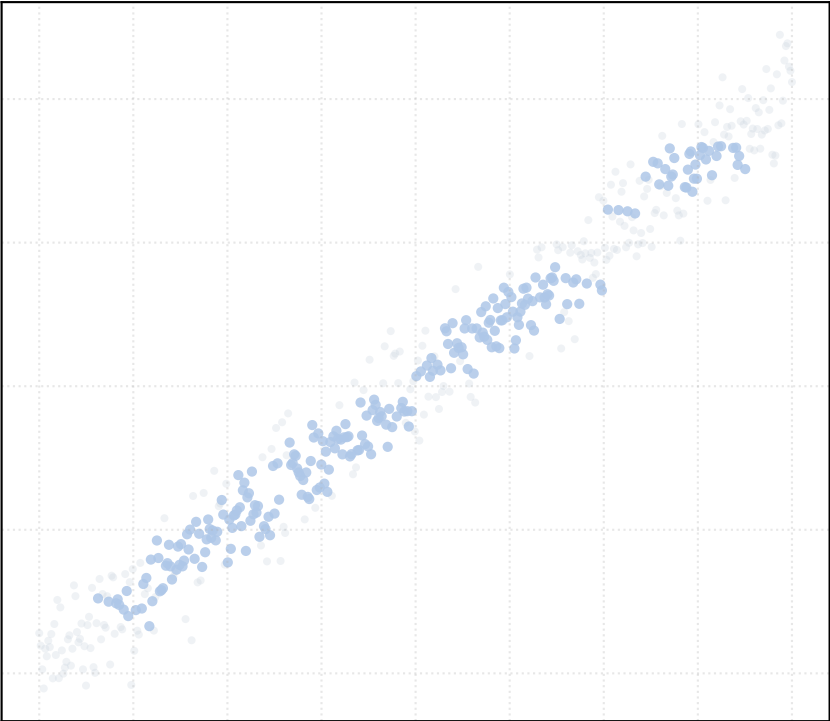


Dataset: linear_noise_high | Search Radius Sweep (r in grow_ensemble)

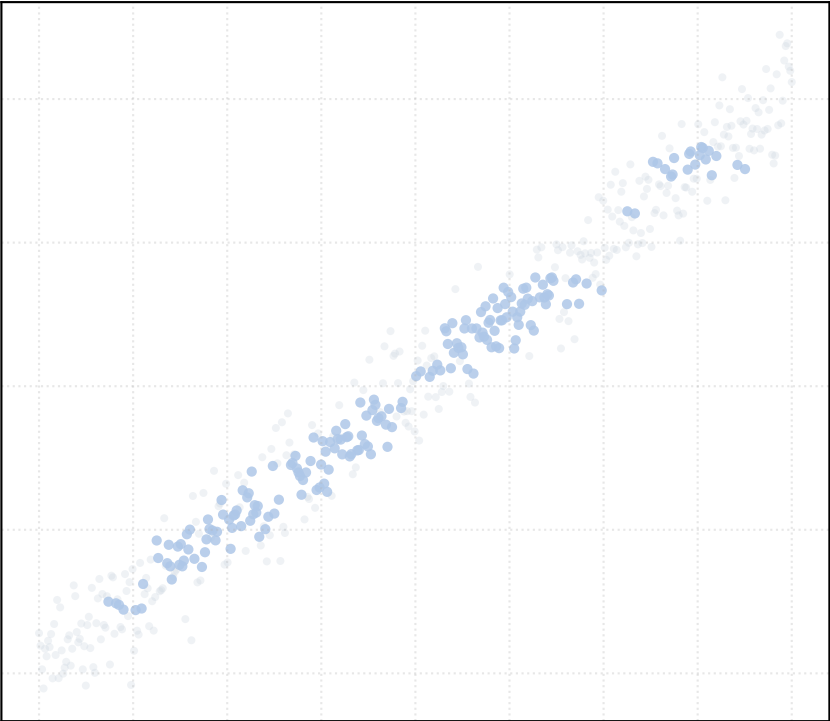
r = Default (Adaptive) [2 Clusters]



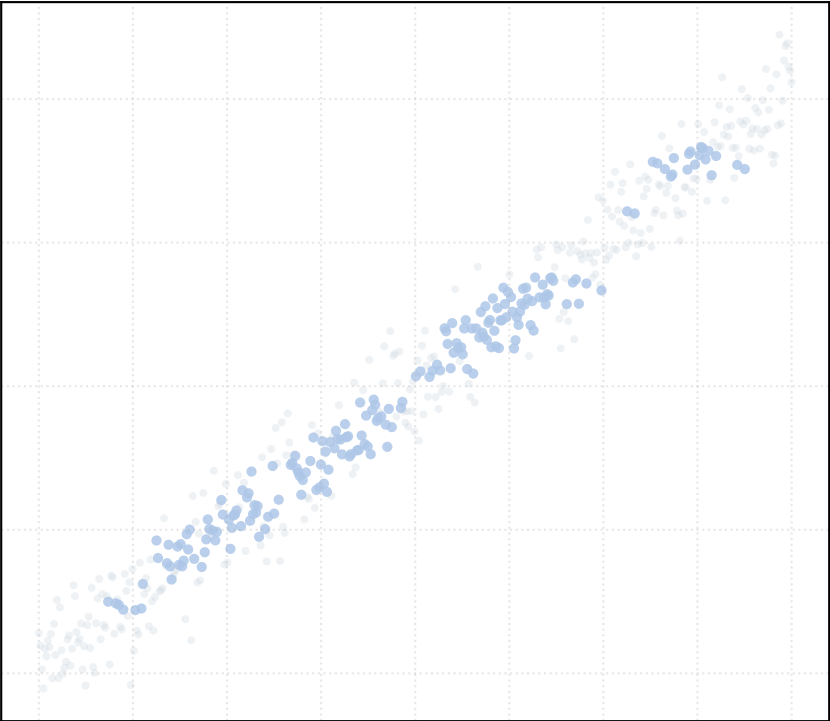
r = 0.5 [1 Clusters]



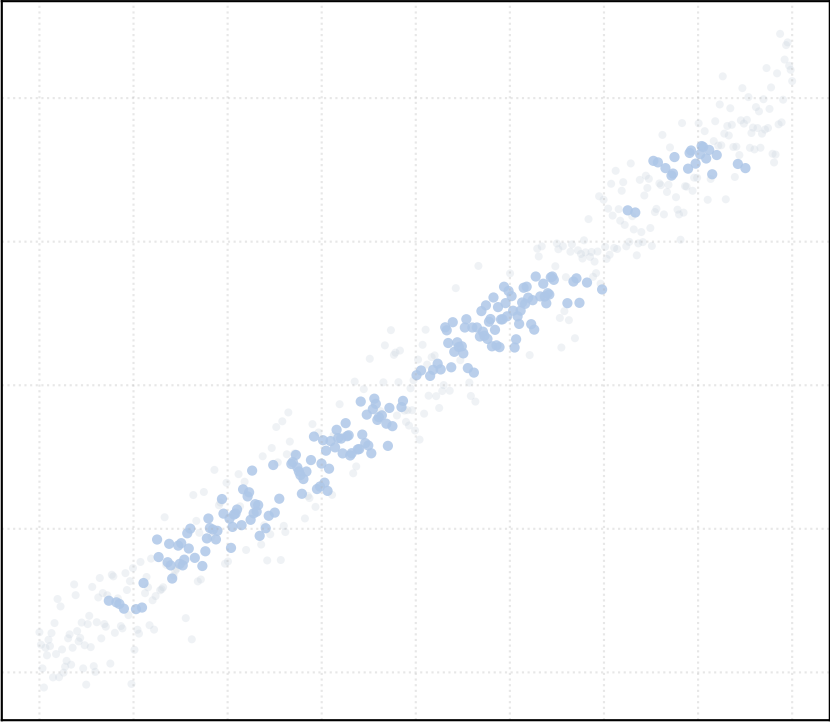
r = 1.0 [1 Clusters]



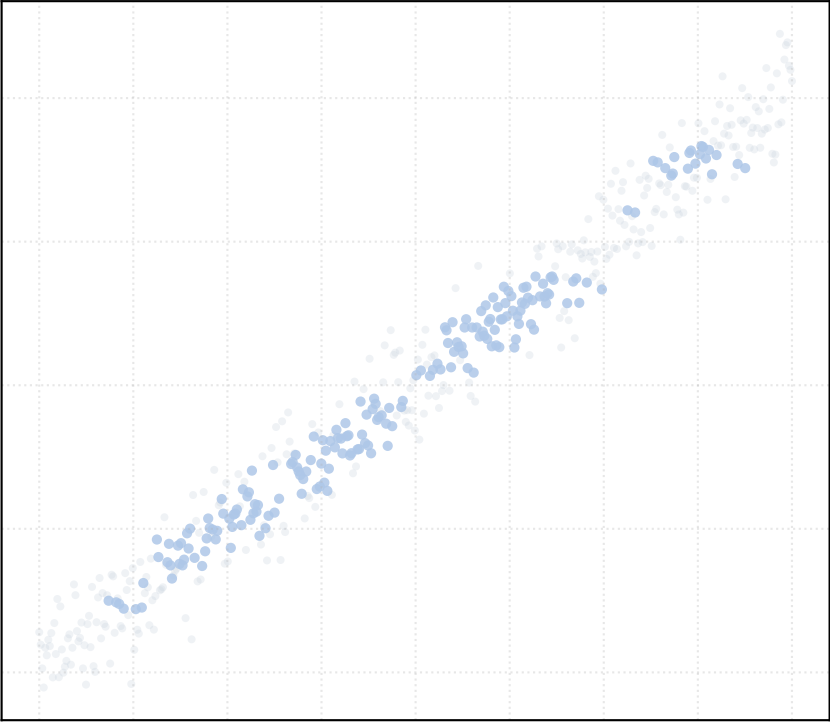
r = 1.5 [1 Clusters]



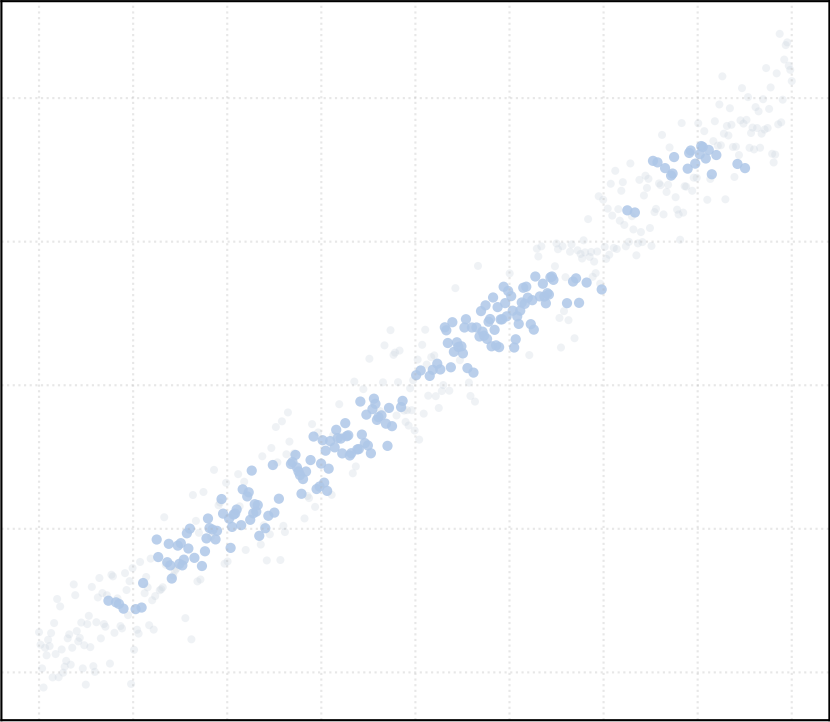
r = 2.0 [1 Clusters]



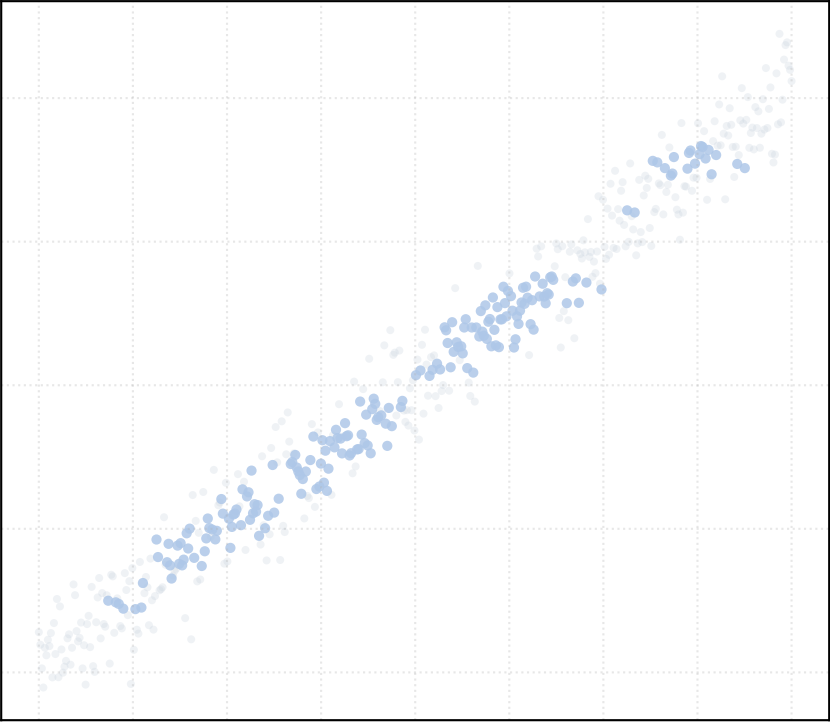
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

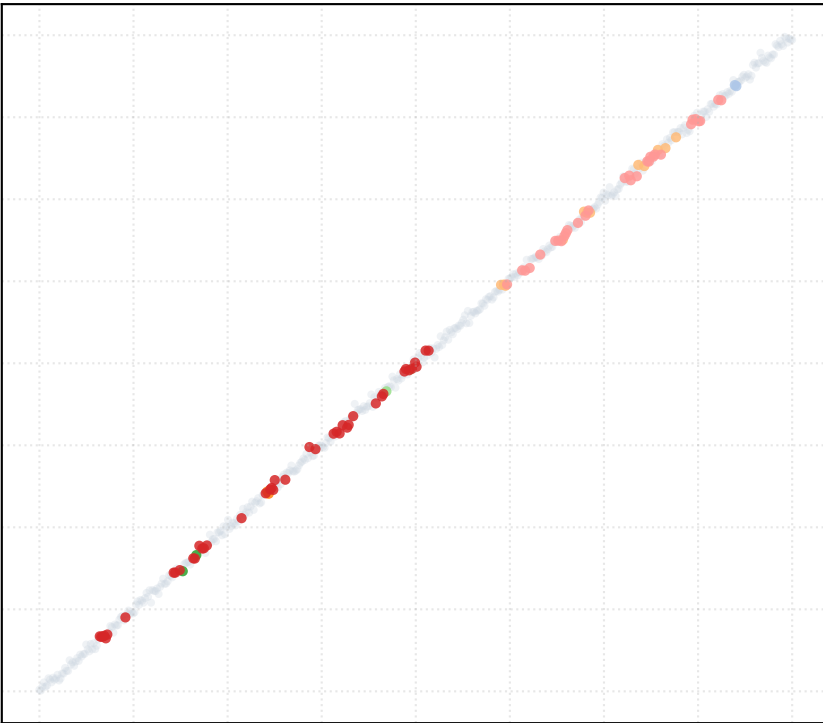


r = 3.5 [1 Clusters]

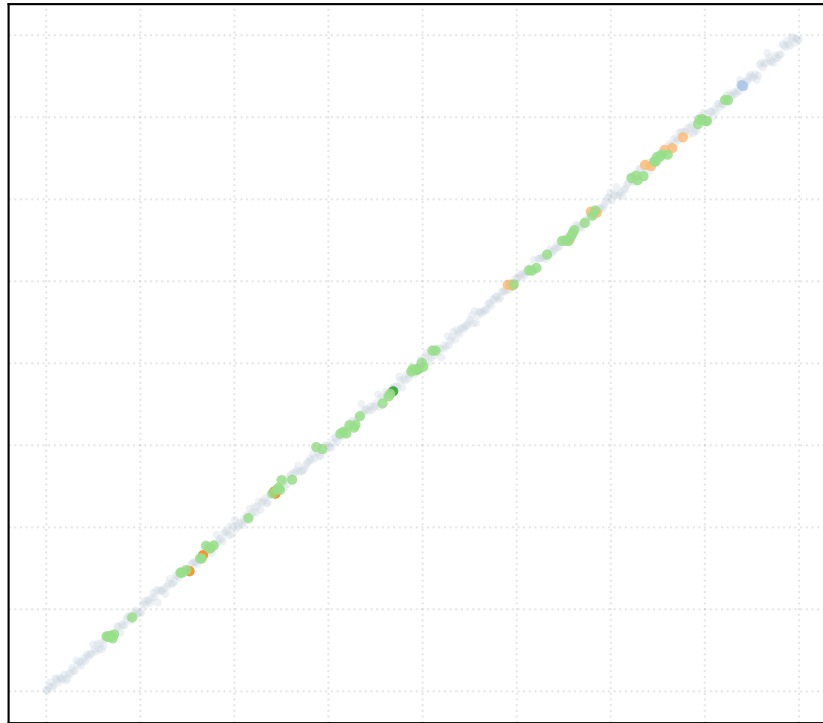


Dataset: linear_noise_low | Search Radius Sweep (r in grow_ensemble)

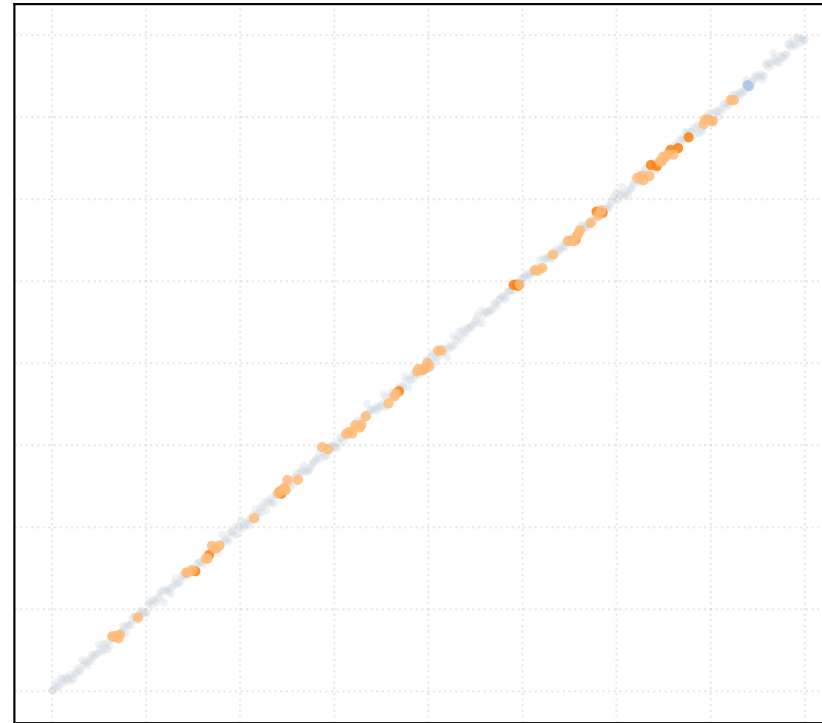
r = Default (Adaptive) [7 Clusters]



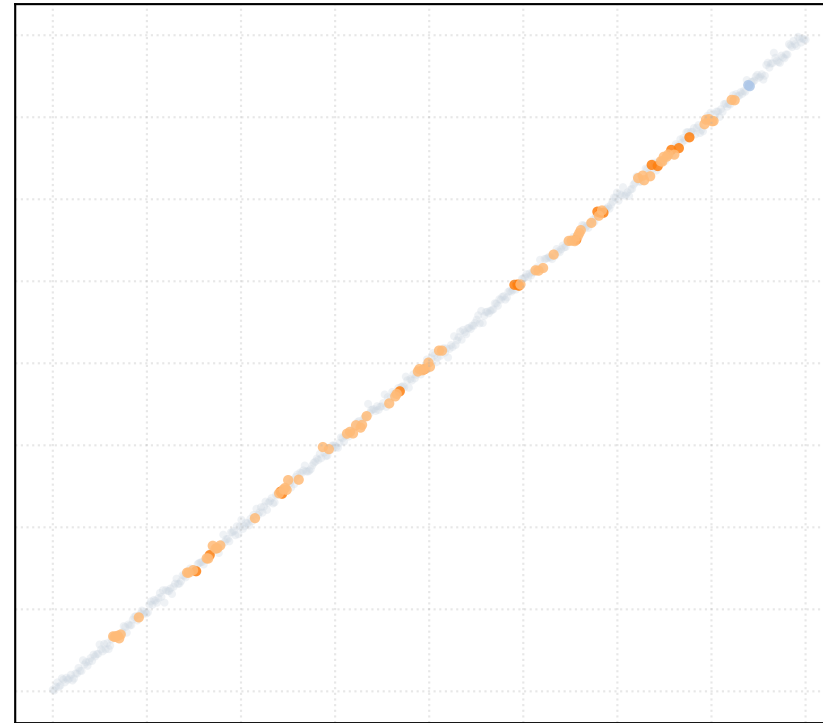
r = 0.5 [5 Clusters]



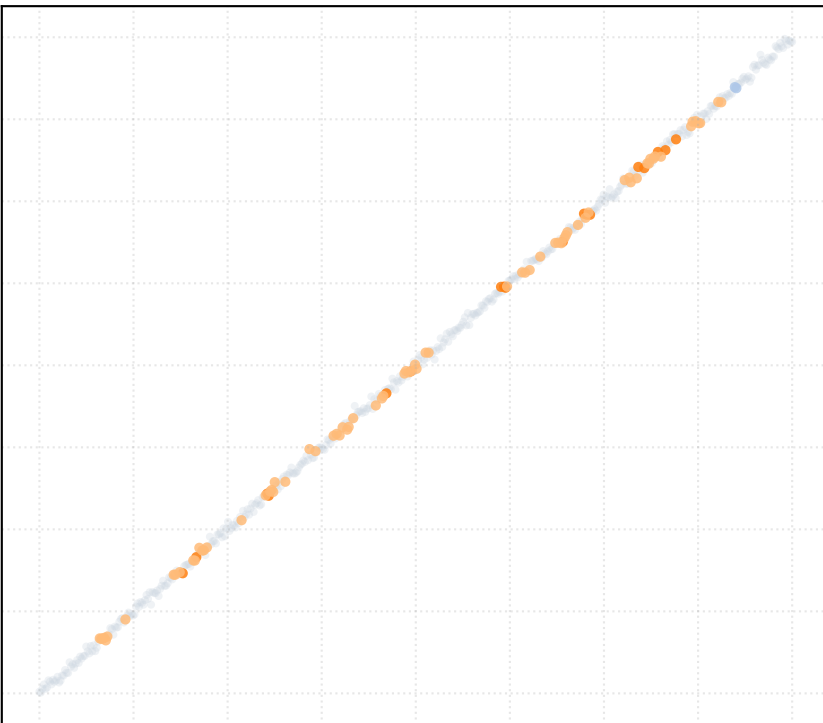
r = 1.0 [3 Clusters]



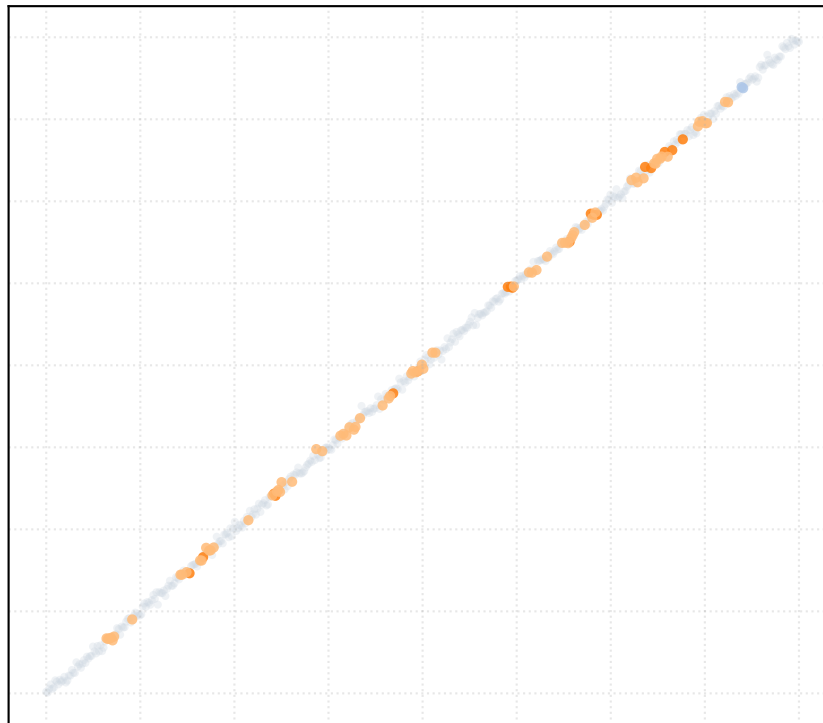
r = 1.5 [3 Clusters]



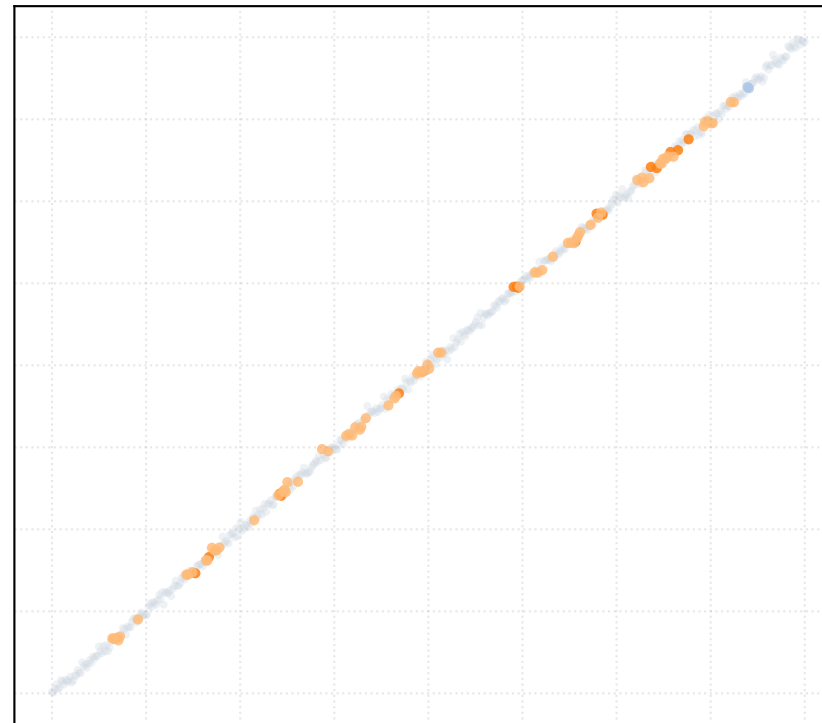
r = 2.0 [3 Clusters]



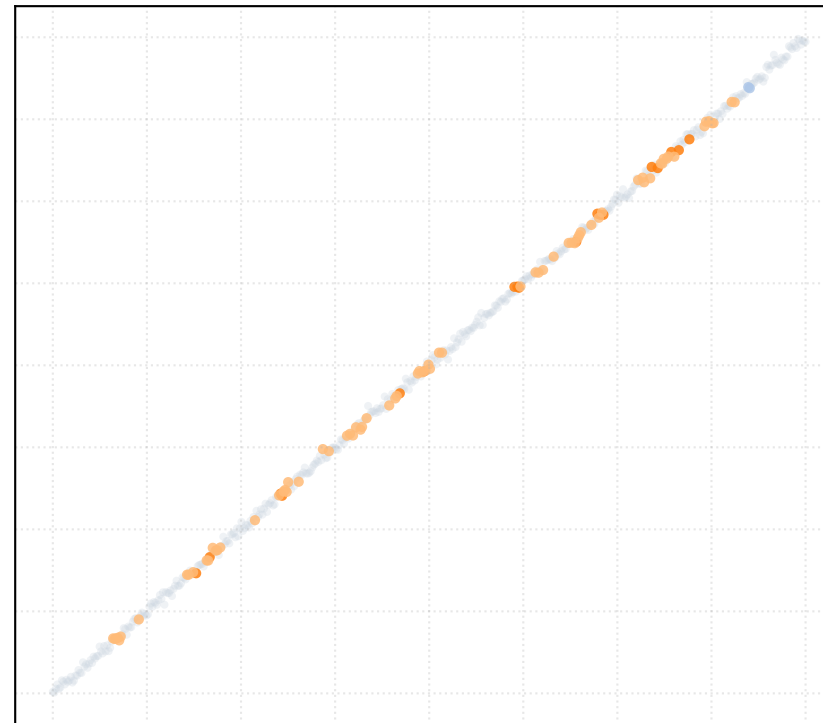
r = 2.5 [3 Clusters]



r = 3.0 [3 Clusters]

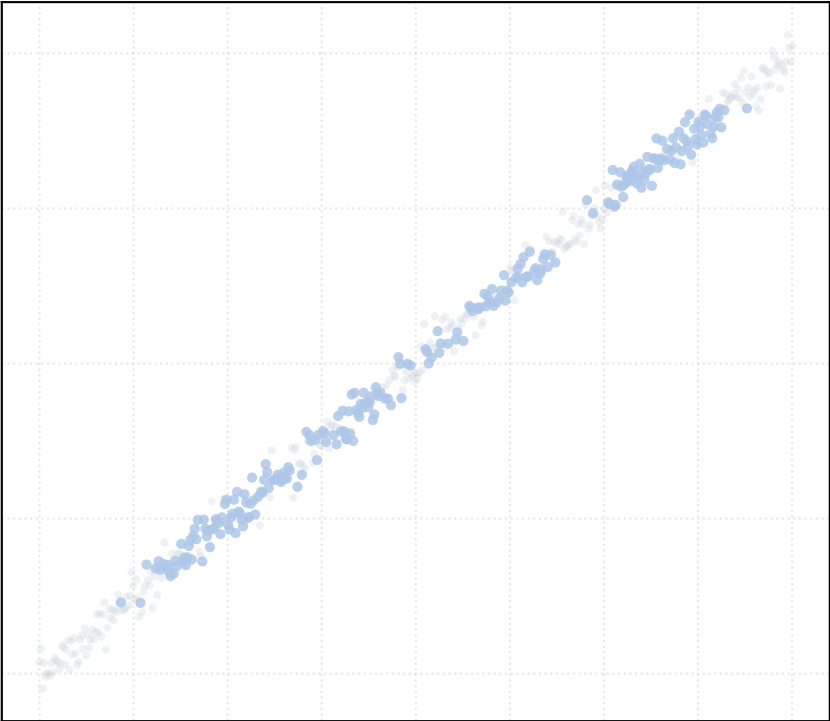


r = 3.5 [3 Clusters]

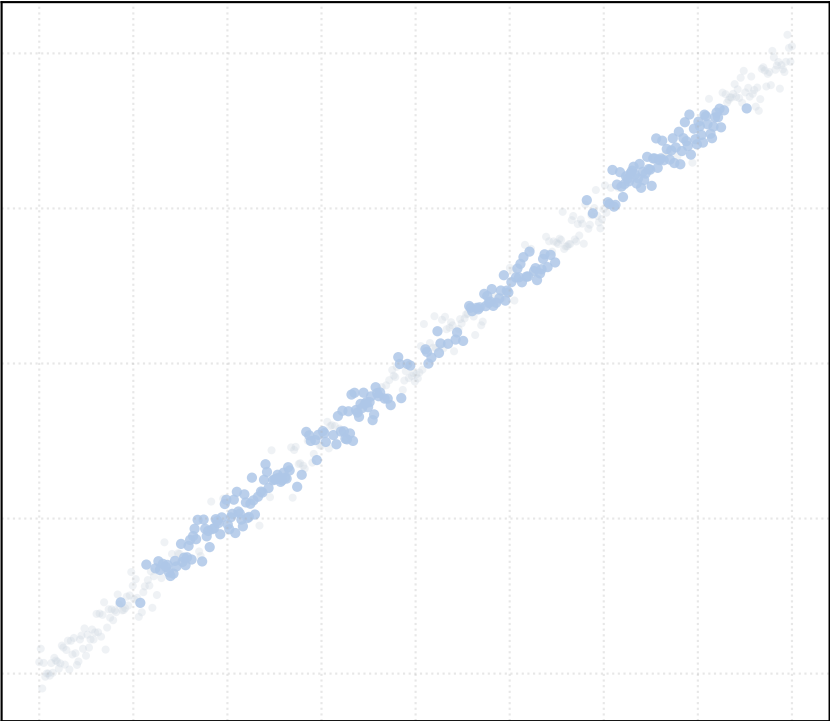


Dataset: linear_noise_medium | Search Radius Sweep (r in grow_ensemble)

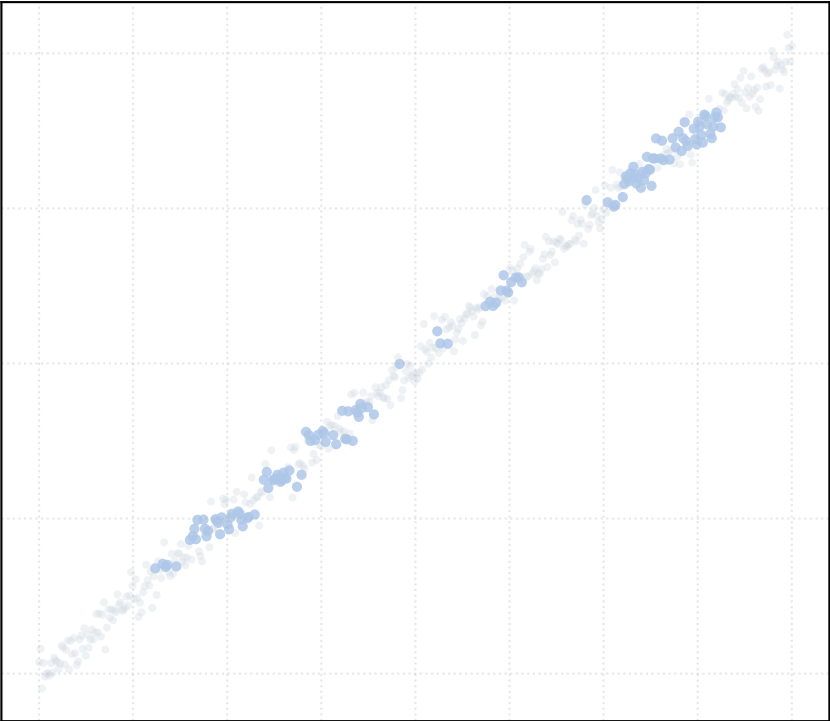
r = Default (Adaptive) [1 Clusters]



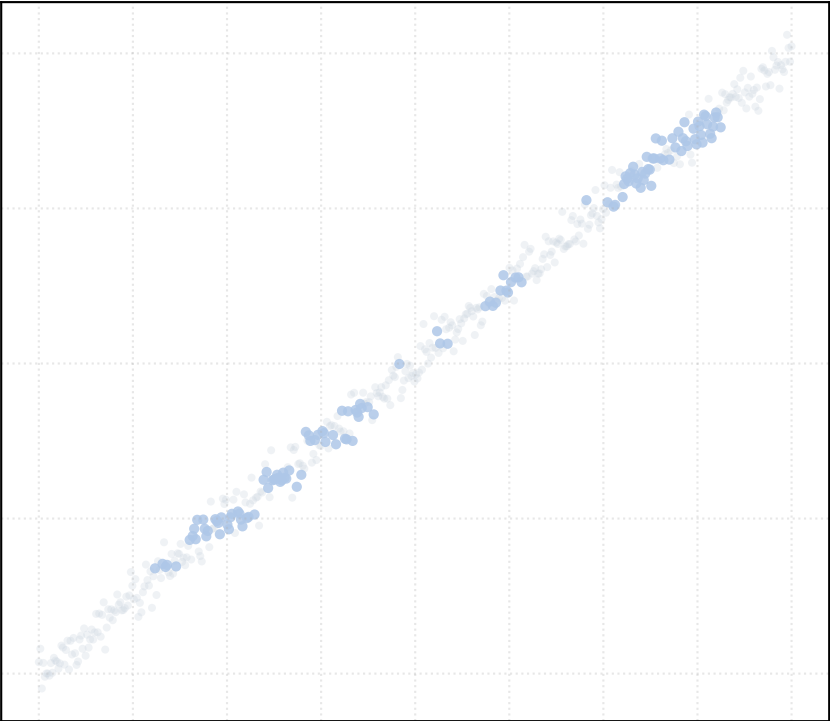
r = 0.5 [1 Clusters]



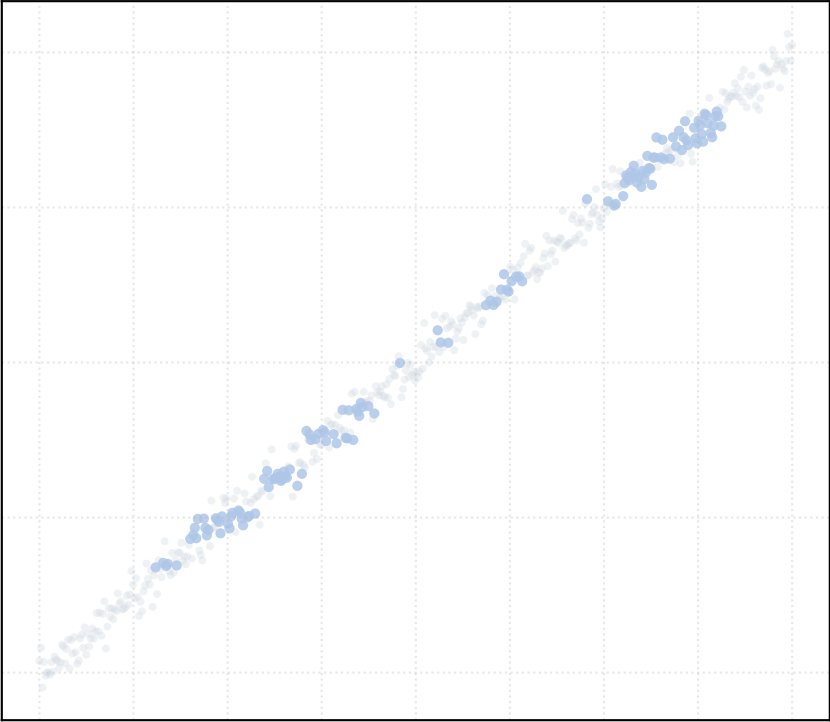
r = 1.0 [1 Clusters]



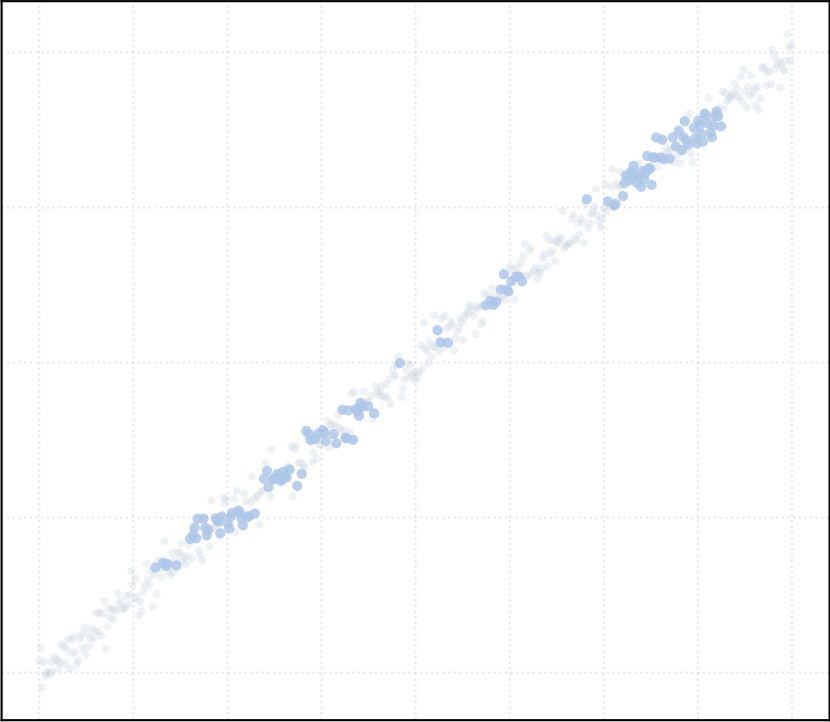
r = 1.5 [1 Clusters]



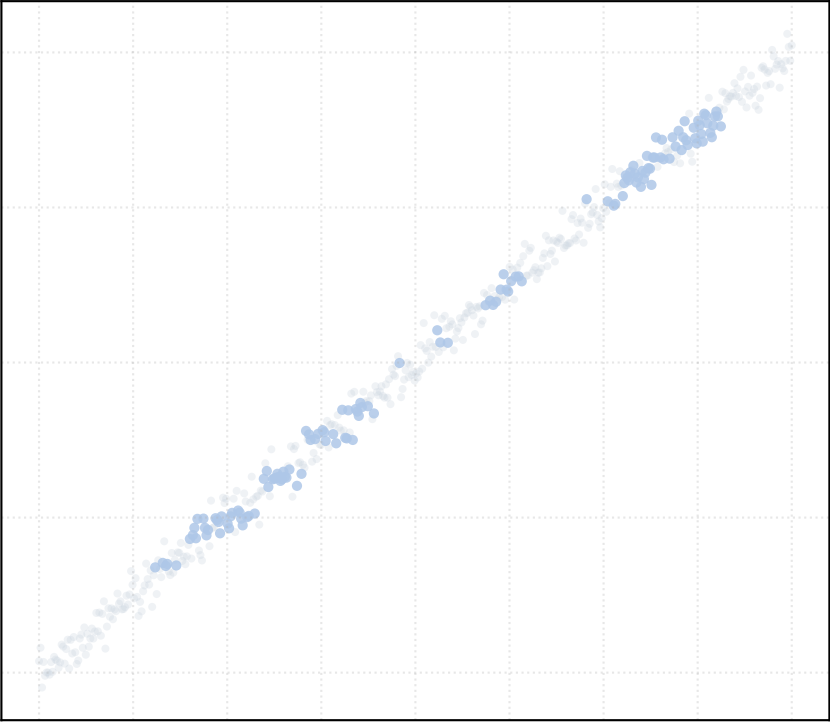
r = 2.0 [1 Clusters]



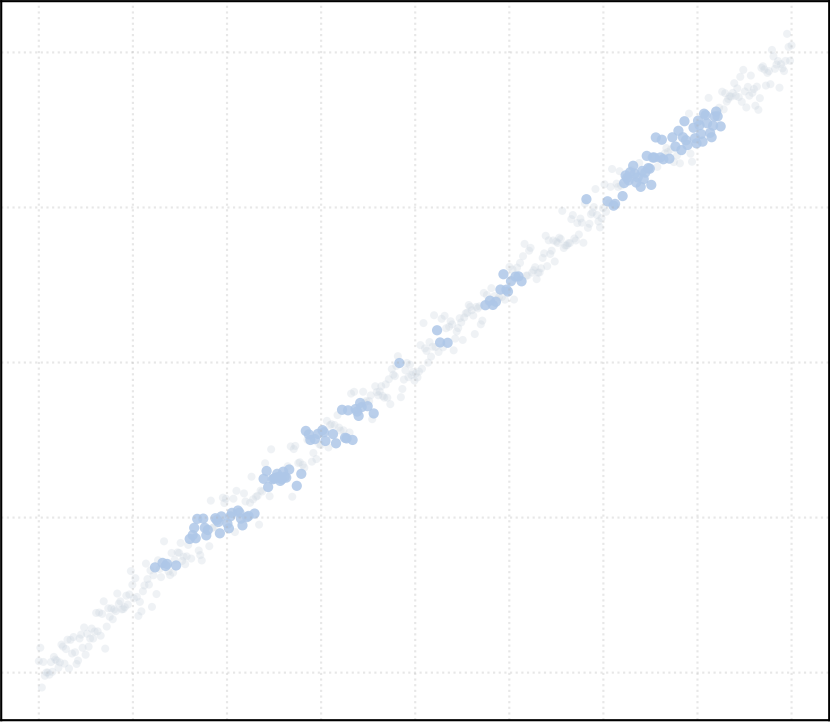
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

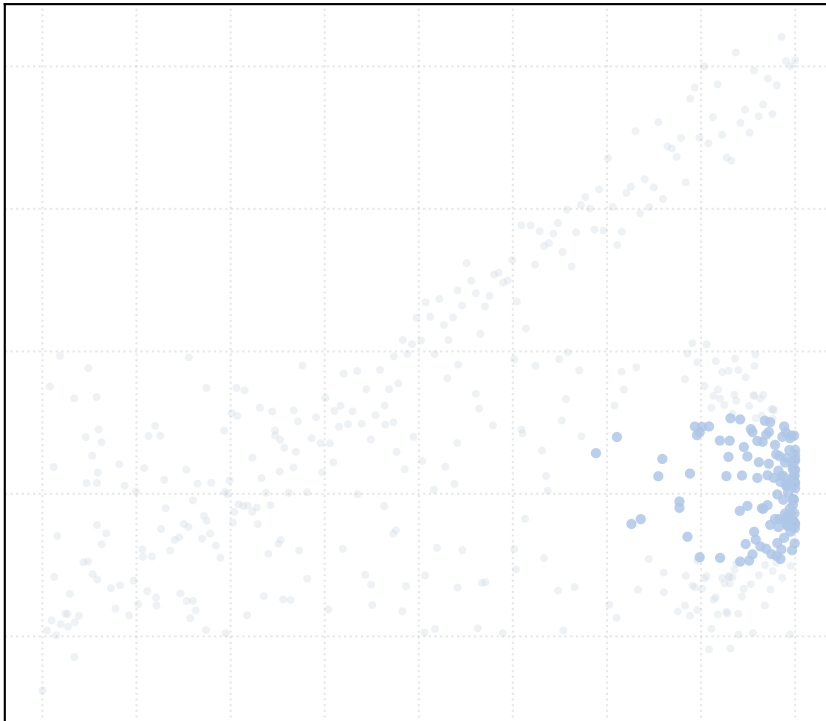


r = 3.5 [1 Clusters]

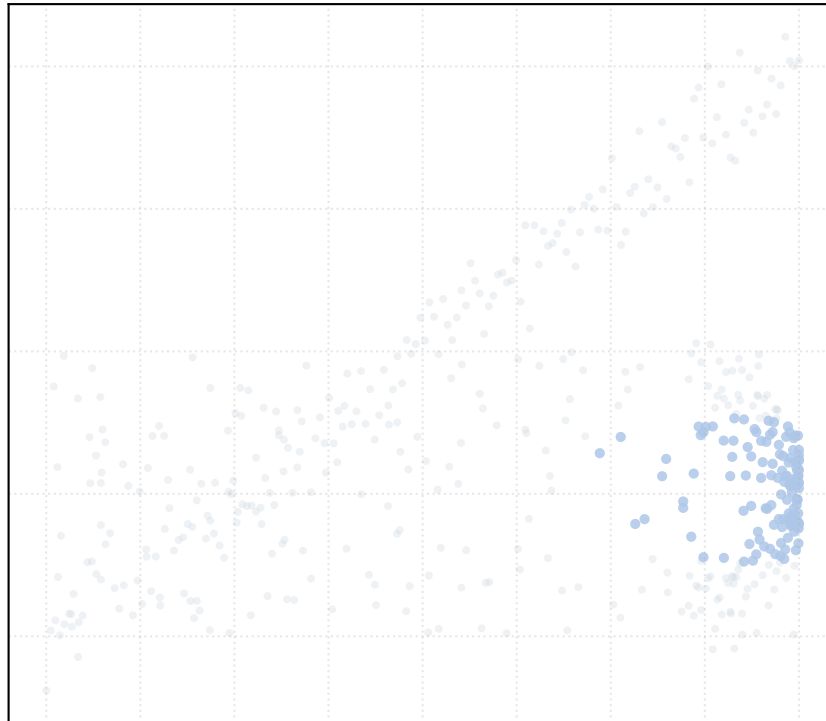


Dataset: mixed_generators_noise_high | Search Radius Sweep (r in grow_ensemble)

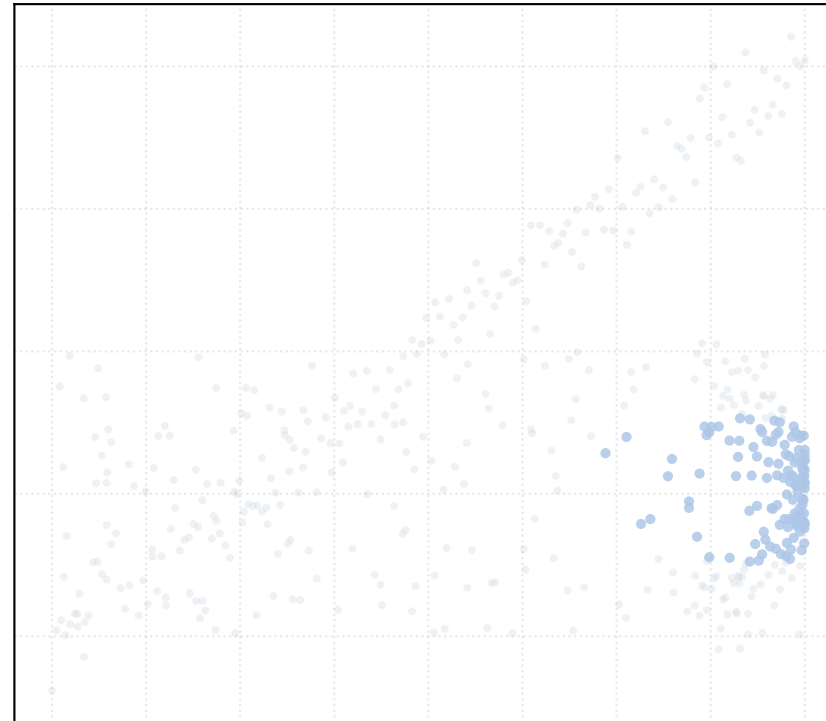
r = Default (Adaptive) [1 Clusters]



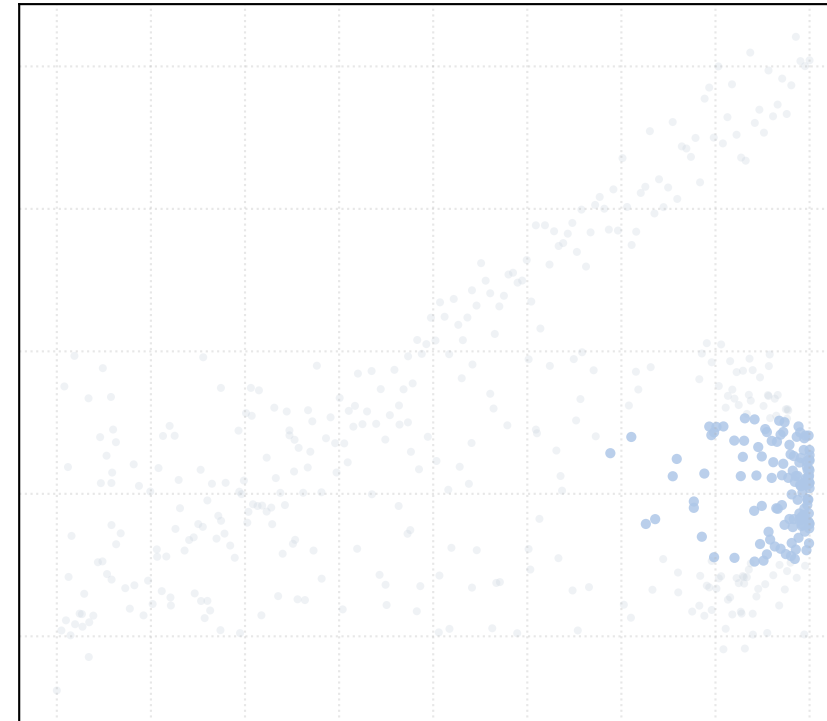
r = 0.5 [1 Clusters]



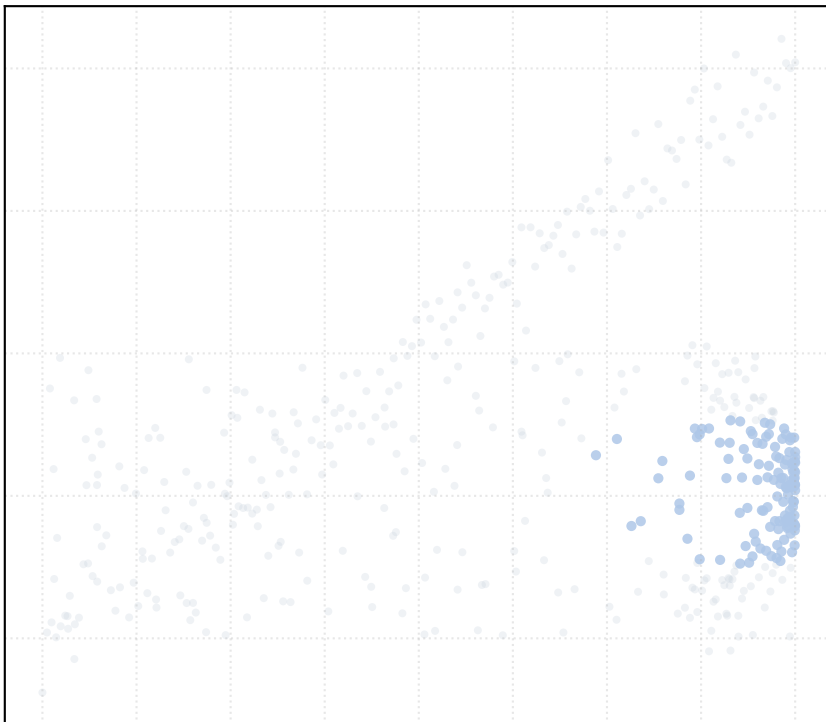
r = 1.0 [1 Clusters]



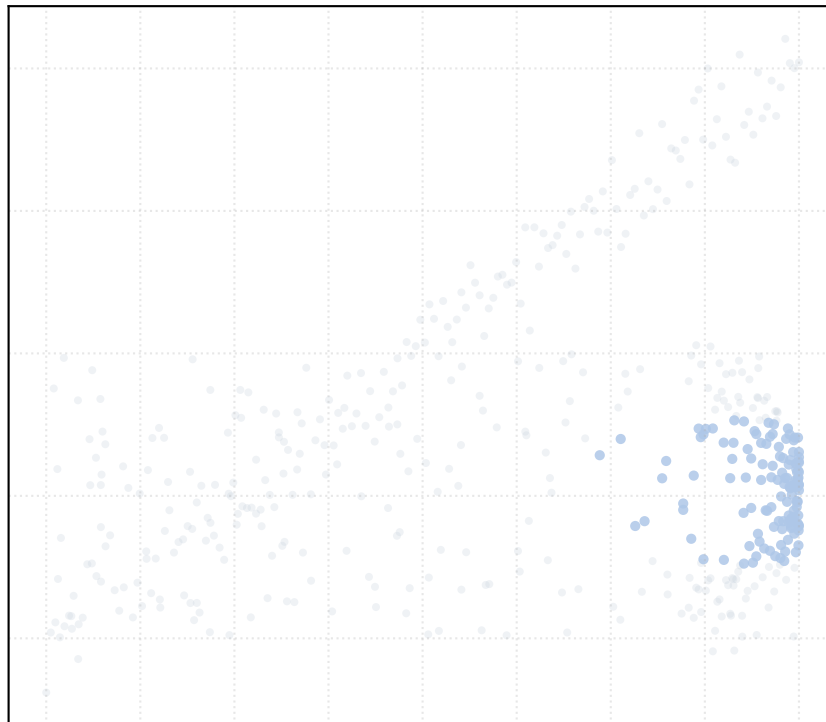
r = 1.5 [1 Clusters]



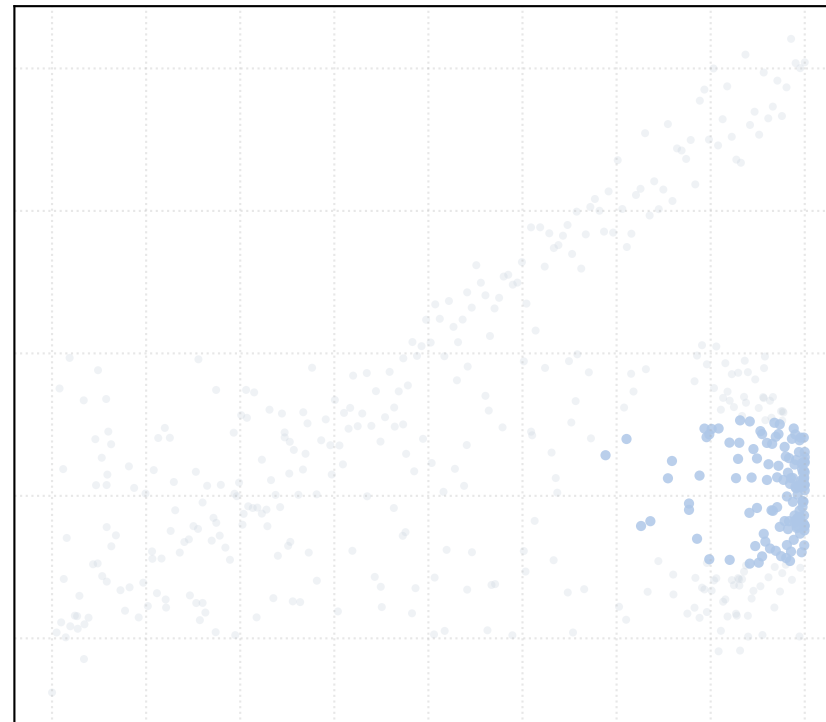
r = 2.0 [1 Clusters]



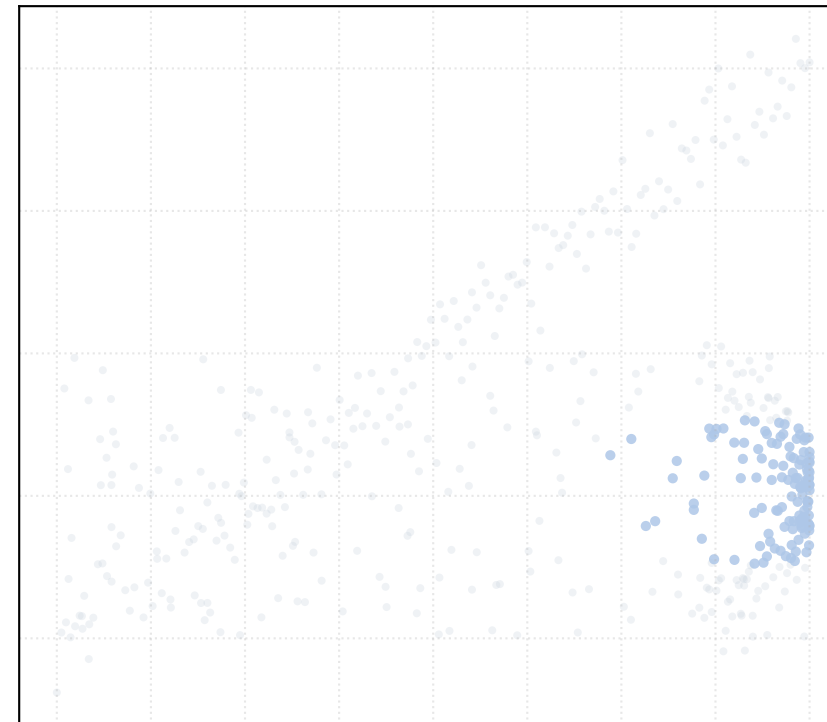
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

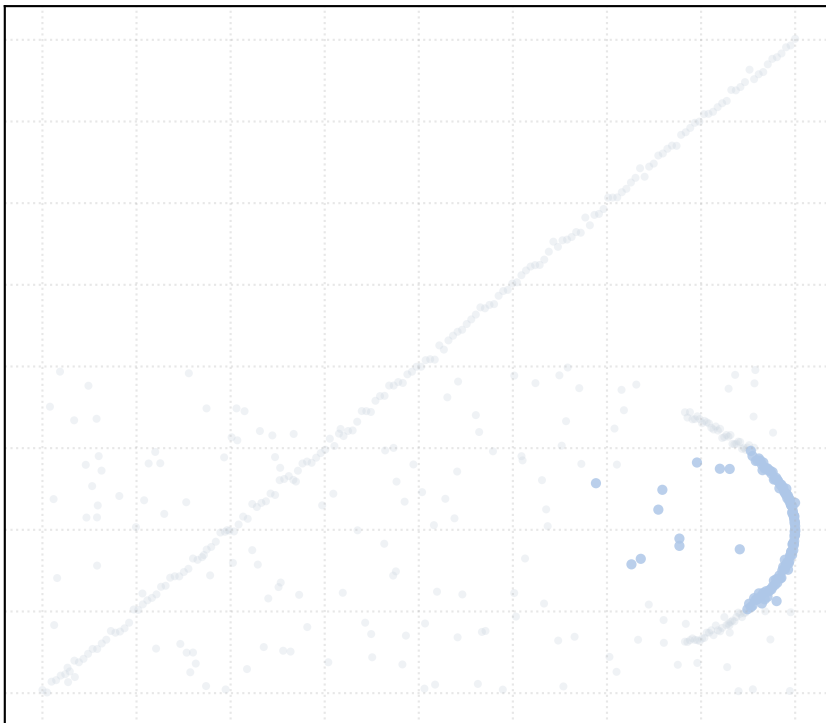


r = 3.5 [1 Clusters]

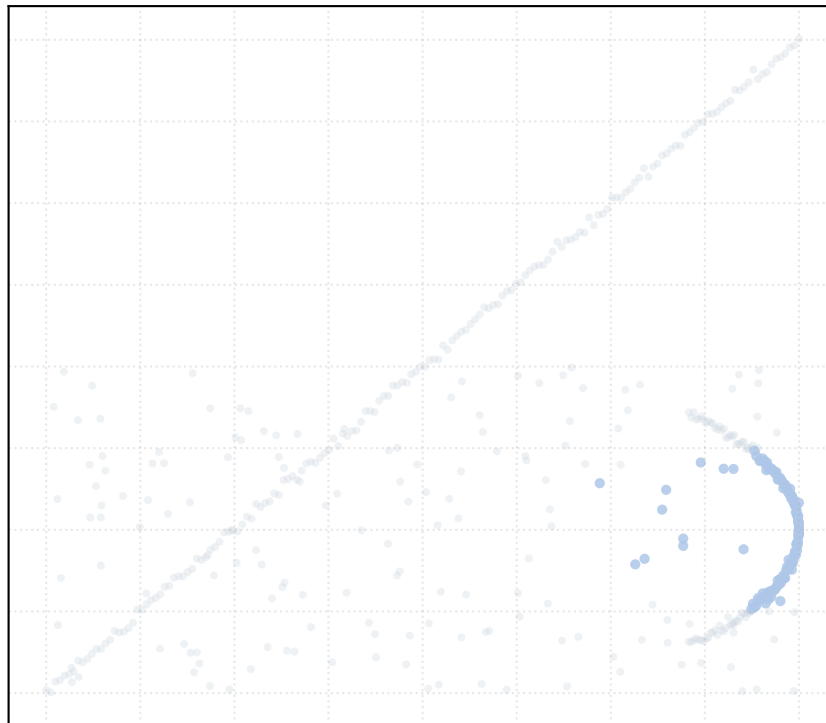


Dataset: mixed_generators_noise_low | Search Radius Sweep (r in grow_ensemble)

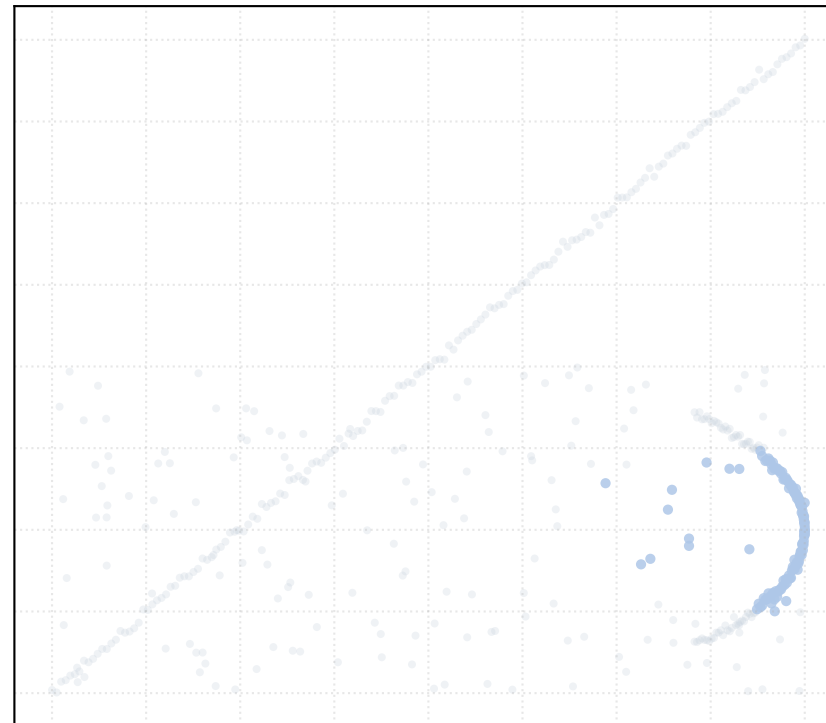
r = Default (Adaptive) [1 Clusters]



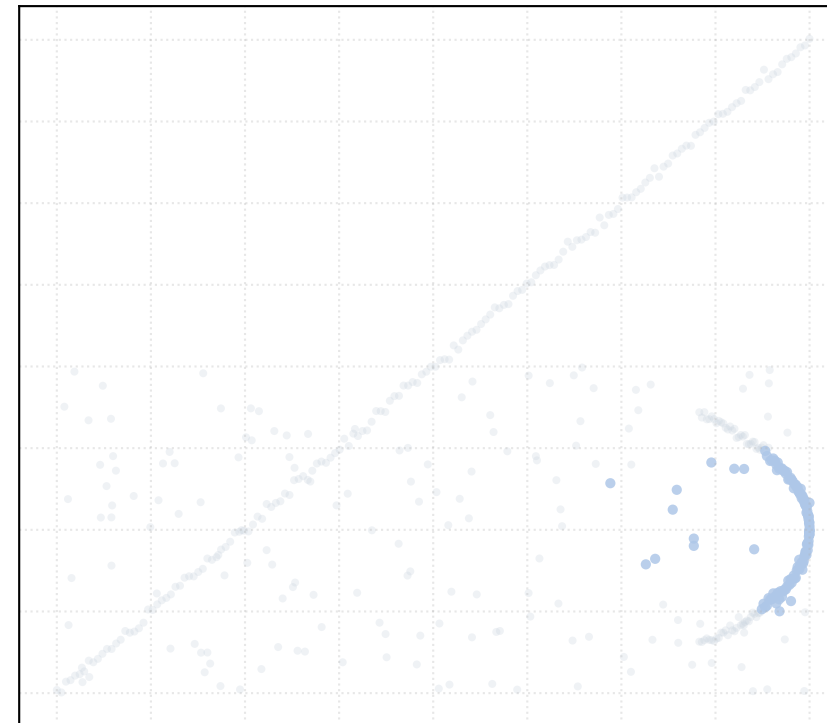
r = 0.5 [1 Clusters]



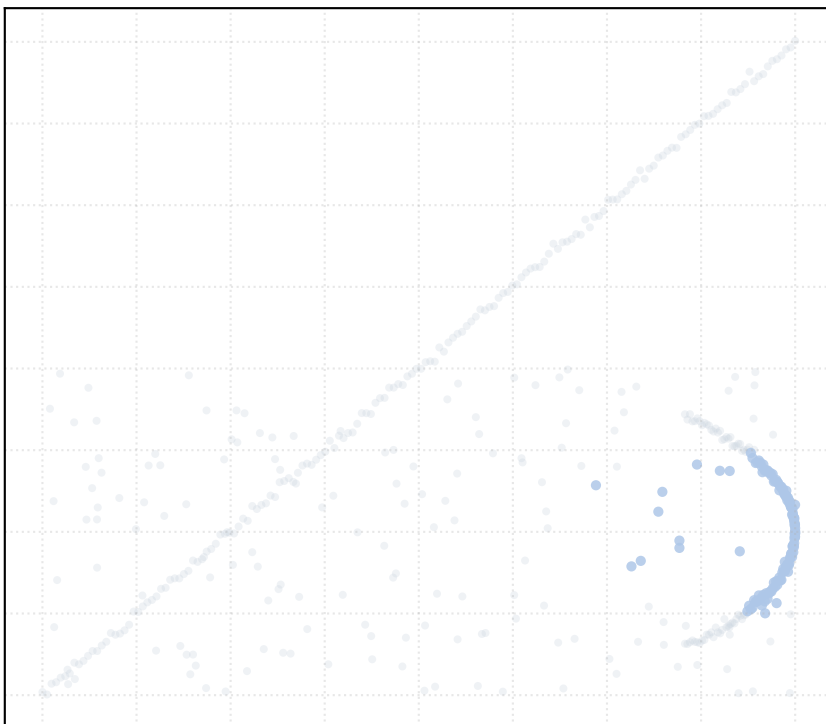
r = 1.0 [1 Clusters]



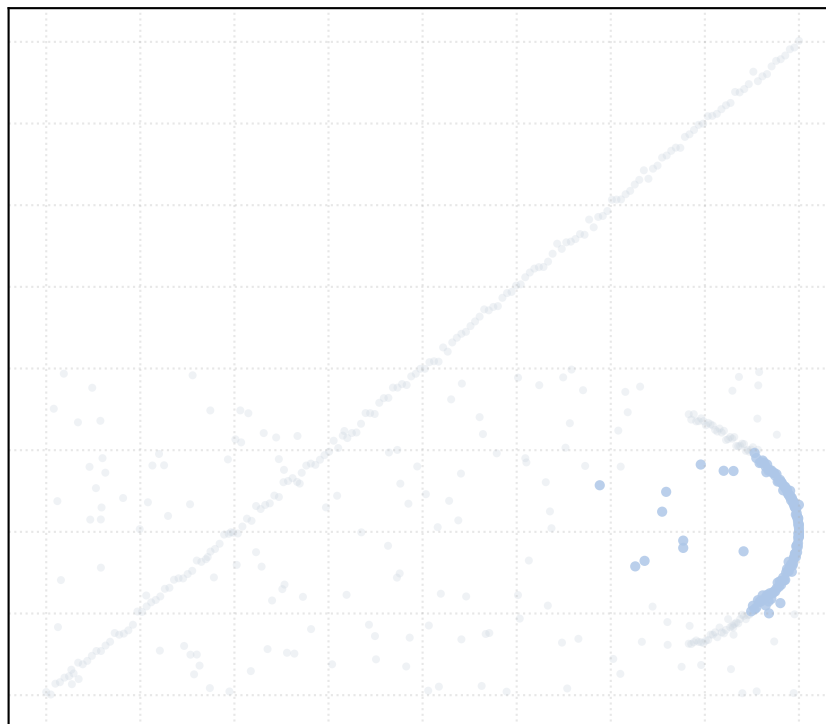
r = 1.5 [1 Clusters]



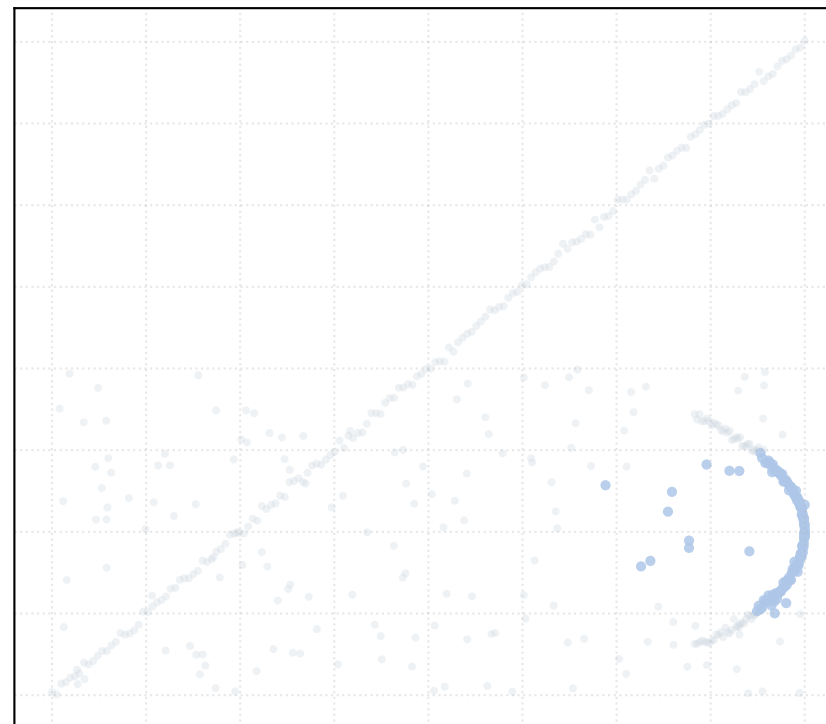
r = 2.0 [1 Clusters]



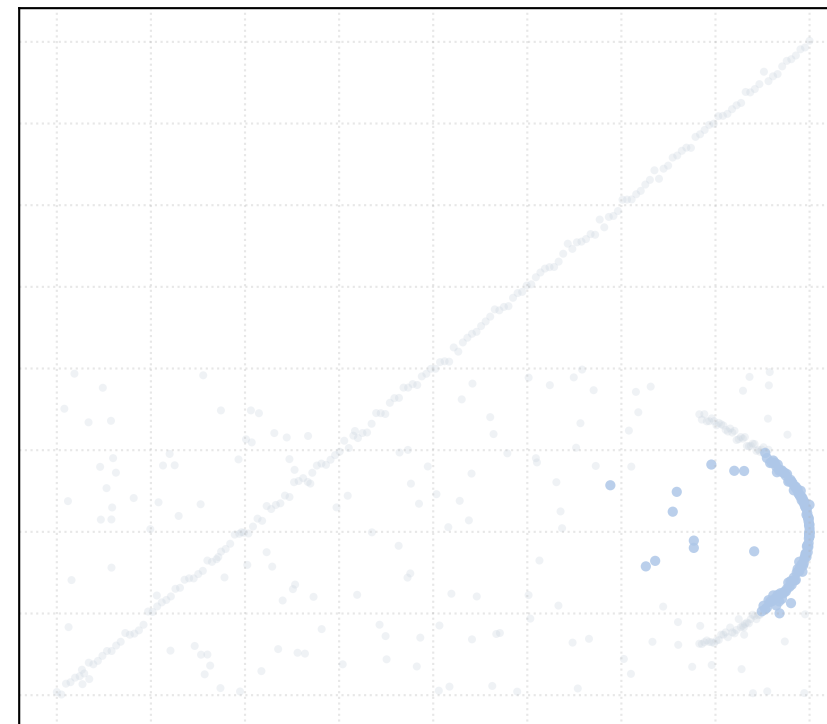
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

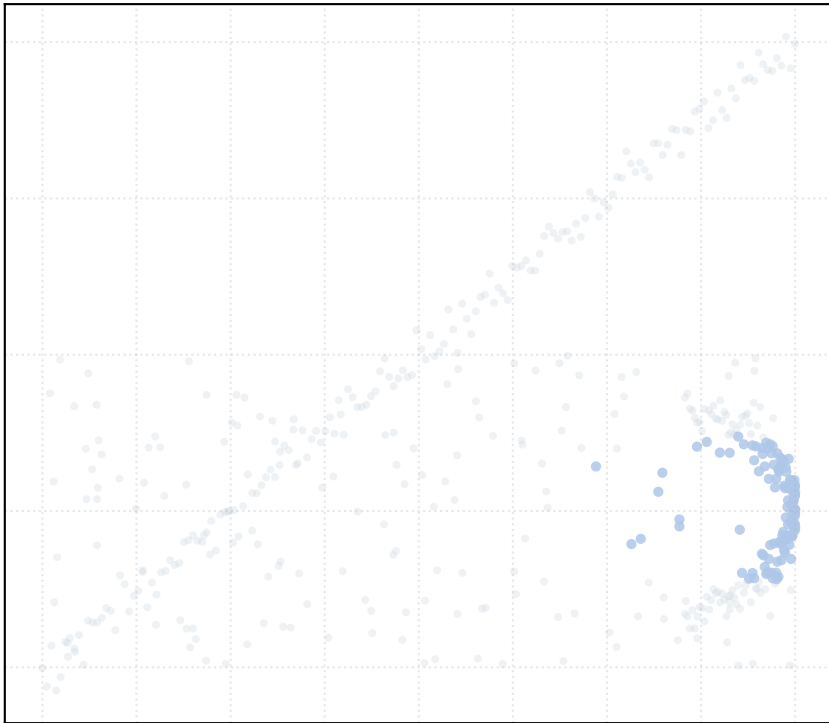


r = 3.5 [1 Clusters]

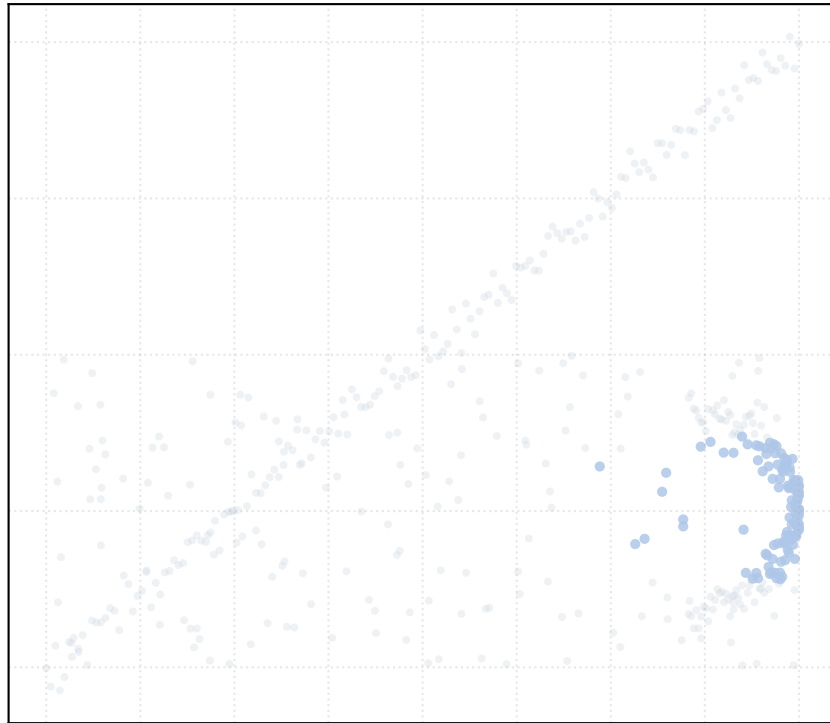


Dataset: mixed_generators_noise_medium | Search Radius Sweep (r in grow_ensemble)

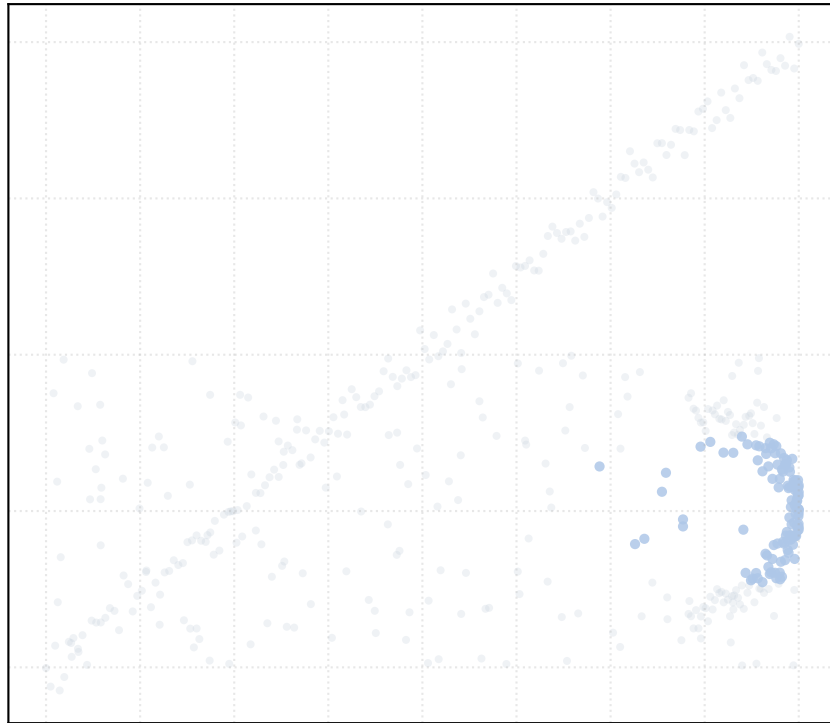
r = Default (Adaptive) [1 Clusters]



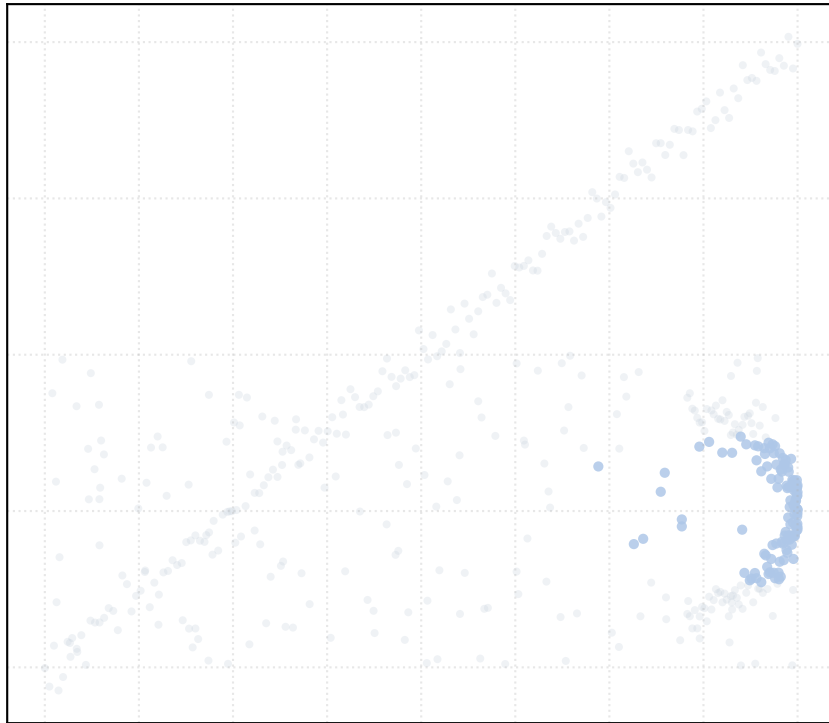
r = 0.5 [1 Clusters]



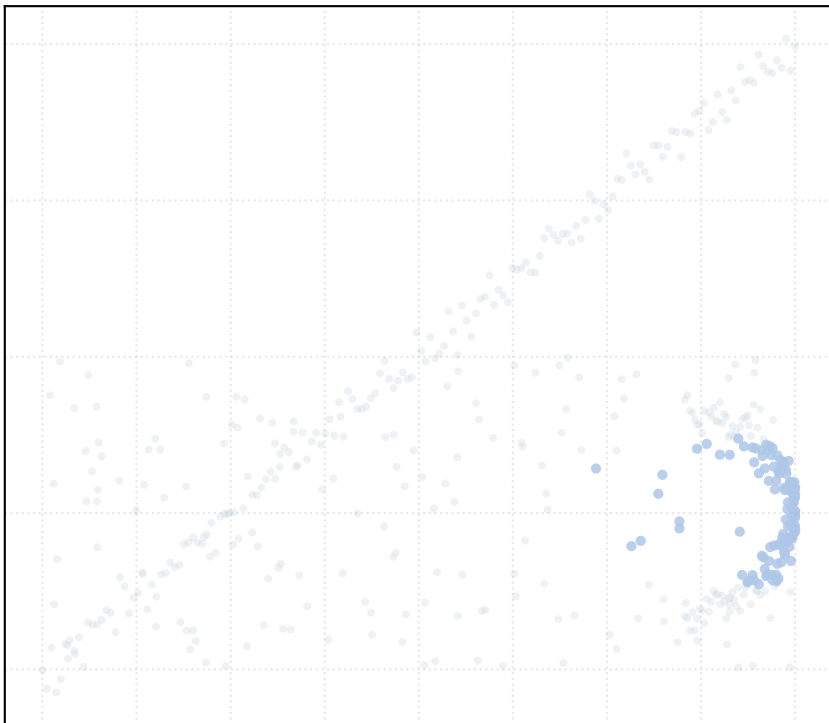
r = 1.0 [1 Clusters]



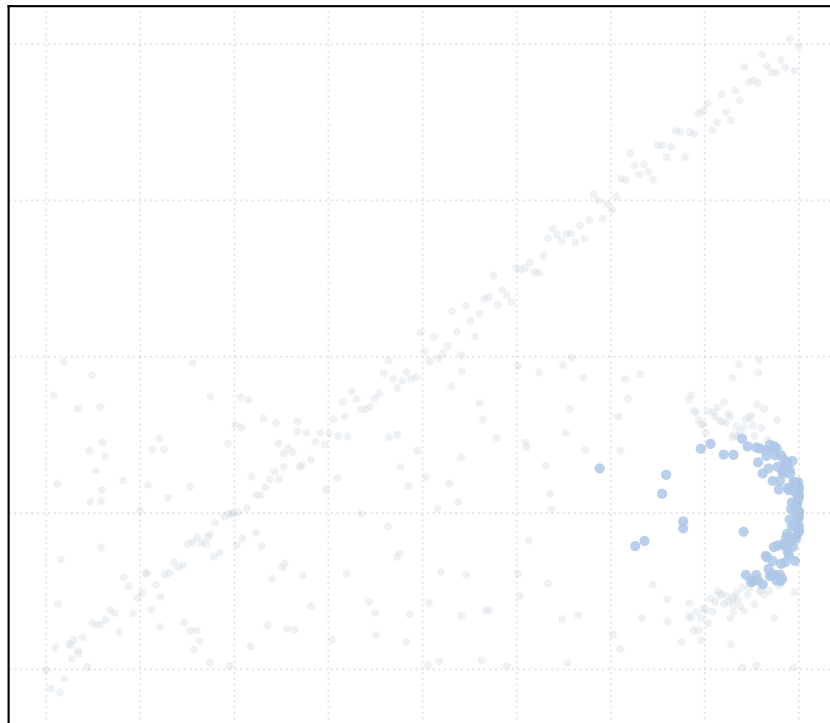
r = 1.5 [1 Clusters]



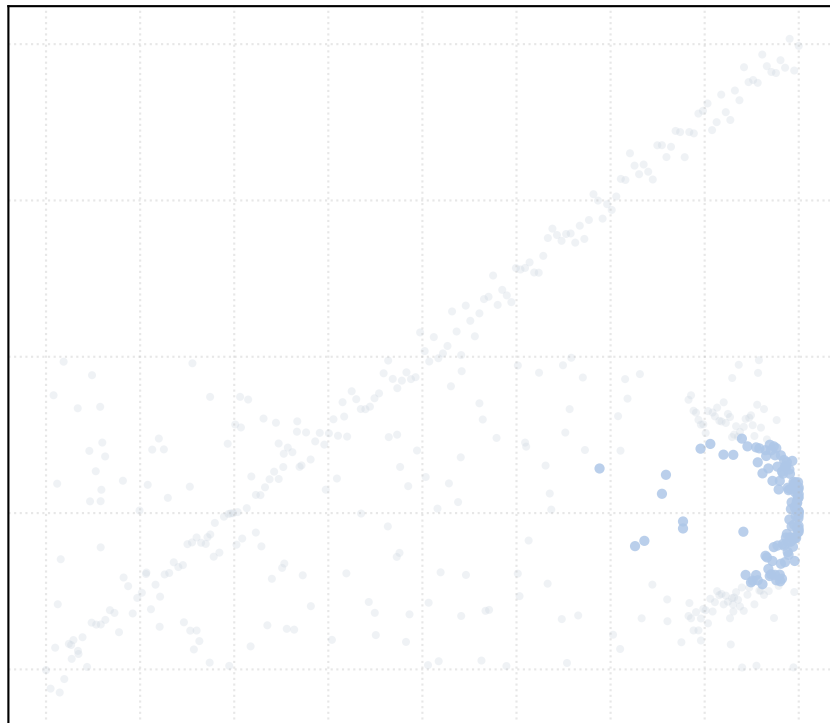
r = 2.0 [1 Clusters]



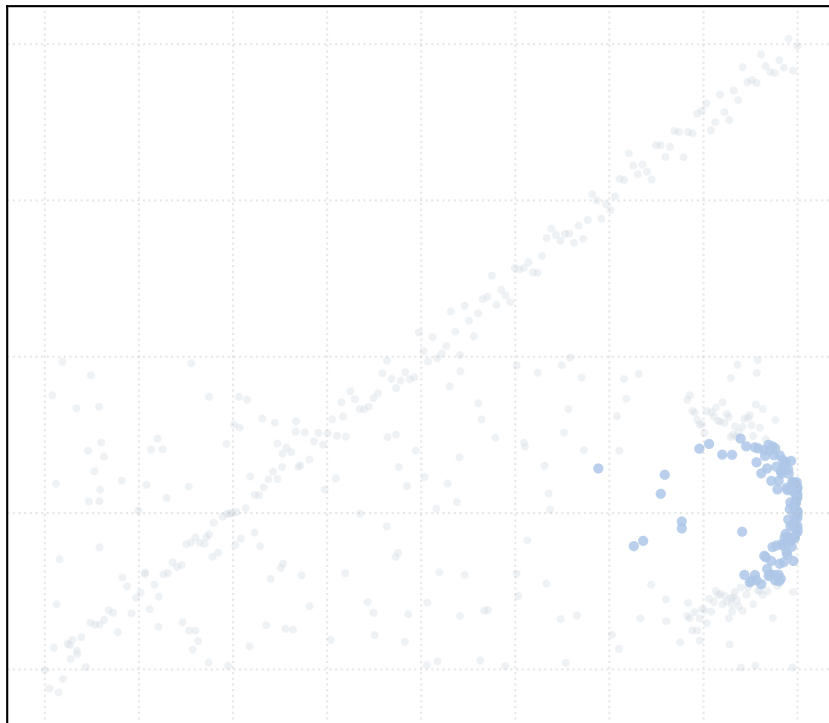
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

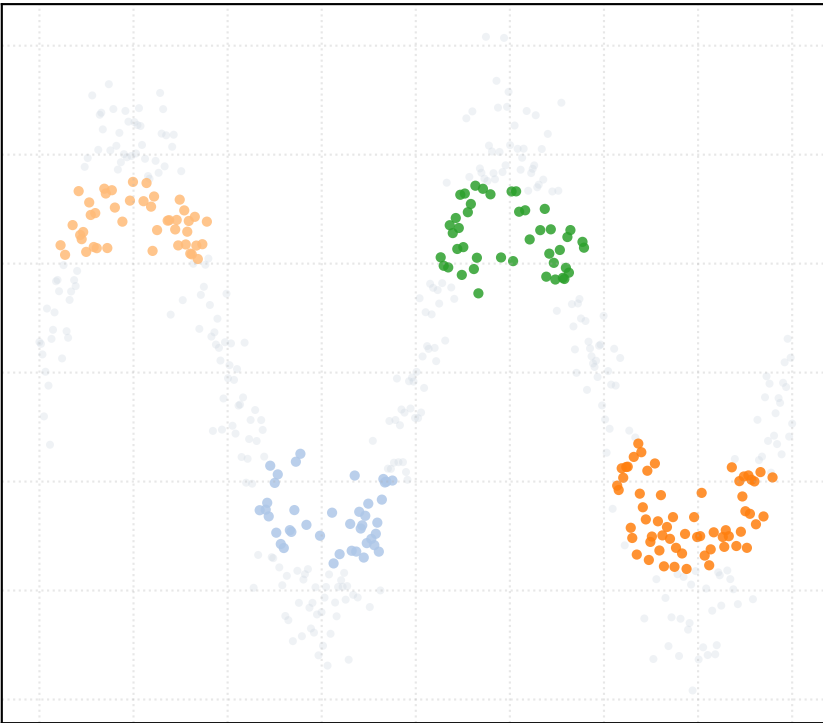


r = 3.5 [1 Clusters]

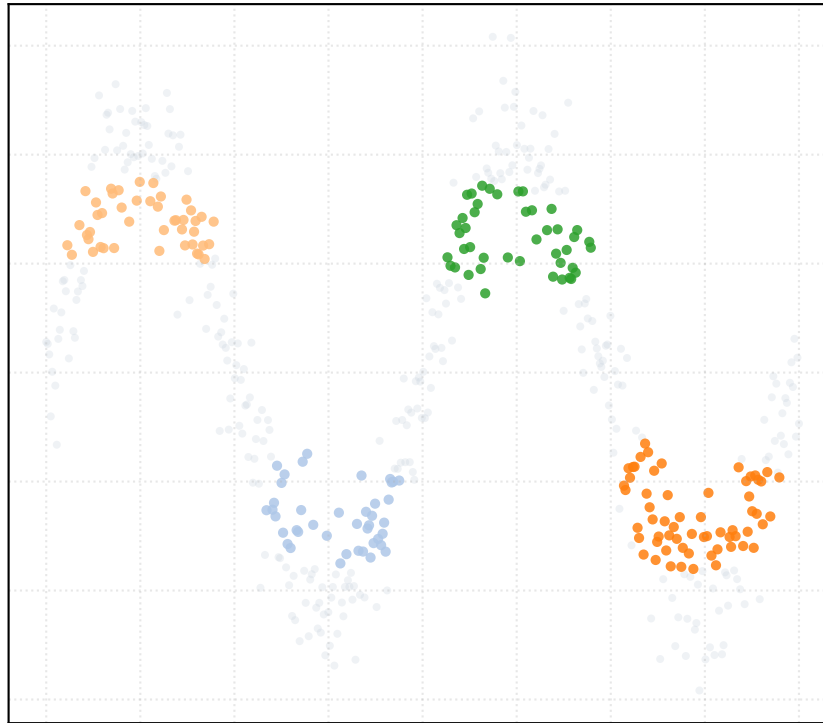


Dataset: s_curve_noise_high | Search Radius Sweep (r in grow_ensemble)

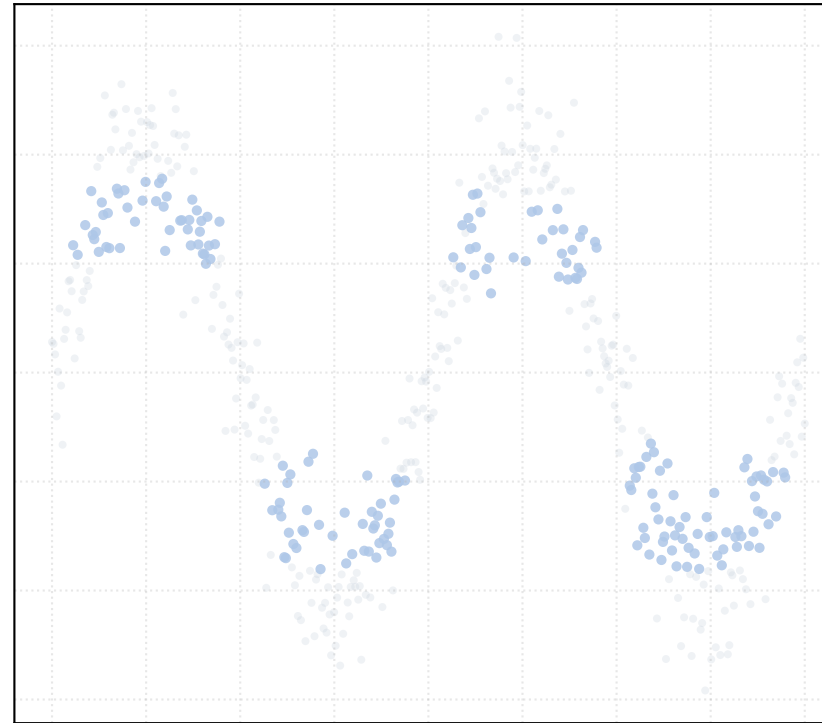
r = Default (Adaptive) [4 Clusters]



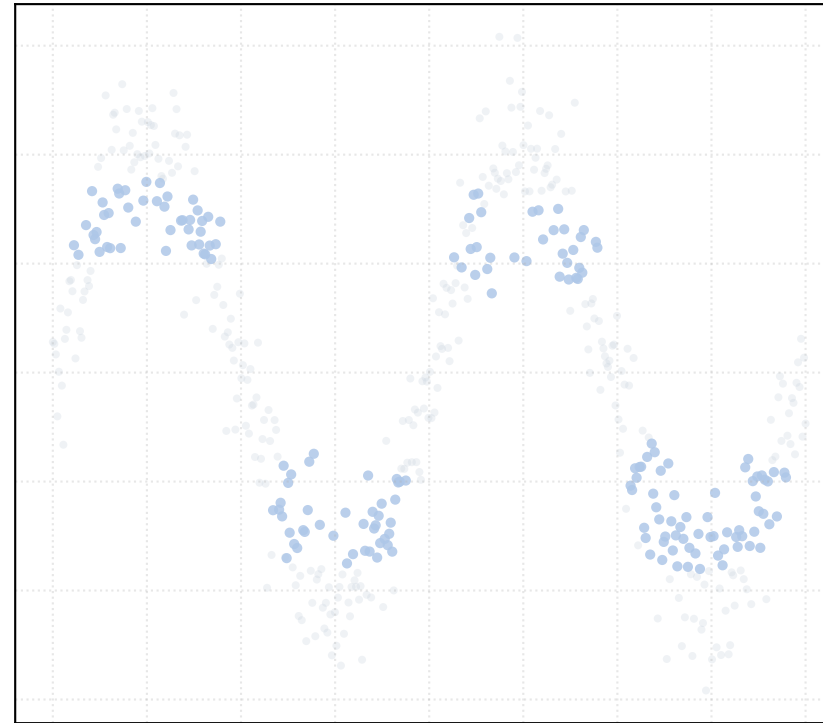
r = 0.5 [4 Clusters]



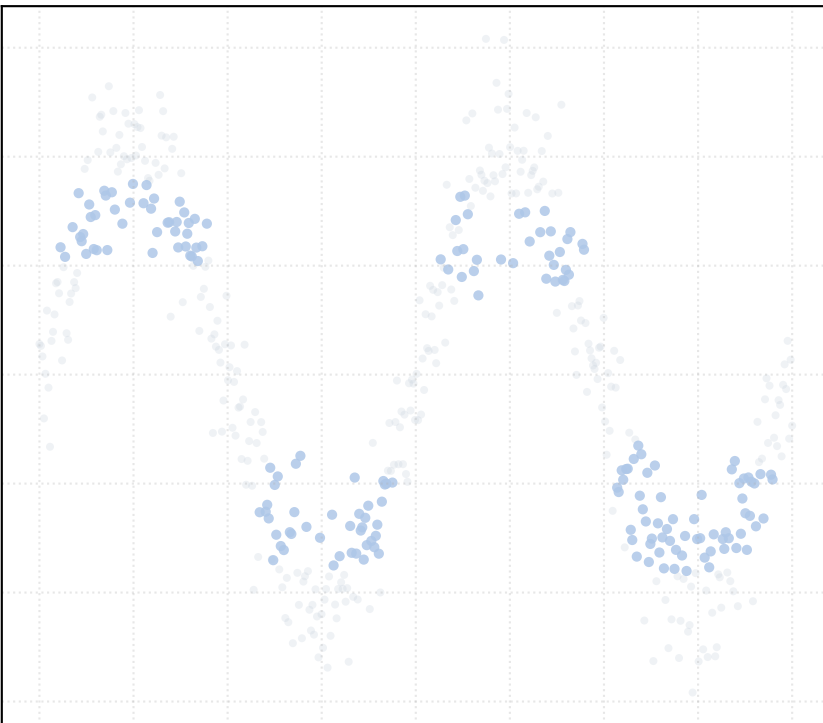
r = 1.0 [1 Clusters]



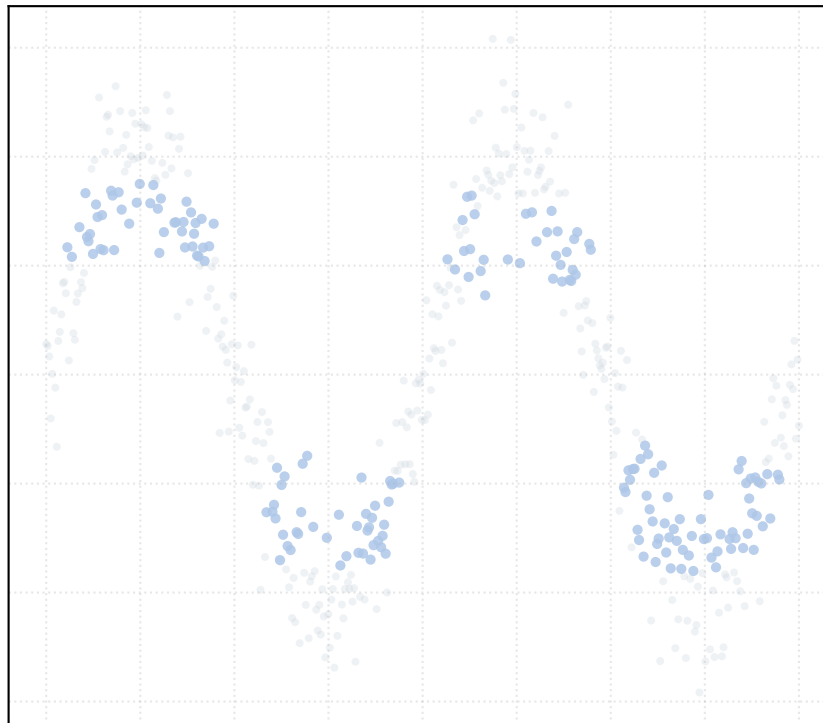
r = 1.5 [1 Clusters]



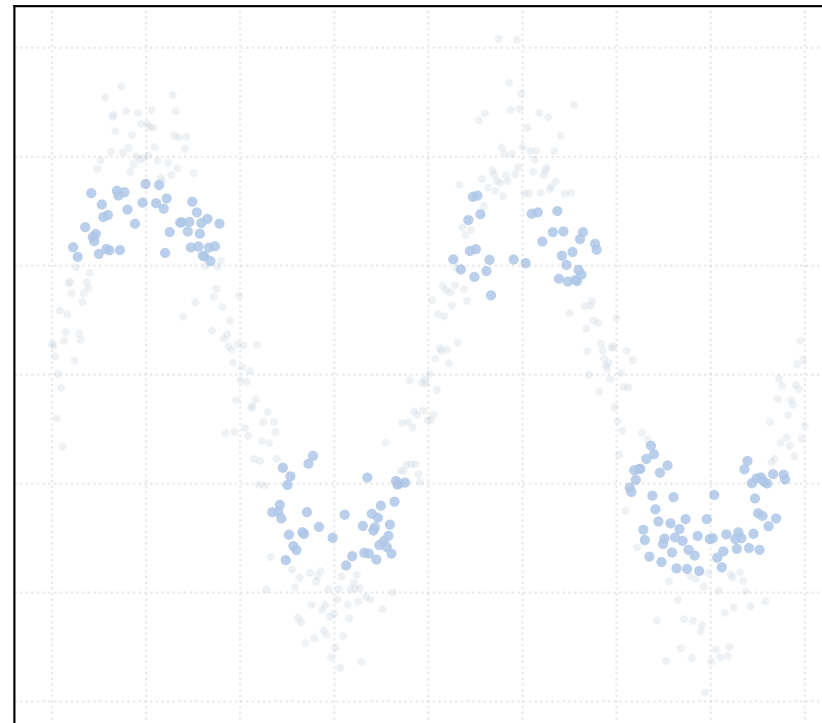
r = 2.0 [1 Clusters]



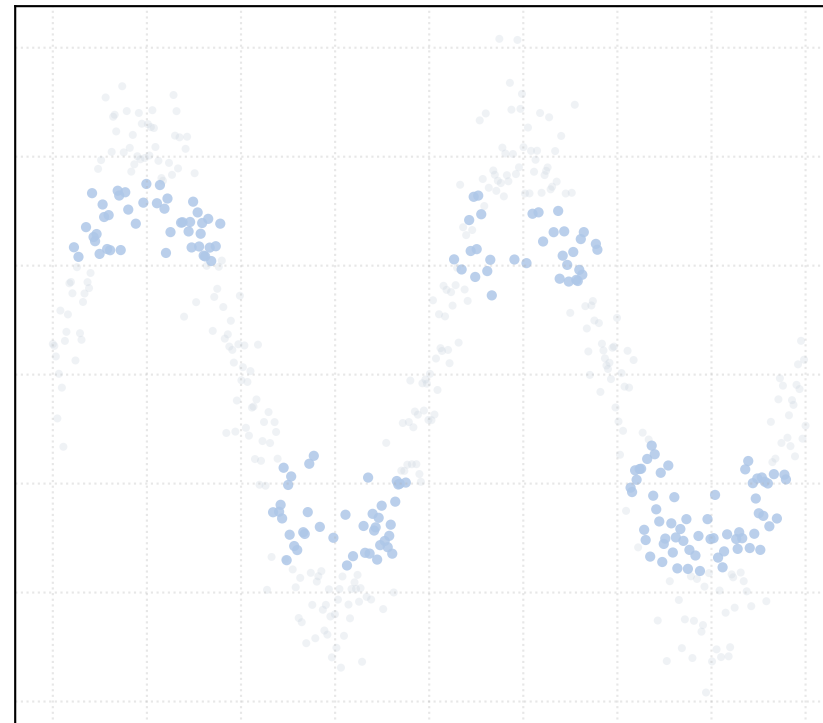
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

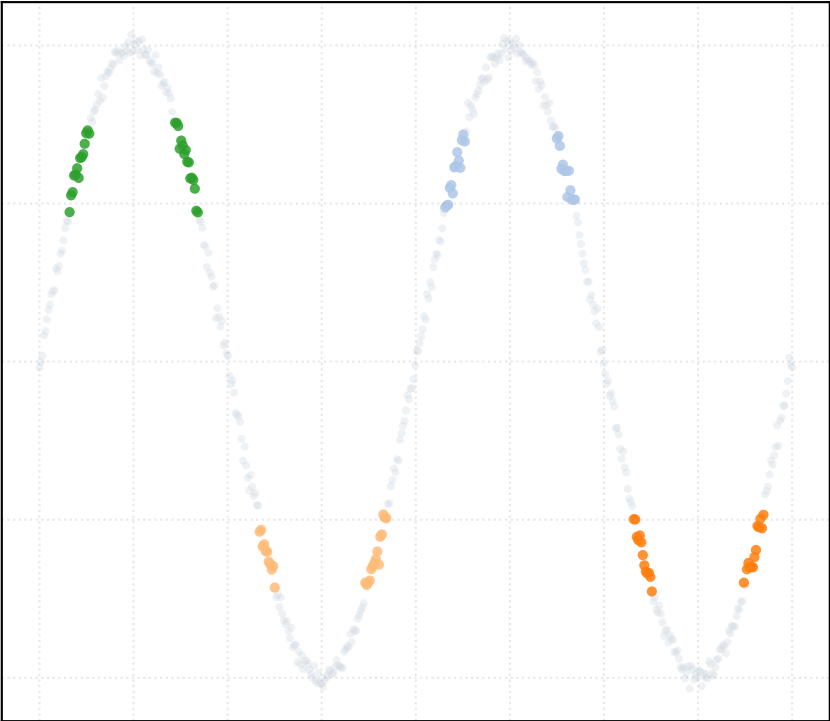


r = 3.5 [1 Clusters]

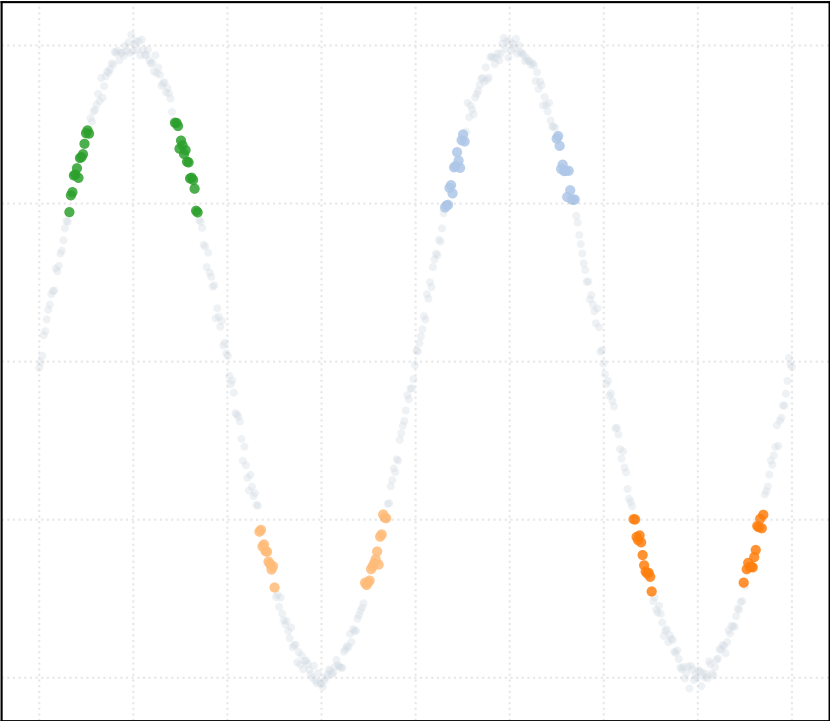


Dataset: s_curve_noise_low | Search Radius Sweep (r in grow_ensemble)

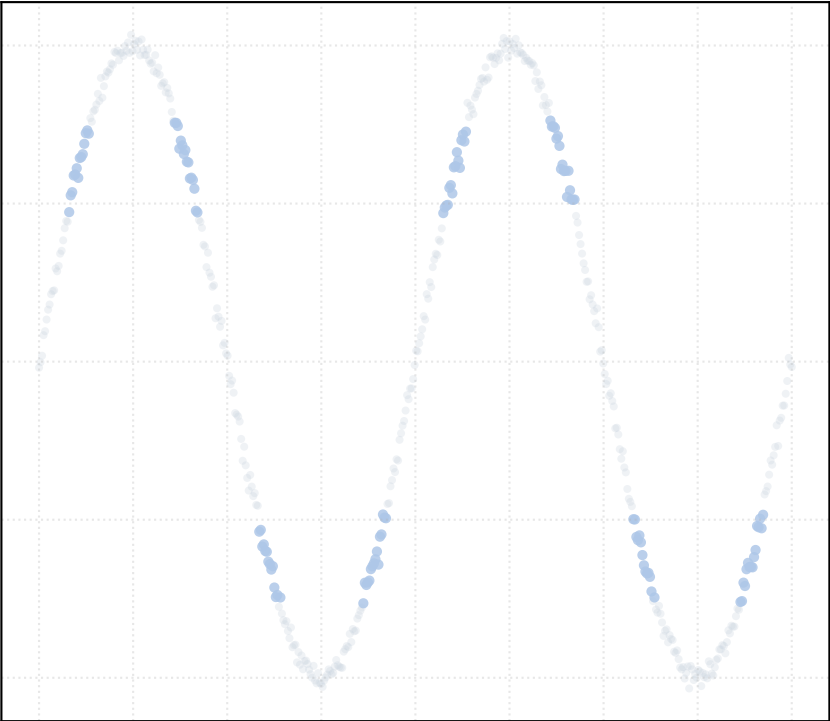
r = Default (Adaptive) [4 Clusters]



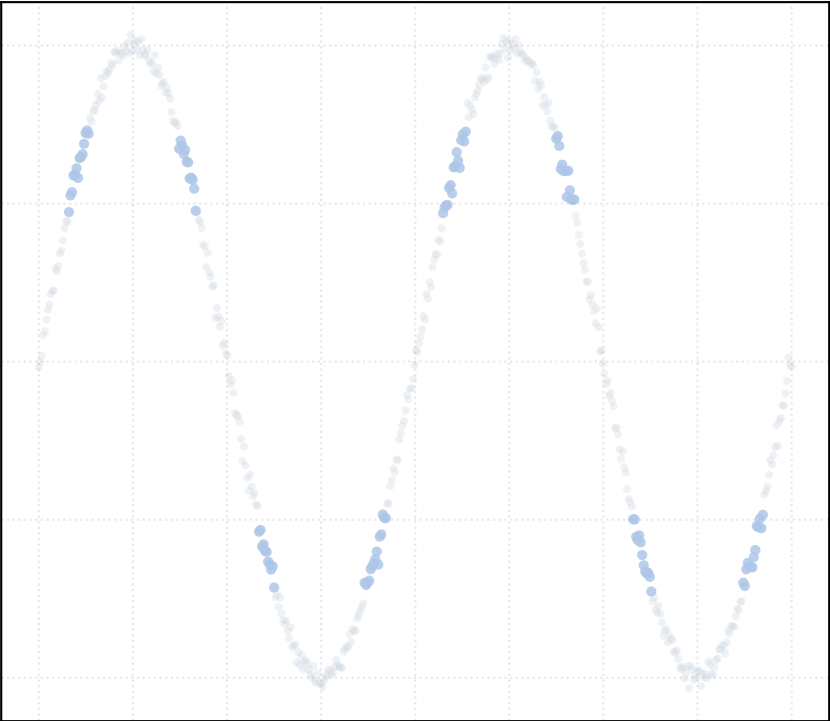
r = 0.5 [4 Clusters]



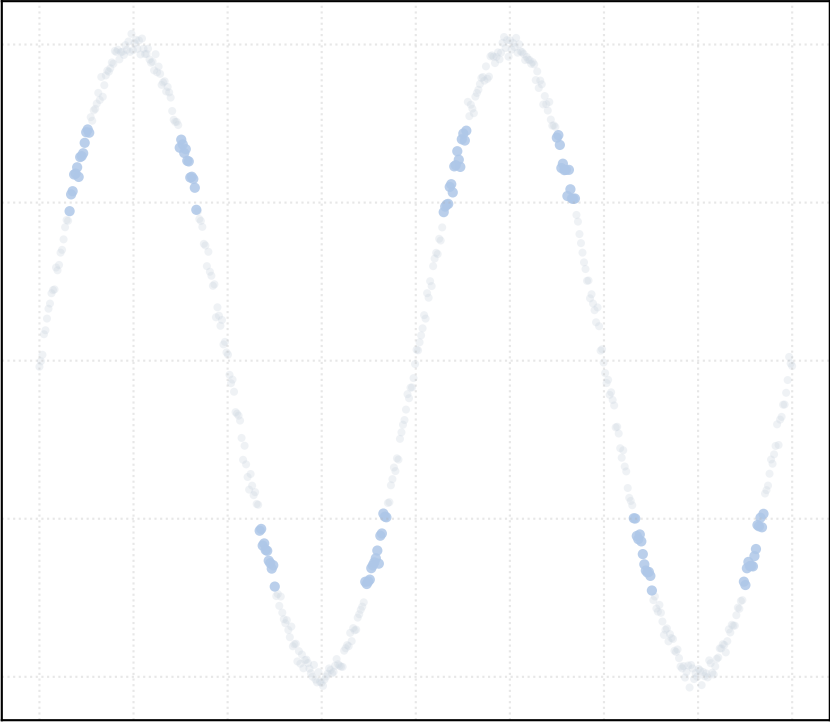
r = 1.0 [1 Clusters]



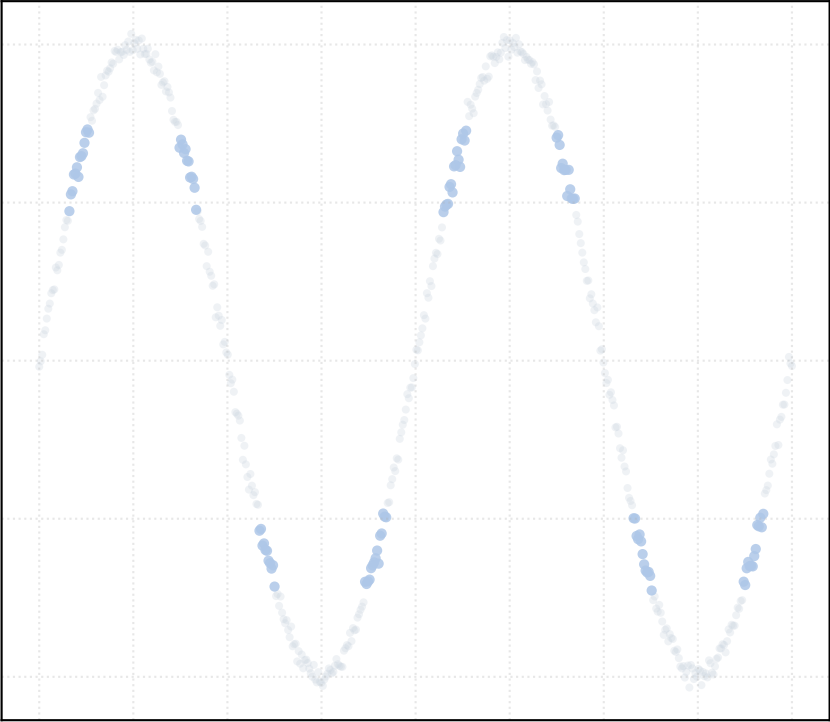
r = 1.5 [1 Clusters]



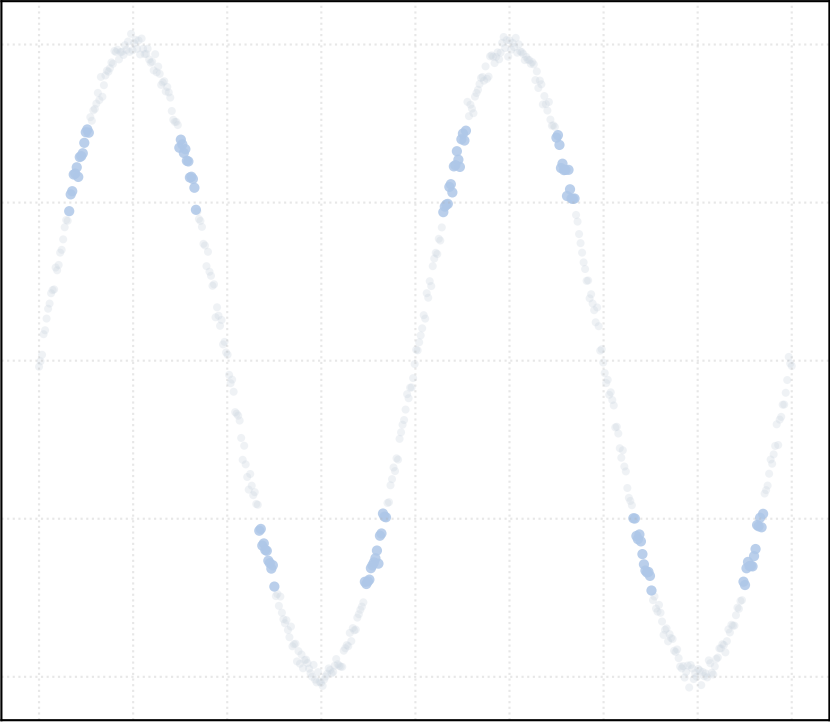
r = 2.0 [1 Clusters]



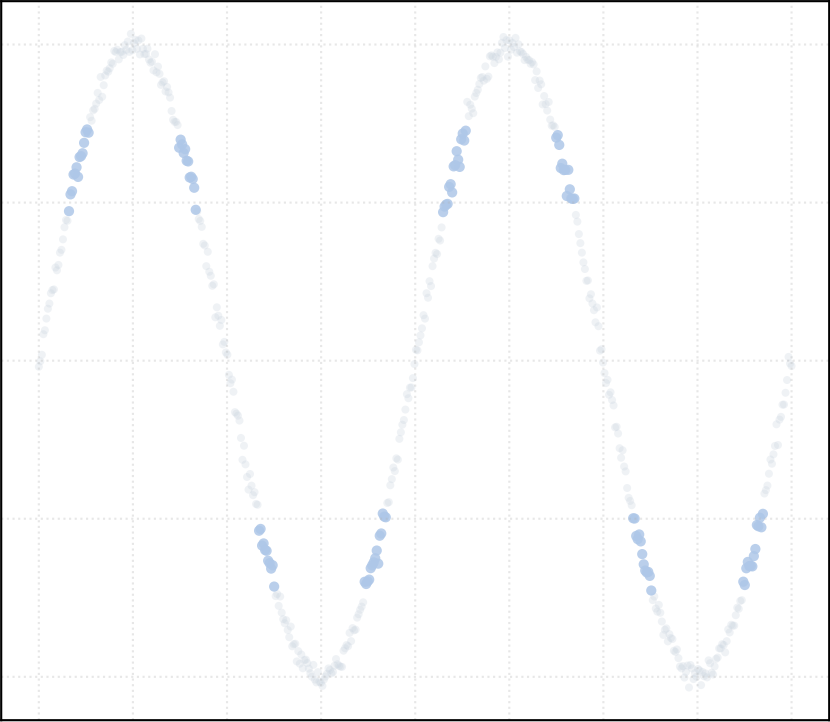
r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]

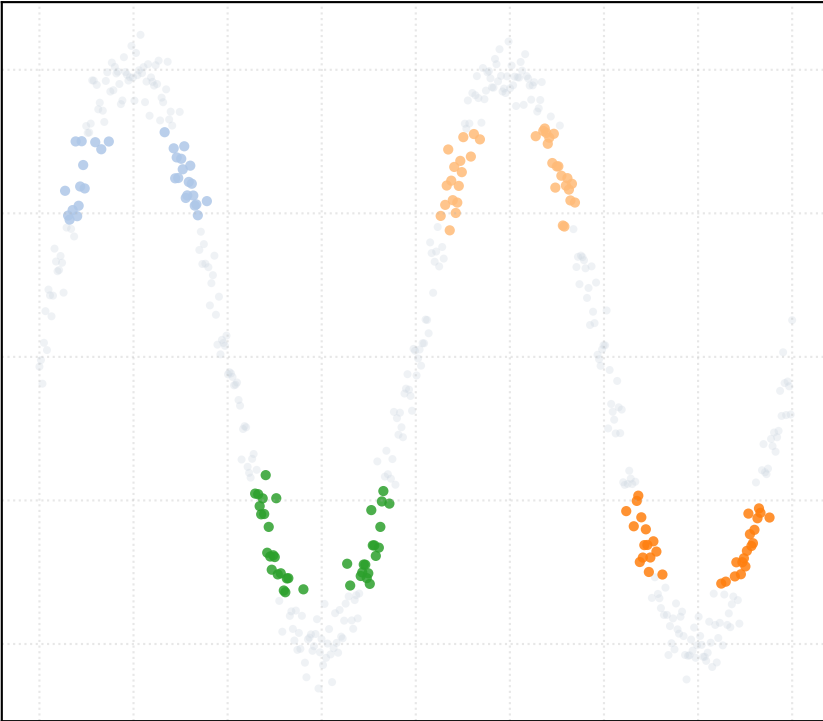


r = 3.5 [1 Clusters]

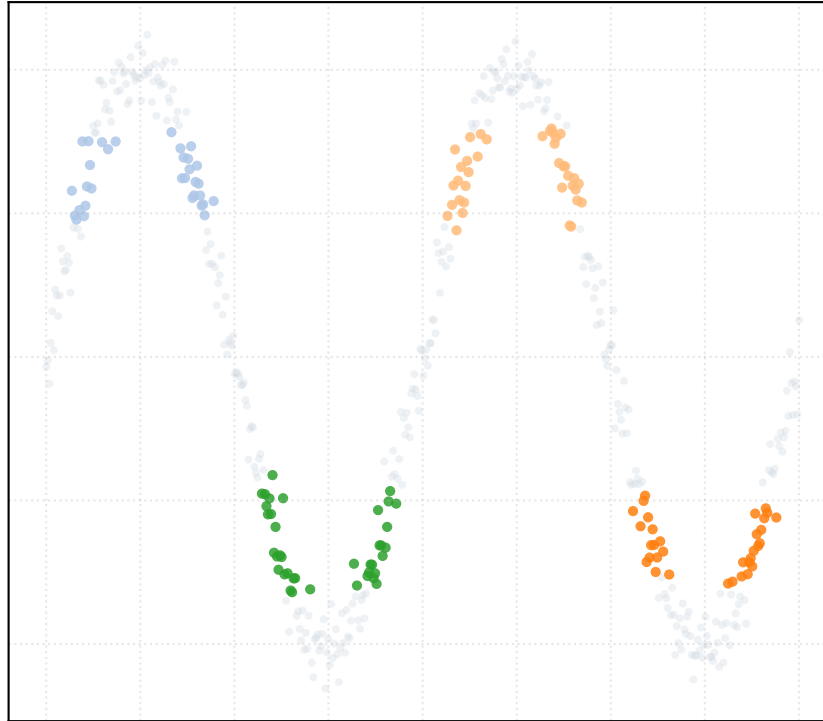


Dataset: s_curve_noise_medium | Search Radius Sweep (r in grow_ensemble)

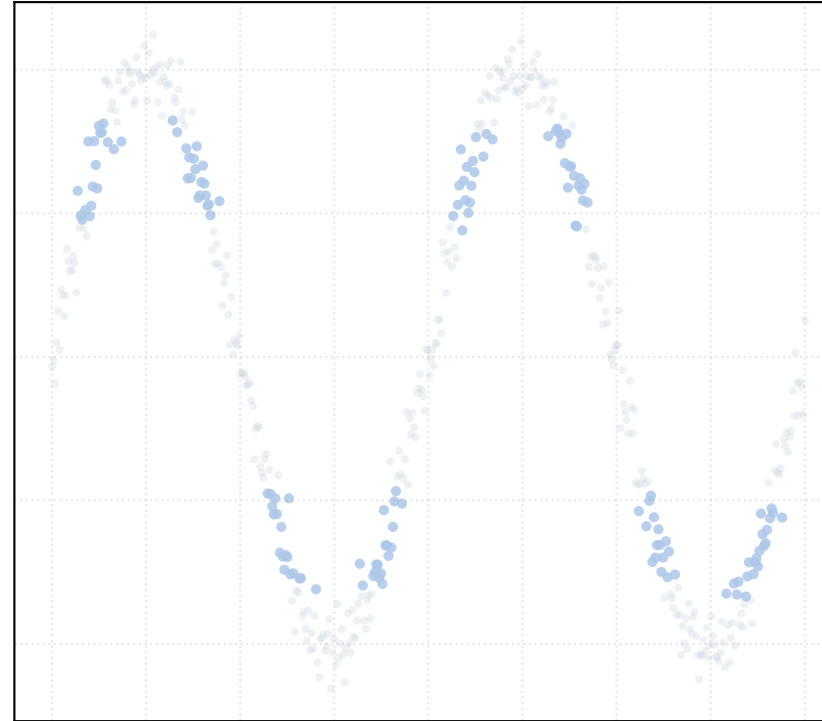
r = Default (Adaptive) [4 Clusters]



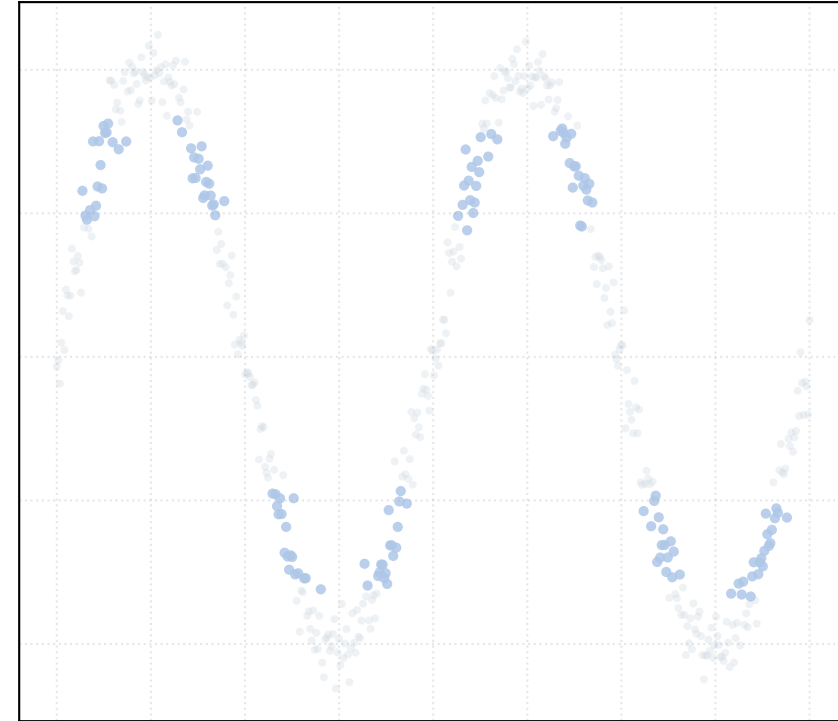
r = 0.5 [4 Clusters]



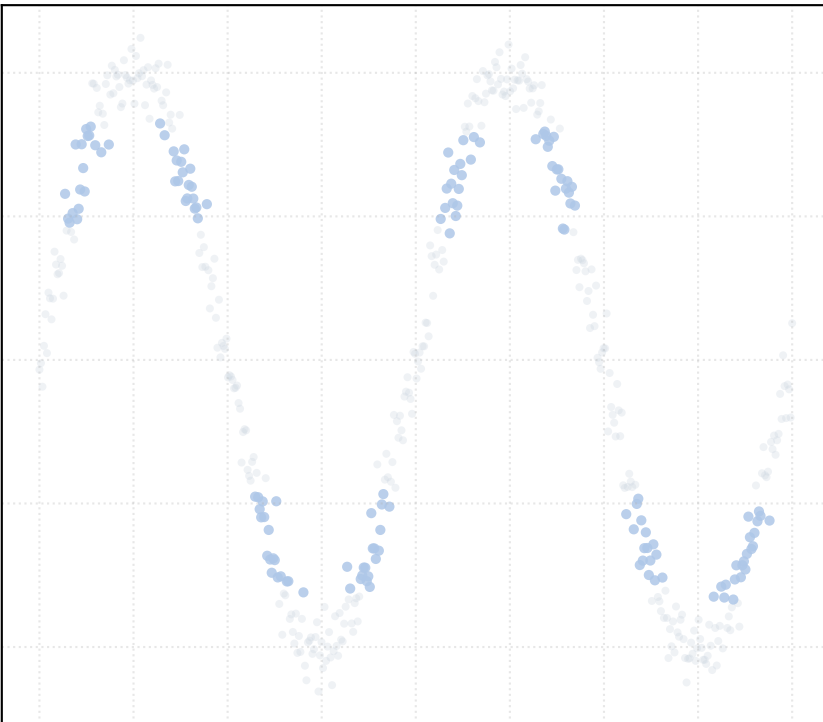
r = 1.0 [1 Clusters]



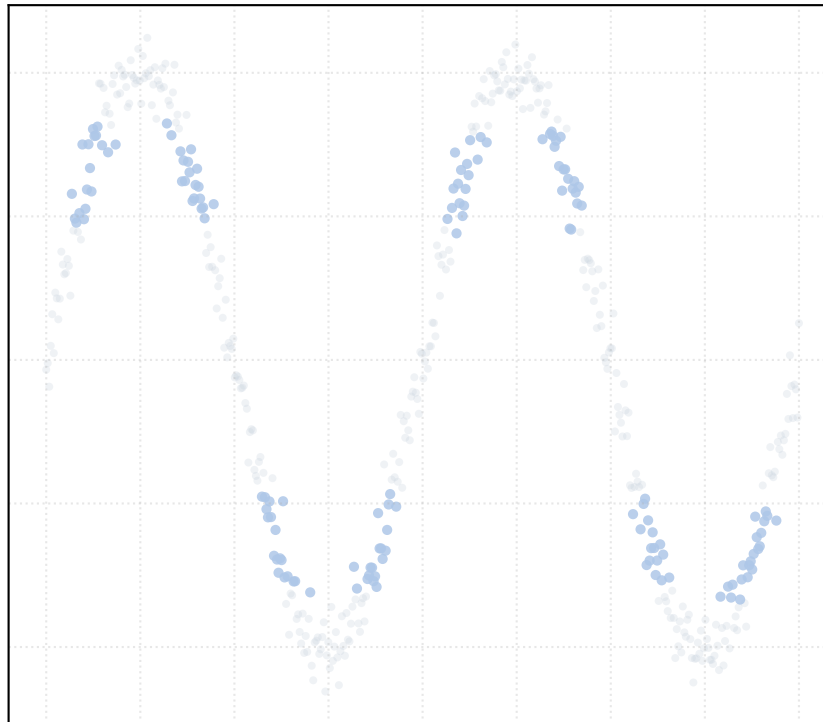
r = 1.5 [1 Clusters]



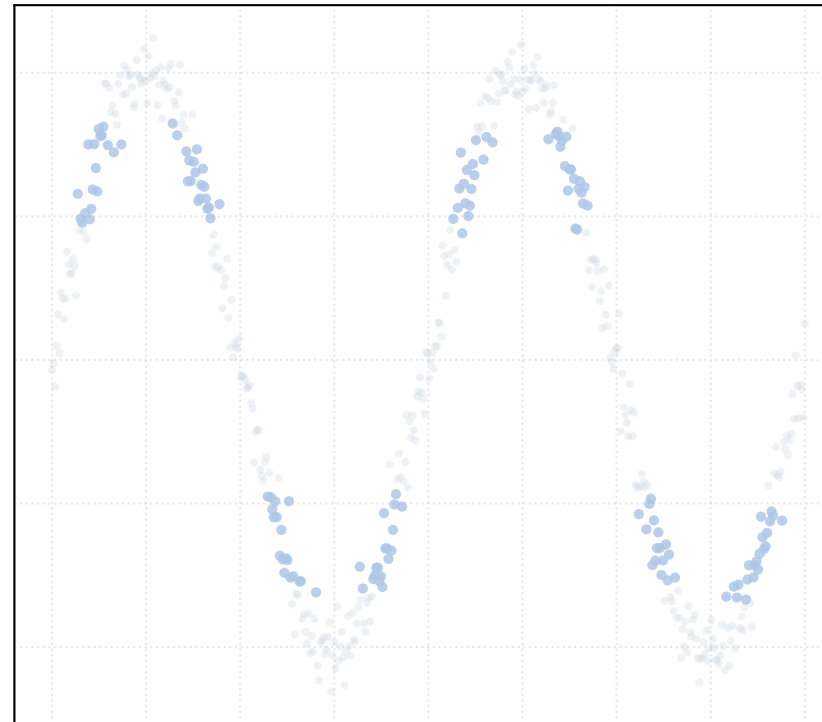
r = 2.0 [1 Clusters]



r = 2.5 [1 Clusters]



r = 3.0 [1 Clusters]



r = 3.5 [1 Clusters]

